

Fall 2024 CS 4641/7641 A: Machine Learning Homework 4

Instructor: Dr. Mahdi Roozbahani

Deadline: Monday, Dec 2nd, 2024 11:59 pm EST

For Homework 4, the December 2nd deadline is a hard and strict deadline. This means that this deadline cannot be even extended for students with GT-approved accommodations.

- Discussion is encouraged on Ed as part of the Q/A. However, all assignments should be done individually.
- Plagiarism is a **serious offense**. You are responsible for completing your own work. You are not allowed to copy and paste, or paraphrase, or submit materials created or published by others, as if you created the materials. All materials submitted must be your own.
- All incidents of suspected dishonesty, plagiarism, or violations of the Georgia Tech Honor Code will be subject to the institute's Academic Integrity procedures. If we observe any (even small) similarities/plagiarisms detected by Gradescope or our TAs, **WE WILL DIRECTLY REPORT ALL CASES TO OSI**, which may, unfortunately, lead to a very harsh outcome. **Consequences can be severe, e.g., academic probation or dismissal, grade penalties, a 0 grade for assignments concerned, and prohibition from withdrawing from the class.**

Instructions for the assignment

- This assignment consists of both programming and theory questions.
- Unless a theory question explicitly states that no work is required to be shown, you must provide an explanation, justification, or calculation for your answer.
- To switch between cell for code and for markdown, see the menu -> Cell -> Cell Type
- You can directly type Latex equations into markdown cells.
- If a question requires a picture, you could use this syntax `<img src="" style="width: 300px;"/>` to include them within your ipython notebook.
- Your write up must be submitted in PDF format. Please ensure all questions are answered within the Jupyter Notebook using either Markdown or LaTeX. **We will NOT accept handwritten work.** Make sure that your work is formatted correctly, for example submit $\sum_{i=0} x_i$ instead of `\text{sum}_{i=0} x_i`
- When submitting the non-programming part of your assignment, you must correctly map pages of your PDF to each question/subquestion to reflect where they appear. **Improperly mapped questions may not be graded correctly and/or will result in point deductions for the error.**

- All assignments should be done individually, and each student must write up and submit their own answers.
- **Graduate Students:** You are required to complete any sections marked as Bonus for Undergrads

Using the autograder

- You will find three assignments (for grads) on Gradescope that correspond to HW4: "Assignment 4 Programming", "Assignment 4 - Non-programming" and "Assignment 4 Programming - Bonus for all". Undergrads will have an additional assignment called "Assignment 4 Programming - Bonus for Undergrads".
- You will submit your code for the autograder in the Assignment 4 Programming sections. Please refer to the Deliverables and Point Distribution section for what parts are considered required, bonus for undergrads, and bonus for all".
- We provided you different .py files and we added libraries in those files please DO NOT remove those lines and add your code after those lines. Note that these are the only allowed libraries that you can use for the homework
- You are allowed to make as many submissions until the deadline as you like. Additionally, note that the autograder tests each function separately, therefore it can serve as a useful tool to help you debug your code if you are not sure of what part of your implementation might have an issue
- **For the "Assignment 4 - Non-programming" part, you will need to submit to Gradescope a PDF copy of your Jupyter Notebook with the cells ran. [See this EdStem Post for multiple ways on to convert your .ipynb into a .pdf file](#).** Please refer to the **Deliverables and Point Distribution** section for an outline of the non-programming questions.
- **When submitting to Gradescope, please make sure to mark the page(s) corresponding to each problem/sub-problem. The pages in the PDF should be of size 8.5" x 11", otherwise there may be a deduction in points for extra long sheets.**
- You **MUST** pass the Autograder Test to gain points for the programming section. There will not be any partial credit or manual grading for this part.

Using the local tests

- For some of the programming questions we have included a local test using a small toy dataset to aid in debugging. The local test sample data and outputs are stored in localtests.py
- There are no points associated with passing or failing the local tests, you must still pass the autograder to get points.
- **It is possible to fail the local test and pass the autograder** since the autograder has a certain allowed error tolerance while the local test allowed error may be smaller. Likewise, passing the local tests does not guarantee passing the autograder.
- **You do not need to pass both local and autograder tests to get points, passing the Gradescope autograder is sufficient for credit.**
- It might be helpful to comment out the tests for functions that have not been completed yet.
- It is recommended to test the functions as it gets completed instead of completing the whole class and then testing. This may help in isolating errors. Do not solely rely on the local tests, continue to test on the autograder regularly as well.

Deliverables and Points Distribution

Q1: Classification with Two Layer NN [80 pts: 55pts + 25pts Grad / 3.3% Undergrad Bonus]

Deliverables: **NN.py**

- **1.1 NN Implementation** [65pts: 50pts + 15pts Grad / 2% **Bonus for Undergrad**] - *programming*
 - SiLU [5pts]
 - Softmax [5pts]
 - Cross Entropy loss [5pts]
 - dropout [5pts]
 - forward propagation and with and without dropout [5pts + 5pts]
 - compute gradients and update weights [2.5pts + 2.5pts]
 - backward without momentum [5pt]
 - Gradient Descent [5pts]
 - Batch Gradient Descent [10pts Grad / 1.3% **Bonus for Undergrad**]
 - Momentum [5pts Grad / 0.7% **Bonus for Undergrad**]
- **1.2 Loss plot and CE for Gradient Descent** [5pts] - *programming*
- **1.3 Loss plot and CE for Batch Gradient Descent** [5pts Grad / 0.7% **Bonus for Undergrad**] - *programming*
- **1.4 Loss plot and CE value for NN with Gradient Descent with Momentum** [5pts Grad / 0.6% **Bonus for Undergrad**] - *programming*

Q2: CNN [20pts Grad / 2.7% Bonus for Undergrad + 1.1% Bonus for All]

Deliverables: **cnn.py, cnn_image_transformations.py and Written Report**

- **2.1 Image Classification using Pytorch CNN** [20pts Grad / 2.7% **Bonus for Undergrad**]
 - 2.1.1 Loading the Model [5pts Grad / 0.7% **Bonus for Undergrad**] - *programming*
 - 2.1.2 Building the Model [5pts Grad / 0.7% **Bonus for Undergrad**] - *non-programming*
 - 2.1.3 Training the Model [8pts Grad / 1% **Bonus for Undergrad**] - *non-programming*
 - 2.1.4 Examining Accuracy and Loss [2pts Grad / 0.3% **Bonus for Undergrad**] - *non-programming*
- **2.2 Exploring Deep CNN Architectures** [1.1% **Bonus for All**] - *non-programming*

Q3: Random Forest [40pts + 2.1% Bonus for All]

Deliverables: **random_forest.py** and **Written Report**

- **3.1 Random Forest Implementation** [35pts] - *programming*
- **3.2 Hyperparameter Tuning with a Random Forest** [5pts] - *programming*
- **3.3 Plotting Feature Importance** [1.1% **Bonus for All**] - *non-programming*
- **3.4 ADABOOST** [1% Bonus for All] - *programming*

Q4: SVM [15 pts]

Deliverables: **Written Report**

- **4.1 Fitting an SVM Classifier** [10 pts]
 - 4.1.1 Fit the SVM Classifier [7 pts] - *non programming*
 - 4.1.2 Plot the SVM Classifier [3 pts] - *non programming*
- **4.2 Using Kernels** [5 pts] - *non programming*

Q5: Next Character Prediction using Recurrent Neural Networks (RNNs) [6.8% Bonus for all]

Deliverables: **rnn.py**, **lstm.py**, **base_sequential_model.py** and **Written Report**

- **5.1: Model Architecture** [4.4% **Bonus for All**] - *programming*
 - 5.1.1: Defining the Simple RNN model [2.2% **Bonus for All**]
 - 5.1.2: Defining the LSTM model [2.2% **Bonus for All**]
- **5.2: Simple RNN vs LSTM Model Text Generation Training Comparison Analysis** [2.4% **Bonus for All**] - *non programming*

Points Totals:

- Total Base: 150 pts for grads / 105 pts for undergrads
- Total Undergrad Bonus: 6%
- Total Bonus for All: 10%

Submission Instructions

Submit the following files to their respective assignments on Gradescope for the programming portions:\n",

- **Assignment 4 Programming**
 - nn.py
 - random_forest.py

- cnn_image_transformations.py (for grads only)
- **Assignment 4 Programming - Bonus for All**
 - random_forest.py
 - lstm.py
 - rnn.py
 - base_sequential_model.py
- **Assignment 4 Programming - Bonus for Undergrad**
 - NN.py
 - cnn_image_transformations.py

Environment Setup

```
In [1]: import random
import sys

import matplotlib
import matplotlib.pyplot as plt
import numpy as np
import requests
from sklearn.model_selection import train_test_split
from utilities.utils import get_housing_dataset

print("Version information")

print("python: {}".format(sys.version))
print("matplotlib: {}".format(matplotlib.__version__))
print("numpy: {}".format(np.__version__))

%load_ext autoreload
%autoreload 2
%reload_ext autoreload
```

Version information

python: 3.11.10 | packaged by Anaconda, Inc. | (main, Oct 3 2024, 07:22:26) [MSC v.1929 64 bit (AMD64)]

matplotlib: 3.9.2

numpy: 1.26.4

Coding and Emissions

Coding and computational research contribute to greenhouse gas emissions. The main source of these emissions is the power draw of computers during compute- and data-intensive computational analyses. In 2020, the sector of information and communication technologies was responsible for between 1.8% and 2.8% of GHG emissions, surprisingly more than the sector of aviation ^[1]. Machine learning models, especially large ones, can consume significant amounts of energy during training and inference, which contributes to greenhouse gas emissions. Artificial intelligence, including large language models, is also a significant emitter of carbon ^[2].

Carbon footprint of coding impacts several Sustainable Development Goals (SDGs), particularly SDG 13 (Climate Action) and SDG 12 (Responsible Consumption and Production).^[3] This means writing clean and efficient code transcends functionality—it's an environmental imperative. As coders, we can play a role in mitigating this impact.

Measuring Our Impact:

CodeCarbon estimates the amount of CO2 produced by the cloud or personal computing resources used to execute the code^[4].

Using CodeCarbon in your upcoming assignment will help you understand the environmental impact of your code and explore ways to reduce it.

```
In [ ]: from codecarbon import EmissionsTracker
```

```
tracker = EmissionsTracker()  
tracker.start()
```

```
[codecarbon INFO @ 14:23:05] [setup] RAM Tracking...  
[codecarbon INFO @ 14:23:05] [setup] GPU Tracking...  
[codecarbon INFO @ 14:23:05] No GPU found.  
[codecarbon INFO @ 14:23:05] [setup] CPU Tracking...  
[codecarbon WARNING @ 14:23:05] No CPU tracking mode found. Falling back on CPU constant mode.  
Windows OS detected: Please install Intel Power Gadget to measure CPU  
  
[codecarbon WARNING @ 14:23:07] We saw that you have a 12th Gen Intel(R) Core(TM) i7-1255U but we  
don't know it. Please contact us.  
[codecarbon INFO @ 14:23:07] CPU Model on constant consumption mode: 12th Gen Intel(R) Core(TM) i  
7-1255U  
[codecarbon INFO @ 14:23:07] >>> Tracker's metadata:  
[codecarbon INFO @ 14:23:07] Platform system: Windows-10-10.0.22631-SP0  
[codecarbon INFO @ 14:23:07] Python version: 3.11.10  
[codecarbon INFO @ 14:23:07] CodeCarbon version: 2.8.0  
[codecarbon INFO @ 14:23:07] Available RAM : 15.718 GB  
[codecarbon INFO @ 14:23:07] CPU count: 12  
[codecarbon INFO @ 14:23:07] CPU model: 12th Gen Intel(R) Core(TM) i7-1255U  
[codecarbon INFO @ 14:23:07] GPU count: None  
[codecarbon INFO @ 14:23:07] GPU model: None  
[codecarbon INFO @ 14:23:10] Saving emissions data to file c:\Users\devin\OneDrive\Documents\Geor  
gia_Tech\CS 4641\HW4\emissions.csv
```

[codecarbon INFO @ 14:23:25] Energy consumed for RAM : 0.000025 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:23:25] Energy consumed for all CPUs : 0.000177 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:23:25] 0.000202 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:23:40] Energy consumed for RAM : 0.000049 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:23:40] Energy consumed for all CPUs : 0.000354 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:23:40] 0.000404 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:23:55] Energy consumed for RAM : 0.000074 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:23:55] Energy consumed for all CPUs : 0.000532 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:23:55] 0.000605 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:24:10] Energy consumed for RAM : 0.000098 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:24:10] Energy consumed for all CPUs : 0.000709 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:24:10] 0.000807 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:24:25] Energy consumed for RAM : 0.000123 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:24:25] Energy consumed for all CPUs : 0.000886 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:24:25] 0.001009 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:24:40] Energy consumed for RAM : 0.000147 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:24:40] Energy consumed for all CPUs : 0.001063 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:24:40] 0.001210 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:24:55] Energy consumed for RAM : 0.000172 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:24:55] Energy consumed for all CPUs : 0.001240 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:24:55] 0.001412 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:25:10] Energy consumed for RAM : 0.000197 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:25:10] Energy consumed for all CPUs : 0.001417 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:25:10] 0.001614 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:25:10] 0.006107 g.CO2eq/s mean an estimation of 192.59095419437946 kg.CO2eq/year

[codecarbon INFO @ 14:25:25] Energy consumed for RAM : 0.000221 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:25:25] Energy consumed for all CPUs : 0.001595 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:25:25] 0.001816 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:25:40] Energy consumed for RAM : 0.000246 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:25:40] Energy consumed for all CPUs : 0.001772 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:25:40] 0.002017 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:25:55] Energy consumed for RAM : 0.000270 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:25:55] Energy consumed for all CPUs : 0.001949 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:25:55] 0.002219 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:26:10] Energy consumed for RAM : 0.000295 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:26:10] Energy consumed for all CPUs : 0.002126 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:26:10] 0.002421 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:26:25] Energy consumed for RAM : 0.000319 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:26:25] Energy consumed for all CPUs : 0.002303 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:26:25] 0.002623 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:26:40] Energy consumed for RAM : 0.000344 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:26:40] Energy consumed for all CPUs : 0.002481 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:26:40] 0.002824 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:26:55] Energy consumed for RAM : 0.000368 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:26:55] Energy consumed for all CPUs : 0.002658 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:26:55] 0.003026 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:27:10] Energy consumed for RAM : 0.000393 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:27:10] Energy consumed for all CPUs : 0.002835 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:27:10] 0.003228 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:27:10] 0.006107 g.CO2eq/s mean an estimation of 192.5836015908991 kg.CO2eq/year

[codecarbon INFO @ 14:27:25] Energy consumed for RAM : 0.000418 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:27:25] Energy consumed for all CPUs : 0.003012 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:27:25] 0.003430 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:27:40] Energy consumed for RAM : 0.000442 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:27:40] Energy consumed for all CPUs : 0.003189 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:27:40] 0.003631 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:27:55] Energy consumed for RAM : 0.000467 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:27:55] Energy consumed for all CPUs : 0.003366 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:27:55] 0.003833 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:28:11] Energy consumed for RAM : 0.000491 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:28:11] Energy consumed for all CPUs : 0.003544 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:28:11] 0.004035 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:28:26] Energy consumed for RAM : 0.000516 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:28:26] Energy consumed for all CPUs : 0.003721 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:28:26] 0.004237 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:28:41] Energy consumed for RAM : 0.000540 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:28:41] Energy consumed for all CPUs : 0.003898 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:28:41] 0.004438 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:28:56] Energy consumed for RAM : 0.000565 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:28:56] Energy consumed for all CPUs : 0.004075 kWh. Total CPU Power : 42.5 W

```
[codecarbon INFO @ 14:28:56] 0.004640 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:29:11] Energy consumed for RAM : 0.000589 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:29:11] Energy consumed for all CPUs : 0.004252 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:29:11] 0.004842 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:29:11] 0.006107 g.CO2eq/s mean an estimation of 192.58662832710516 kg.CO2e
q/year
[codecarbon INFO @ 14:29:26] Energy consumed for RAM : 0.000614 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:29:26] Energy consumed for all CPUs : 0.004429 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:29:26] 0.005043 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:29:41] Energy consumed for RAM : 0.000639 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:29:41] Energy consumed for all CPUs : 0.004606 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:29:41] 0.005245 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:29:56] Energy consumed for RAM : 0.000663 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:29:56] Energy consumed for all CPUs : 0.004783 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:29:56] 0.005447 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:30:11] Energy consumed for RAM : 0.000688 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:30:11] Energy consumed for all CPUs : 0.004961 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:30:11] 0.005648 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:30:26] Energy consumed for RAM : 0.000712 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:30:26] Energy consumed for all CPUs : 0.005138 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:30:26] 0.005850 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:30:41] Energy consumed for RAM : 0.000737 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:30:41] Energy consumed for all CPUs : 0.005315 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:30:41] 0.006052 kWh of electricity used since the beginning.
```

1: Two Layer Neural Network [80 pts; 55pts + 25pts Grad / 3.3% Undergrad Bonus] [\[P\]](#)

1.1 NN Implementation [65pts; 50pts + 15pts Grad / 2% Bonus for Undergrad] [\[P\]](#)

In this section, you will implement a two layer fully connected neural network to perform a Classification Task. You will also experiment with different activation functions and optimization techniques. We provide two activation functions here - SiLU and Softmax. You will implement a neural network where the first hidden layer uses a SiLU activation and the output layer uses Softmax.

You'll also implement Gradient Descent (GD) and Batch Gradient Descent (BGD) algorithms for training these neural nets. **GD is mandatory for all. BGD is bonus for undergraduate students but mandatory for graduate students.**

In the **NN.py** file, complete the following functions:

- **silu**
- **derivative_silu**
- **softmax**
- **cross_entropy_loss**
- **_dropout**
- **forward**
- **compute_gradients**
- **update_weights**
- **backward**
- **gradient_descent**
- **batch_gradient_descent**: **Mandatory for graduate students, bonus for undergraduate students.**

Please batch your data in a wraparound manner. For example, given a dataset of 9 numbers, [1, 2, 3, 4, 5, 6, 7, 8, 9], and a batch size of 6, the first iteration batch will be [1, 2, 3, 4, 5, 6], the second iteration batch will be [7, 8, 9, 1, 2, 3], the third iteration batch will be [4, 5, 6, 7, 8, 9], etc...

We'll train this neural network on sklearn's California Housing dataset.

Activation Function

There are many activation functions that are used for various purposes. For this question, we use SiLU and the softmax activation functions. We encourage you to explore the plethora of options, many of which are listed on [Wikipedia](#).

Sigmoid

The sigmoid function is a non-linear function with an S-shaped curve and is regarded as a foundational activation function. Its output is in the range $(0, 1)$, making it the function to use for binary classification output. The function is expressed as

$$o = \phi(u) = \frac{1}{1 + e^{-u}}$$

The derivation of the sigmoid function is given by

$$o' = \phi'(u) = \frac{1}{1 + e^{-u}} \left(1 - \frac{1}{1 + e^{-u}} \right) = o(1 - o)$$

Note: We do not use sigmoid in this homework; it is only included for the sake of completeness.



Softmax

Softmax is a common activation function used in neural networks, especially for multiclass classification problems like the one we are tackling. It is used to convert a vector of raw outputs from the last layer of the

Neural Network into a probability distribution over multiple classes. The softmax function takes as input a vector of real numbers and transforms them into a probability distribution, ensuring that the probabilities sum to 1.

Mathematically, given an input vector of $[x_1, x_2, \dots, x_n]$, the softmax function calculates the probability $p(y=i)$ for each class i as follows:

$$p(y=i) = e^{x_i} / (e^{x_1} + e^{x_2} + \dots + e^{x_n})$$



As discussed in class, the equation that we will use in this Neural network accounts for both the x values and the weights:



TODO: Implement the function **softmax** in **NN.py**.

```
In [ ]: from utilities.localtests import TestNN
```

```
TestNN("test_softmax").test_softmax()
```

```
[codecarbon INFO @ 14:30:56] Energy consumed for RAM : 0.000762 kWh. RAM Power : 5.8941850662231445 W
```

```
[codecarbon INFO @ 14:30:56] Energy consumed for all CPUs : 0.005494 kWh. Total CPU Power : 42.5 W
```

```
[codecarbon INFO @ 14:30:56] 0.006256 kWh of electricity used since the beginning.  
test_softmax passed!
```

```
[codecarbon INFO @ 14:31:11] Energy consumed for RAM : 0.000786 kWh. RAM Power : 5.8941850662231445 W
```

```
[codecarbon INFO @ 14:31:11] Energy consumed for all CPUs : 0.005671 kWh. Total CPU Power : 42.5 W
```

```
[codecarbon INFO @ 14:31:11] 0.006457 kWh of electricity used since the beginning.
```

```
[codecarbon INFO @ 14:31:11] 0.006106 g.CO2eq/s mean an estimation of 192.57430267010506 kg.CO2eq/year
```

SiLU (Sigmoid Linear Unit)

The Sigmoid Linear Unit (SiLU), also known as the Swish activation function, is defined as:

$$o = \phi(u) = u \cdot \sigma(u)$$

where $\sigma(u)$ is the sigmoid function:

$$\sigma(u) = \frac{1}{1 + e^{-u}}$$



The derivative of SiLU, $\phi'(u)$, is given by:

$$\phi'(u) = \sigma(u) \cdot (1 + u \cdot (1 - \sigma(u)))$$

Unlike ReLU, SiLU is a smooth and non-linear activation function that retains gradients for negative inputs, which helps during training by improving gradient flow and enabling better convergence.

In this homework, we implement SiLU.

 Derivate SiLu

TODO: Implement the function **silu** and **derivative_silu** in **NN.py**.

```
In [ ]: from utilities.localtests import TestNN
```

```
TestNN("test_silu").test_silu()  
TestNN("test_d_silu").test_d_silu()
```

test_silu passed!

test_d_silu passed!

[codecarbon INFO @ 14:31:26] Energy consumed for RAM : 0.000811 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:31:26] Energy consumed for all CPUs : 0.005848 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:31:26] 0.006659 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:31:41] Energy consumed for RAM : 0.000835 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:31:41] Energy consumed for all CPUs : 0.006026 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:31:41] 0.006861 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:31:56] Energy consumed for RAM : 0.000860 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:31:56] Energy consumed for all CPUs : 0.006203 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:31:56] 0.007063 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:32:11] Energy consumed for RAM : 0.000884 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:32:11] Energy consumed for all CPUs : 0.006380 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:32:11] 0.007264 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:32:26] Energy consumed for RAM : 0.000909 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:32:26] Energy consumed for all CPUs : 0.006557 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:32:26] 0.007466 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:32:41] Energy consumed for RAM : 0.000934 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:32:41] Energy consumed for all CPUs : 0.006734 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:32:41] 0.007668 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:32:56] Energy consumed for RAM : 0.000958 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:32:56] Energy consumed for all CPUs : 0.006911 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:32:56] 0.007870 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:33:11] Energy consumed for RAM : 0.000983 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:33:11] Energy consumed for all CPUs : 0.007089 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:33:11] 0.008071 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:33:11] 0.006107 g.CO2eq/s mean an estimation of 192.59136786162387 kg.CO2eq/year

[codecarbon INFO @ 14:33:26] Energy consumed for RAM : 0.001007 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:33:26] Energy consumed for all CPUs : 0.007266 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:33:26] 0.008273 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:33:41] Energy consumed for RAM : 0.001032 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:33:41] Energy consumed for all CPUs : 0.007443 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:33:41] 0.008475 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:33:56] Energy consumed for RAM : 0.001056 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:33:56] Energy consumed for all CPUs : 0.007620 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:33:56] 0.008676 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:34:11] Energy consumed for RAM : 0.001081 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:34:11] Energy consumed for all CPUs : 0.007797 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:34:11] 0.008878 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:34:26] Energy consumed for RAM : 0.001106 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:34:26] Energy consumed for all CPUs : 0.007974 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:34:26] 0.009080 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:34:41] Energy consumed for RAM : 0.001130 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:34:41] Energy consumed for all CPUs : 0.008151 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:34:41] 0.009282 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:34:56] Energy consumed for RAM : 0.001155 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:34:56] Energy consumed for all CPUs : 0.008329 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:34:56] 0.009483 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:35:11] Energy consumed for RAM : 0.001179 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:35:11] Energy consumed for all CPUs : 0.008506 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:35:11] 0.009685 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:35:11] 0.006107 g.CO2eq/s mean an estimation of 192.59747448541196 kg.CO2eq/year

[codecarbon INFO @ 14:35:26] Energy consumed for RAM : 0.001204 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:35:26] Energy consumed for all CPUs : 0.008683 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:35:26] 0.009887 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:35:41] Energy consumed for RAM : 0.001228 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:35:41] Energy consumed for all CPUs : 0.008860 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:35:41] 0.010088 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:35:56] Energy consumed for RAM : 0.001253 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:35:56] Energy consumed for all CPUs : 0.009037 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:35:56] 0.010290 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:36:11] Energy consumed for RAM : 0.001277 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:36:11] Energy consumed for all CPUs : 0.009214 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:36:11] 0.010492 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:36:26] Energy consumed for RAM : 0.001302 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:36:26] Energy consumed for all CPUs : 0.009392 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:36:26] 0.010694 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:36:41] Energy consumed for RAM : 0.001327 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:36:41] Energy consumed for all CPUs : 0.009569 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:36:41] 0.010895 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:36:56] Energy consumed for RAM : 0.001351 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 14:36:56] Energy consumed for all CPUs : 0.009746 kWh. Total CPU Power : 42.5 W

```

[codecarbon INFO @ 14:36:56] 0.011097 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:37:11] Energy consumed for RAM : 0.001376 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:37:11] Energy consumed for all CPUs : 0.009923 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:37:11] 0.011299 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:37:11] 0.006107 g.CO2eq/s mean an estimation of 192.593503144339 kg.CO2eq/y
ear
[codecarbon INFO @ 14:37:26] Energy consumed for RAM : 0.001400 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:37:26] Energy consumed for all CPUs : 0.010100 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:37:26] 0.011500 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:37:41] Energy consumed for RAM : 0.001425 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:37:41] Energy consumed for all CPUs : 0.010277 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:37:41] 0.011702 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:37:56] Energy consumed for RAM : 0.001449 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:37:56] Energy consumed for all CPUs : 0.010454 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:37:56] 0.011904 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:38:11] Energy consumed for RAM : 0.001474 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:38:11] Energy consumed for all CPUs : 0.010632 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:38:11] 0.012106 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:38:26] Energy consumed for RAM : 0.001499 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:38:26] Energy consumed for all CPUs : 0.010809 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:38:26] 0.012307 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:38:41] Energy consumed for RAM : 0.001523 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:38:41] Energy consumed for all CPUs : 0.010986 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:38:41] 0.012509 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:38:56] Energy consumed for RAM : 0.001548 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:38:56] Energy consumed for all CPUs : 0.011163 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:38:56] 0.012711 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:39:11] Energy consumed for RAM : 0.001572 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:39:11] Energy consumed for all CPUs : 0.011340 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:39:11] 0.012913 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:39:11] 0.006107 g.CO2eq/s mean an estimation of 192.58014014204127 kg.CO2e
q/year

```

Perceptron

A single layer perceptron can be thought of as a linear hyperplane as in logistic regression followed by a non-linear activation function.

$$u_i = \sum_{j=1}^d \theta_{ij} x_j + b_i$$

$$o_i = \phi \left(\sum_{j=1}^d \theta_{ij} x_j + b_i \right) = \phi(\theta_i^T x + b_i)$$

where x is a d -dimensional vector i.e. $x \in \mathbb{R}^d$. It is one datapoint with d features. $\theta_i \in \mathbb{R}^d$ is the weight vector for the i^{th} hidden unit, $b_i \in \mathbb{R}$ is the bias element for the i^{th} hidden unit and $\phi(\cdot)$ is a non-linear activation function that has been described below. u_i is a linear combination of the features in x_j weighted by θ_i whereas o_i is the i^{th} output unit from the activation layer.

Fully connected Layer

Typically, a modern neural network contains millions of perceptrons as the one shown in the previous image. Perceptrons interact in different configurations such as cascaded or parallel. In this part, we describe a fully connected layer configuration in a neural network which comprises multiple parallel perceptrons forming one layer.

We extend the previous notation to describe a fully connected layer. Each layer in a fully connected network has a number of input/hidden/output units cascaded in parallel. Let us define a single layer of the neural net as follows:

m denotes the number of hidden units in a single layer l whereas n denotes the number of units in the previous layer $l - 1$.

$$u^{[l]} = \theta^{[l]} o^{[l-1]} + b^{[l]}$$

where $u^{[l]} \in \mathbb{R}^m$ is a m -dimensional vector pertaining to the hidden units of the l^{th} layer of the neural network after applying linear operations. Similarly, $o^{[l-1]} \in \mathbb{R}^n$ is the n -dimensional output vector corresponding to the hidden units of the $(l - 1)^{th}$ activation layer. $\theta^{[l]} \in \mathbb{R}^{m \times n}$ is the weight matrix of the l^{th} layer where each row of $\theta^{[l]}$ is analogous to θ_i described in the previous section i.e. each row corresponds to one hidden unit of the l^{th} layer. $b^{[l]} \in \mathbb{R}^m$ is the bias vector of the layer where each element of b pertains to one hidden unit of the l^{th} layer. This is followed by element wise non-linear activation function $o^{[l]} = \phi(u^{[l]})$. The whole operation can be summarized as,

$$o^{[l]} = \phi(\theta^{[l]} o^{[l-1]} + b^{[l]})$$

where $o^{[l-1]}$ is the output of the previous layer.

Dropout

A dropout layer is a regularization technique used in neural networks to reduce overfitting. During training, a dropout layer looks at each input unit and randomly decide if it will be dropped (set to zero) with some given probability p . The decision for each unit is made independently. Formally, given an input of shape $N \times K$ (where N is the number of data points and K is the number of features), it samples from $\text{Bernoulli}(p)$ for each unit, resulting in an output where approximately pNK of the units are zero (in expectation). This forces the network to learn more robust and generalizable features, since it cannot rely too much on any particular input. During inference, the dropout layer is turned off, and the full network is used to make predictions.

The dropout probability p is a hyperparameter than can be tuned to adjust the strength of regularization. Setting $p = 0$ is equivalent to no dropout.

Note that the derivative of $\text{dropout}(u)$ with respect to u has the same shape as u . The values of the derivative depend on the random mask.

Use [this](#) as a reference for your implementation.

Note that after applying the mask, we must scale the result by a factor of $1/(1 - p)$. Why is this necessary?

TODO: Implement the `_dropout` function in `NN.py`.

```
In [ ]: from utilities.localtests import TestNN
```

```
TestNN("test_dropout").test_dropout()
```

test_dropout passed!

```
[codecarbon INFO @ 14:57:12] Energy consumed for RAM : 0.003340 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:57:12] Energy consumed for all CPUs : 0.024095 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:57:12] 0.027436 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:57:12] 0.006107 g.CO2eq/s mean an estimation of 192.58175082703164 kg.CO2e
q/year
[codecarbon INFO @ 14:57:27] Energy consumed for RAM : 0.003365 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:57:27] Energy consumed for all CPUs : 0.024273 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:57:27] 0.027638 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:57:42] Energy consumed for RAM : 0.003390 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:57:42] Energy consumed for all CPUs : 0.024450 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:57:42] 0.027839 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:57:57] Energy consumed for RAM : 0.003414 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:57:57] Energy consumed for all CPUs : 0.024627 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:57:57] 0.028041 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:58:12] Energy consumed for RAM : 0.003439 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:58:12] Energy consumed for all CPUs : 0.024804 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:58:12] 0.028243 kWh of electricity used since the beginning.
[codecarbon INFO @ 14:58:27] Energy consumed for RAM : 0.003463 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 14:58:27] Energy consumed for all CPUs : 0.024981 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 14:58:27] 0.028444 kWh of electricity used since the beginning.
```

Cross Entropy Loss

Cross-Entropy Loss is a widely used loss function in machine learning and deep learning, especially for classification tasks. It measures the dissimilarity between the predicted probability distribution and the true probability distribution of a classification problem. If it is closer to zero, the better the learnt function is.

Implementation details

For classification problems as in this exercise, we compute the loss as follows:

$$CE = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log(\hat{y}_i))$$

where y_i is the true label and $\hat{y}_i$ is the estimated label.

TODO: Implement the **cross_entropy_loss** function in **NN.py**.

```
In [6]: from utilities.localtests import TestNN  
  
TestNN("test_loss").test_loss()
```

test_loss passed!

Neural Network Architecture

The architecture of our neural network.



The above diagram shows the dimensions of the neural network you will implement, along with the relationships between the quantities. Note that the neural network consists of two hidden linear layers, each followed by a SiLU activation function. The logits outputted by the second hidden linear layer are then passed through the softmax function, which turns them into probability distributions over the 3 classes.

Here is a helpful [guide](#) that walks through the matrix multiplication operations and shapes involved in a forward and backward pass.

Note: Implement drop out function only on the first hidden layer!

```
In [12]: from utilities.localtests import TestNN  
  
TestNN("test_forward_without_dropout").test_forward_without_dropout()  
TestNN("test_forward").test_forward()
```

test_forward_without_dropout passed!
test_forward passed!

Backward Propagation: Update Weights and Compute Gradients

After the forward pass, we do back propagation to update the weights and biases in the direction of the negative gradient of the loss function.

Update Weights

So, we update the weights and biases using the following formulas

$$\theta^{[3]} := \theta^{[3]} - lr \times \frac{\partial l}{\partial \theta^{[3]}}$$

$$b^{[3]} := b^{[3]} - lr \times \frac{\partial l}{\partial b^{[3]}}$$

$$\theta^{[2]} := \theta^{[2]} - lr \times \frac{\partial l}{\partial \theta^{[2]}}$$

$$b^{[2]} := b^{[2]} - lr \times \frac{\partial l}{\partial b^{[2]}}$$

$$\theta^{[1]} := \theta^{[1]} - lr \times \frac{\partial l}{\partial \theta^{[1]}}$$

$$b^{[1]} := b^{[1]} - lr \times \frac{\partial l}{\partial b^{[1]}}$$

where lr is the learning rate. It decides the step size we want to take in the direction of the negative gradient.

TODO: Implement the **update_weights** function in **NN.py** with `use_momentum=False`.

Hint: Refer to this [guide](#) for more detail on the backward pass.

```
In [ ]: from utilities.localtests import TestNN

TestNN("test_update_weights").test_update_weights()
```

test_update_weights passed!

[codecarbon INFO @ 14:58:42] Energy consumed for RAM : 0.003488 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:58:42] Energy consumed for all CPUs : 0.025158 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:58:42] 0.028646 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:58:57] Energy consumed for RAM : 0.003512 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:58:57] Energy consumed for all CPUs : 0.025335 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:58:57] 0.028848 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:59:12] Energy consumed for RAM : 0.003537 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:59:12] Energy consumed for all CPUs : 0.025512 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:59:12] 0.029049 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:59:12] 0.006107 g.CO2eq/s mean an estimation of 192.585617669106 kg.CO2eq/year

[codecarbon INFO @ 14:59:27] Energy consumed for RAM : 0.003562 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:59:27] Energy consumed for all CPUs : 0.025690 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:59:27] 0.029251 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:59:42] Energy consumed for RAM : 0.003586 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:59:42] Energy consumed for all CPUs : 0.025867 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:59:42] 0.029453 kWh of electricity used since the beginning.

[codecarbon INFO @ 14:59:57] Energy consumed for RAM : 0.003611 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 14:59:57] Energy consumed for all CPUs : 0.026044 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 14:59:57] 0.029655 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:00:12] Energy consumed for RAM : 0.003635 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:00:12] Energy consumed for all CPUs : 0.026221 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:00:12] 0.029856 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:00:27] Energy consumed for RAM : 0.003660 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:00:27] Energy consumed for all CPUs : 0.026398 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:00:27] 0.030058 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:00:42] Energy consumed for RAM : 0.003684 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:00:42] Energy consumed for all CPUs : 0.026576 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:00:42] 0.030260 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:00:57] Energy consumed for RAM : 0.003709 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:00:57] Energy consumed for all CPUs : 0.026753 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:00:57] 0.030462 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:01:12] Energy consumed for RAM : 0.003733 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:01:12] Energy consumed for all CPUs : 0.026930 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:01:12] 0.030663 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:01:12] 0.006107 g.CO2eq/s mean an estimation of 192.58836283162935 kg.CO2eq/year

[codecarbon INFO @ 15:01:27] Energy consumed for RAM : 0.003758 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:01:27] Energy consumed for all CPUs : 0.027107 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:01:27] 0.030865 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:01:42] Energy consumed for RAM : 0.003783 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:01:42] Energy consumed for all CPUs : 0.027284 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:01:42] 0.031067 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:01:57] Energy consumed for RAM : 0.003807 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:01:57] Energy consumed for all CPUs : 0.027461 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:01:57] 0.031268 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:02:12] Energy consumed for RAM : 0.003832 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:02:12] Energy consumed for all CPUs : 0.027638 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:02:12] 0.031470 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:02:27] Energy consumed for RAM : 0.003856 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:02:27] Energy consumed for all CPUs : 0.027815 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:02:27] 0.031672 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:02:42] Energy consumed for RAM : 0.003881 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:02:42] Energy consumed for all CPUs : 0.027993 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:02:42] 0.031873 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:02:57] Energy consumed for RAM : 0.003905 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:02:57] Energy consumed for all CPUs : 0.028170 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:02:57] 0.032075 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:03:12] Energy consumed for RAM : 0.003930 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:03:12] Energy consumed for all CPUs : 0.028347 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:03:12] 0.032277 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:03:12] 0.006107 g.CO2eq/s mean an estimation of 192.58662603412503 kg.CO2eq/year

[codecarbon INFO @ 15:03:27] Energy consumed for RAM : 0.003954 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:03:27] Energy consumed for all CPUs : 0.028524 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:03:27] 0.032478 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:03:42] Energy consumed for RAM : 0.003979 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:03:42] Energy consumed for all CPUs : 0.028701 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:03:42] 0.032680 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:03:57] Energy consumed for RAM : 0.004004 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:03:57] Energy consumed for all CPUs : 0.028878 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:03:57] 0.032882 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:04:12] Energy consumed for RAM : 0.004028 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 15:04:12] Energy consumed for all CPUs : 0.029055 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:04:12] 0.033083 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:04:27] Energy consumed for RAM : 0.004053 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:04:27] Energy consumed for all CPUs : 0.029232 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:04:27] 0.033285 kWh of electricity used since the beginning.
```

Update Weights with Momentum [Bonus for Undergrad]

Gradient descent does a generally good job of facilitating the convergence of the model's parameters to minimize the loss function. However, the process of doing so can be slow and/or noisy. **Momentum** is a technique used to stabilize this convergence.

As a reminder, vanilla gradient descent applies the following update function to the parameters:

$$\theta_{t+1} = \theta_t - \alpha \nabla f(\theta_t) \quad (1)$$

where θ_t represents the parameters at time t , α represents the learning rate, and f is the loss function.

Momentum proposes the following tweak to our parameter update function:

$$\begin{aligned} z_{t+1} &= \beta z_t + \nabla f(\theta_t) \\ \theta_{t+1} &= \theta_t - \alpha z_{t+1} \end{aligned}$$

where $\beta \in [0, 1]$ is the momentum constant and z_t represents the momentum records at time t .

You can think of momentum as taking our previous changes into consideration. If we've been moving in a certain direction recently, it's likely we should keep moving in that direction. The recurrence relation given shows that we use an exponentially-weighted average of the previous updates for our current update.

A useful analogy about momentum from [this great article on Distill](#):

Here's a popular story about momentum: gradient descent is a man walking down a hill. He follows the steepest path downwards; his progress is slow, but steady. Momentum is a heavy ball rolling down the same hill. The added inertia acts both as a smoother and an accelerator, dampening oscillations and causing us to barrel through narrow valleys, small humps and local minima.

TODO: Implement the **update_weights** function in **NN.py** with `use_momentum=True`.

HINT: z is stored in `self.change`

```
In [ ]: from utilities.localtests import TestNN
```

```
TestNN("test_update_weights_with_momentum").test_update_weights_with_momentum()
```

test_update_weights_with_momentum passed!

```
[codecarbon INFO @ 15:04:42] Energy consumed for RAM : 0.004077 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:04:42] Energy consumed for all CPUs : 0.029410 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:04:42] 0.033487 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:04:57] Energy consumed for RAM : 0.004102 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:04:57] Energy consumed for all CPUs : 0.029586 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:04:57] 0.033688 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:05:12] Energy consumed for RAM : 0.004126 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:05:12] Energy consumed for all CPUs : 0.029764 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:05:12] 0.033890 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:05:12] 0.006106 g.CO2eq/s mean an estimation of 192.57123220067072 kg.CO2e
q/year
[codecarbon INFO @ 15:05:27] Energy consumed for RAM : 0.004151 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:05:27] Energy consumed for all CPUs : 0.029941 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:05:27] 0.034092 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:05:42] Energy consumed for RAM : 0.004175 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:05:42] Energy consumed for all CPUs : 0.030118 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:05:42] 0.034293 kWh of electricity used since the beginning.
```

Compute Gradients

In order to compute the gradients of the loss with respect to each parameter, we use the equations that make up the forward pass:

$$\begin{aligned}
 u_1 &= \theta_1 X + b_1 \\
 o_1 &= \text{silu}(u_1) \\
 u_2 &= \theta_2 o_1 + b_2 \\
 o_2 &= \text{silu}(u_2) \\
 u_3 &= \theta_3 o_2 + b_3 \\
 o_3 &= \text{softmax}(u_3) \\
 l &= \text{cross_entropy}(o_3)
 \end{aligned}$$

When computing gradients, we travel backwards from the loss all the way back to the input. We first seek to obtain the derivative of the loss l with respect to the logits u_3 . Note that they have the relation

$$l = \text{cross_entropy}(\text{softmax}(u_3))$$

Computing the derivative of this seems very involved, but it actually has a very elegant result:

$$\frac{\partial l}{\partial u_3} = \text{softmax}(u_3) - y = o_3 - y = \hat{y} - y.$$

where $\hat{y}$ is predicted y or o_3 .

While this is given to you, we encourage you to derive it for yourself! You can find a great explanation of the derivation [in this article](#).

Now that we have $\frac{\partial l}{\partial u_3}$, we seek to move further back and compute $\frac{\partial l}{\partial \theta_3}$ and $\frac{\partial l}{\partial b_3}$. This is done using the chain rule:

$$\begin{aligned}\frac{\partial l}{\partial \theta_3} &= \frac{\partial l}{\partial u_3} \cdot \frac{\partial u_3}{\partial \theta_3} \\ \frac{\partial l}{\partial b_3} &= \frac{\partial l}{\partial u_3} \cdot \frac{\partial u_3}{\partial b_3}.\end{aligned}$$

The quantities $\frac{\partial u_3}{\partial \theta_3}$ and $\frac{\partial u_3}{\partial b_3}$ are easy to derive from the relation $u_3 = \theta_3 o_2 + b_3$. We see that

$$\begin{aligned}\frac{\partial l}{\partial \theta_3} &= \frac{\partial l}{\partial u_3} \cdot o_2 \\ \frac{\partial l}{\partial b_3} &= \frac{\partial l}{\partial u_3} \cdot 1.\end{aligned}$$

Note that the derivative involves o_2 , which we computed during the forward pass. Fortunately, we saved that value in `self.cache`, so we don't need to compute it again!

The same procedure is repeated to obtain the gradients for the upstream parameters θ_2 and b_2 . We must first perform the intermediate steps of computing the derivative of the loss with respect to o_2 and then u_2 . These are given by

$$\begin{aligned}\frac{\partial l}{\partial o_2} &= \frac{\partial l}{\partial u_3} \cdot \theta_3 \\ \frac{\partial l}{\partial u_2} &= \frac{\partial l}{\partial o_2} \cdot \frac{\partial \text{SiLu}}{\partial u_2}.\end{aligned}$$

The same procedure is repeated to obtain the gradients for the upstream parameters θ_1 and b_1 . We must first perform the intermediate steps of computing the derivative of the loss with respect to o_1 and then u_1 . These are given by

$$\begin{aligned}\frac{\partial l}{\partial o_1} &= \frac{\partial l}{\partial u_2} \cdot \theta_2 \\ \frac{\partial l}{\partial u_1} &= \frac{\partial l}{\partial o_1} \cdot \frac{\partial \text{SiLu}}{\partial u_1}.\end{aligned}$$

In the second relation, we must consider our use of dropout! If we applied dropout on a particular neuron, it should not be adjusted. To account for this, in the case of `use_dropout=True`, we must instead use

$$\frac{\partial l}{\partial u_1} = \frac{\partial l}{\partial o_1} \cdot \frac{\partial \text{SiLu}}{\partial u_1} \cdot \text{dropout_mask} \cdot \frac{1}{1 - p},$$

where $1/(1 - p)$ is the scaling factor and `dropout_mask` is stored in `self.cache`.

The final step! We can use these values to compute the gradients for θ_1 and b_1 , using the relation $u_1 = \theta_1 X + b_1$, which are given by

$$\begin{aligned}\frac{\partial l}{\partial \theta_1} &= \frac{\partial l}{\partial u_1} \cdot X \\ \frac{\partial l}{\partial b_1} &= \frac{\partial l}{\partial u_1} \cdot 1.\end{aligned}$$

Implementation Tips

The above equations are given in matrix notation. When implementing these computations in code, the easiest way to make sure you are calculating the values correctly and in the right order is to check shapes. Any time you are doing a matrix/vector operation in NumPy, **check the shapes**.

Since we are computing these gradients over N data points, we must divide the gradients by N to take the *average* gradient. Make sure you are dividing by N exactly once, no more and no less!

TODO: Implement the **compute_gradients** function in **NN.py**.

Note: Implement drop out function only on the first hidden layer!

Hint: Refer to this [guide](#) for more detail on computing gradients.

```
In [ ]: from utilities.localtests import TestNN

TestNN(
    "test_compute_gradients_without_dropout"
).test_compute_gradients_without_dropout()
TestNN("test_compute_gradients").test_compute_gradients()

test_compute_gradients_without_dropout passed!
test_compute_gradients passed!
```

[codecarbon INFO @ 15:08:27] Energy consumed for RAM : 0.004446 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:08:27] Energy consumed for all CPUs : 0.032067 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:08:27] 0.036512 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:08:42] Energy consumed for RAM : 0.004470 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:08:42] Energy consumed for all CPUs : 0.032244 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:08:42] 0.036714 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:08:57] Energy consumed for RAM : 0.004495 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:08:57] Energy consumed for all CPUs : 0.032421 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:08:57] 0.036916 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:09:12] Energy consumed for RAM : 0.004519 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:09:12] Energy consumed for all CPUs : 0.032598 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:09:12] 0.037117 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:09:12] 0.006107 g.CO2eq/s mean an estimation of 192.5834839370464 kg.CO2eq/year

[codecarbon INFO @ 15:09:27] Energy consumed for RAM : 0.004544 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:09:27] Energy consumed for all CPUs : 0.032775 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:09:27] 0.037319 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:09:42] Energy consumed for RAM : 0.004568 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:09:42] Energy consumed for all CPUs : 0.032953 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:09:42] 0.037521 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:09:57] Energy consumed for RAM : 0.004593 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:09:57] Energy consumed for all CPUs : 0.033130 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:09:57] 0.037723 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:10:12] Energy consumed for RAM : 0.004618 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:10:12] Energy consumed for all CPUs : 0.033307 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:10:12] 0.037924 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:10:27] Energy consumed for RAM : 0.004642 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:10:27] Energy consumed for all CPUs : 0.033484 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:10:27] 0.038126 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:10:42] Energy consumed for RAM : 0.004667 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:10:42] Energy consumed for all CPUs : 0.033661 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:10:42] 0.038328 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:10:57] Energy consumed for RAM : 0.004691 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:10:57] Energy consumed for all CPUs : 0.033838 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:10:57] 0.038529 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:11:12] Energy consumed for RAM : 0.004716 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 15:11:12] Energy consumed for all CPUs : 0.034015 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:11:12] 0.038731 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:11:12] 0.006107 g.CO2eq/s mean an estimation of 192.57586347663724 kg.CO2eq/year
[codecarbon INFO @ 15:11:27] Energy consumed for RAM : 0.004740 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:11:27] Energy consumed for all CPUs : 0.034192 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:11:27] 0.038933 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:11:42] Energy consumed for RAM : 0.004765 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:11:42] Energy consumed for all CPUs : 0.034370 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:11:42] 0.039134 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:11:57] Energy consumed for RAM : 0.004789 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:11:57] Energy consumed for all CPUs : 0.034547 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:11:57] 0.039336 kWh of electricity used since the beginning.
```

1.1.1 Local Test: Gradient Descent

You may test your implementation of the GD function contained in **NN.py** in the cell below. See [Using the Local Tests](#) for more details. Look at the function documentation in gradient_descent for guidance.

```
In [17]: #####
### DO NOT CHANGE THIS CELL ###
#####
from utilities.localtests import TestNN

TestNN("test_gradient_descent").test_gradient_descent()
```

```
Loss after iteration 0: 1.086276
Loss after iteration 1: 1.086233
Loss after iteration 2: 1.086189
```

Your GD losses works within the expected range: True

1.1.2 Local Test: Batch Gradient Descent [No Points]

You may test your implementation of the BGD function contained in **NN.py** in the cell below. See [Using the Local Tests](#) for more details. Look at the function documentation in gradient_descent for guidance.

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####

from utilities.localtests import TestNN

TestNN("test_batch_gradient_descent").test_batch_gradient_descent()
```

```
Loss after iteration 0: 1.085044
Loss after iteration 1: 1.110789
Loss after iteration 2: 1.109005
```

Your BGD losses works within the expected range: True
Your batch_y works within the expected range: True

```
[codecarbon INFO @ 15:12:12] Energy consumed for RAM : 0.004814 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:12:12] Energy consumed for all CPUs : 0.034724 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:12:12] 0.039538 kWh of electricity used since the beginning.
```

1.1.3 Local Test: Gradient Descent with Momentum

You may test your implementation of the GD function with momentum contained in **NN.py** in the cell below. See [Using the Local Tests](#) for more details. Revisit your implementation for update_weights.

```
In [19]: #####
### DO NOT CHANGE THIS CELL ###
#####

from utilities.localtests import TestNN

TestNN("test_gradient_descent_with_momentum").test_gradient_descent_with_momentum()
```

```
Loss after iteration 0: 1.086276
Loss after iteration 1: 1.086233
Loss after iteration 2: 1.086168
```

Your GD losses works within the expected range: True

1.2 Loss plot and cross-entropy(CE) value for NN with Gradient Descent [5pts] [P]

Train your neural network implementation with gradient descent and print out the loss at every 1000th iteration (starting at iteration 0). The following cells will plot the loss vs epoch graph and calculate the final test cross-entropy(CE).

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####

from NN import NeuralNet
from sklearn.metrics import ConfusionMatrixDisplay, confusion_matrix

x_train, y_train, x_test, y_test = get_housing_dataset()

nn = NeuralNet(
    y_train, lr=0.01, use_dropout=False, use_momentum=False
) # initialize neural net class
nn.gradient_descent(x_train, y_train, iter=60000) # train
```

Loss after iteration 0: 1.086276
Loss after iteration 1000: 1.032790
Loss after iteration 2000: 0.903132
Loss after iteration 3000: 0.728430
Loss after iteration 4000: 0.656171
Loss after iteration 5000: 0.621733
Loss after iteration 6000: 0.598812
Loss after iteration 7000: 0.582288
Loss after iteration 8000: 0.570177
Loss after iteration 9000: 0.561003
Loss after iteration 10000: 0.553710
Loss after iteration 11000: 0.547769
Loss after iteration 12000: 0.543023
Loss after iteration 13000: 0.539342
Loss after iteration 14000: 0.536473
Loss after iteration 15000: 0.534156
Loss after iteration 16000: 0.532210
Loss after iteration 17000: 0.530515

[codecarbon INFO @ 15:12:27] Energy consumed for RAM : 0.004839 kWh. RAM Power : 5.8941850662231445 W

Loss after iteration 18000: 0.528996

[codecarbon INFO @ 15:12:27] Energy consumed for all CPUs : 0.034901 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:12:27] 0.039739 kWh of electricity used since the beginning.

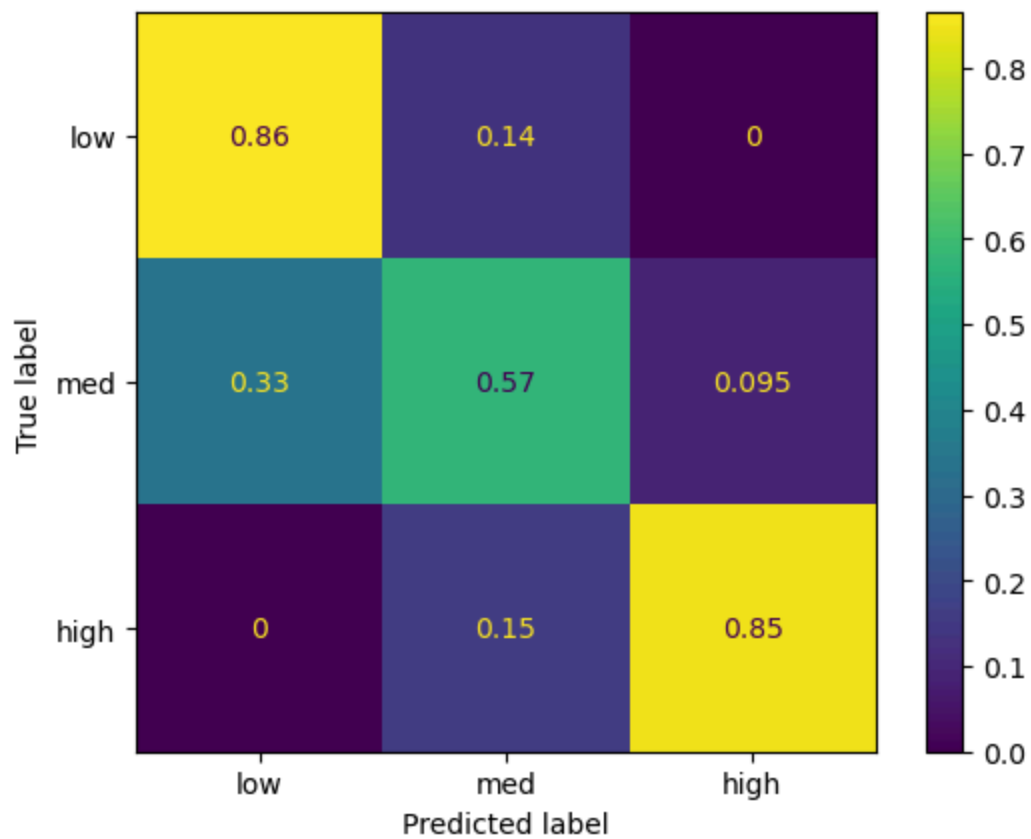
```
Loss after iteration 19000: 0.527603
Loss after iteration 20000: 0.526301
Loss after iteration 21000: 0.525063
Loss after iteration 22000: 0.523866
Loss after iteration 23000: 0.522691
Loss after iteration 24000: 0.521519
Loss after iteration 25000: 0.520332
Loss after iteration 26000: 0.519114
Loss after iteration 27000: 0.517849
Loss after iteration 28000: 0.516524
Loss after iteration 29000: 0.515125
Loss after iteration 30000: 0.513643
Loss after iteration 31000: 0.512071
Loss after iteration 32000: 0.510410
Loss after iteration 33000: 0.508666
Loss after iteration 34000: 0.506852
Loss after iteration 35000: 0.504993
Loss after iteration 36000: 0.503116
Loss after iteration 37000: 0.501250
Loss after iteration 38000: 0.499419
Loss after iteration 39000: 0.497640
Loss after iteration 40000: 0.495918
Loss after iteration 41000: 0.494254
Loss after iteration 42000: 0.492645
Loss after iteration 43000: 0.491086
Loss after iteration 44000: 0.489571
Loss after iteration 45000: 0.488099
Loss after iteration 46000: 0.486665
Loss after iteration 47000: 0.485271
Loss after iteration 48000: 0.483918
Loss after iteration 49000: 0.482609
Loss after iteration 50000: 0.481346
Loss after iteration 51000: 0.480134
Loss after iteration 52000: 0.478976
Loss after iteration 53000: 0.477877
Loss after iteration 54000: 0.476838
Loss after iteration 55000: 0.475859
Loss after iteration 56000: 0.474941
Loss after iteration 57000: 0.474082
Loss after iteration 58000: 0.473277
Loss after iteration 59000: 0.472523
```

```
[codecarbon INFO @ 15:12:42] Energy consumed for RAM : 0.004863 kWh. RAM Power : 5.8941850662231445 W
```

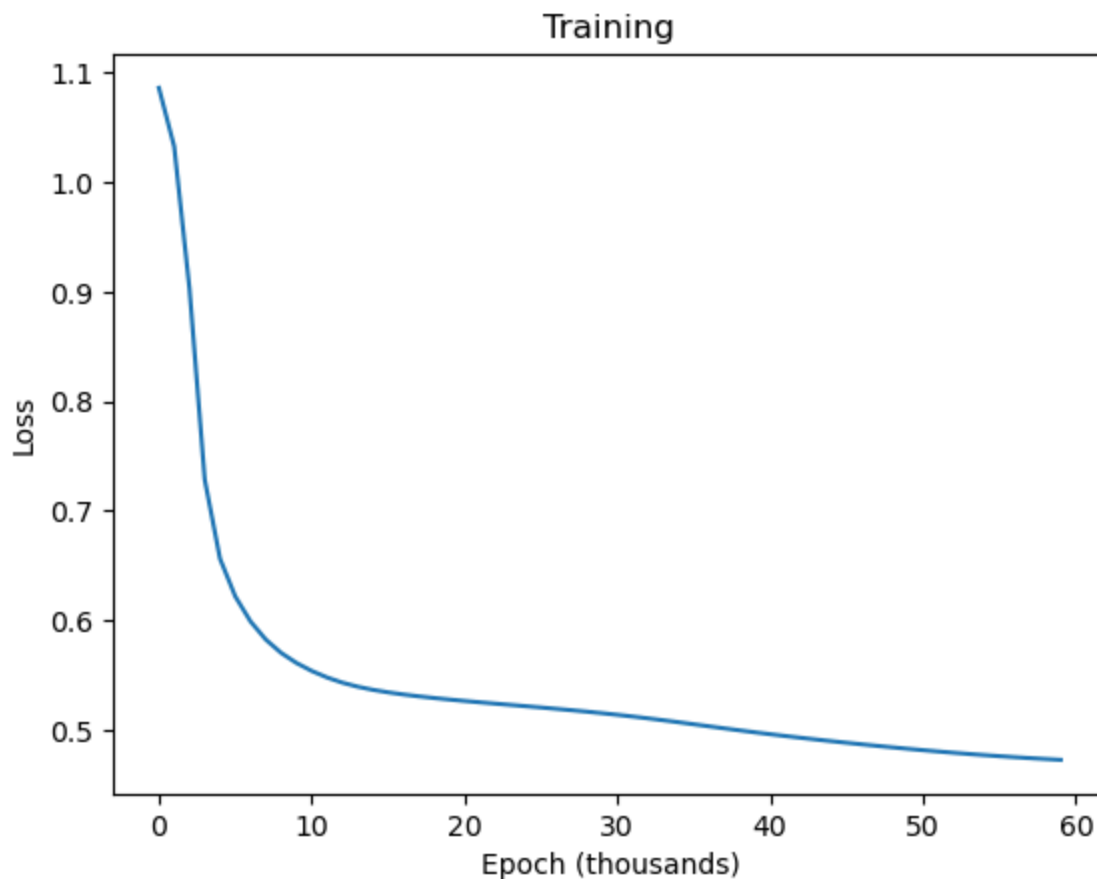
```
[codecarbon INFO @ 15:12:42] Energy consumed for all CPUs : 0.035078 kWh. Total CPU Power : 42.5 W
```

```
[codecarbon INFO @ 15:12:42] 0.039941 kWh of electricity used since the beginning.
```

```
In [21]: # Plot confusion matrix
y_true = np.argmax(y_test, axis=1)
y_pred = nn.predict(x_test)
display_labels = ["low", "med", "high"]
ConfusionMatrixDisplay.from_predictions(
    y_true, y_pred, normalize="true", display_labels=display_labels
)
plt.show()
```



```
In [ ]: # Plot training loss
fig = plt.plot(np.array(nn.loss).squeeze())
plt.title(f"Training")
plt.xlabel("Epoch (thousands)")
plt.ylabel("Loss")
plt.show()
```



```
[codecarbon INFO @ 15:12:57] Energy consumed for RAM : 0.004888 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:12:57] Energy consumed for all CPUs : 0.035255 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:12:57] 0.040143 kWh of electricity used since the beginning.
```

```
In [23]: # Total loss
y_hat = nn.forward(x_test, use_dropout=False)
print("Cross entropy loss:", round(nn.cross_entropy_loss(y_test, y_hat), 3))
```

Cross entropy loss: 0.729

1.3 Loss plot and CE value for NN with BGD [5pts Grad / 0.7% Bonus for Undergrad] [\[P\]](#)

Train your neural network implementation with batch gradient descent and print out the loss at every 1000th iteration (starting at iteration 0). The following cells will plot the loss vs epoch graph and calculate the final test CE.

```
In [24]: #####
### DO NOT CHANGE THIS CELL ###
#####
from NN import NeuralNet
from sklearn.metrics import ConfusionMatrixDisplay, confusion_matrix

x_train, y_train, x_test, y_test = get_housing_dataset()

nn = NeuralNet(
    y_train, lr=0.01, use_dropout=True, use_momentum=False
```

```
) # initialize neural net class
nn.batch_gradient_descent(x_train, y_train, iter=60000, use_momentum=False)
```

```
Loss after iteration 0: 1.092869
Loss after iteration 1000: 1.048985
Loss after iteration 2000: 0.932282
Loss after iteration 3000: 0.869664
Loss after iteration 4000: 0.896471
Loss after iteration 5000: 0.688057
Loss after iteration 6000: 0.680472
Loss after iteration 7000: 0.686783
Loss after iteration 8000: 0.531214
Loss after iteration 9000: 0.586094
Loss after iteration 10000: 0.759610
Loss after iteration 11000: 0.551759
Loss after iteration 12000: 0.517801
Loss after iteration 13000: 0.781921
Loss after iteration 14000: 0.502840
Loss after iteration 15000: 0.560831
Loss after iteration 16000: 0.679815
Loss after iteration 17000: 0.491290
Loss after iteration 18000: 0.471381
Loss after iteration 19000: 0.720984
Loss after iteration 20000: 0.386434
Loss after iteration 21000: 0.500964
Loss after iteration 22000: 0.667739
Loss after iteration 23000: 0.450866
Loss after iteration 24000: 0.485587
Loss after iteration 25000: 0.625858
Loss after iteration 26000: 0.425065
Loss after iteration 27000: 0.455403
Loss after iteration 28000: 0.637729
Loss after iteration 29000: 0.464308
Loss after iteration 30000: 0.496887
Loss after iteration 31000: 0.672519
Loss after iteration 32000: 0.402011
Loss after iteration 33000: 0.487205
```

```
[codecarbon INFO @ 15:13:12] Energy consumed for RAM : 0.004912 kWh. RAM Power : 5.89418506622314
45 W
```

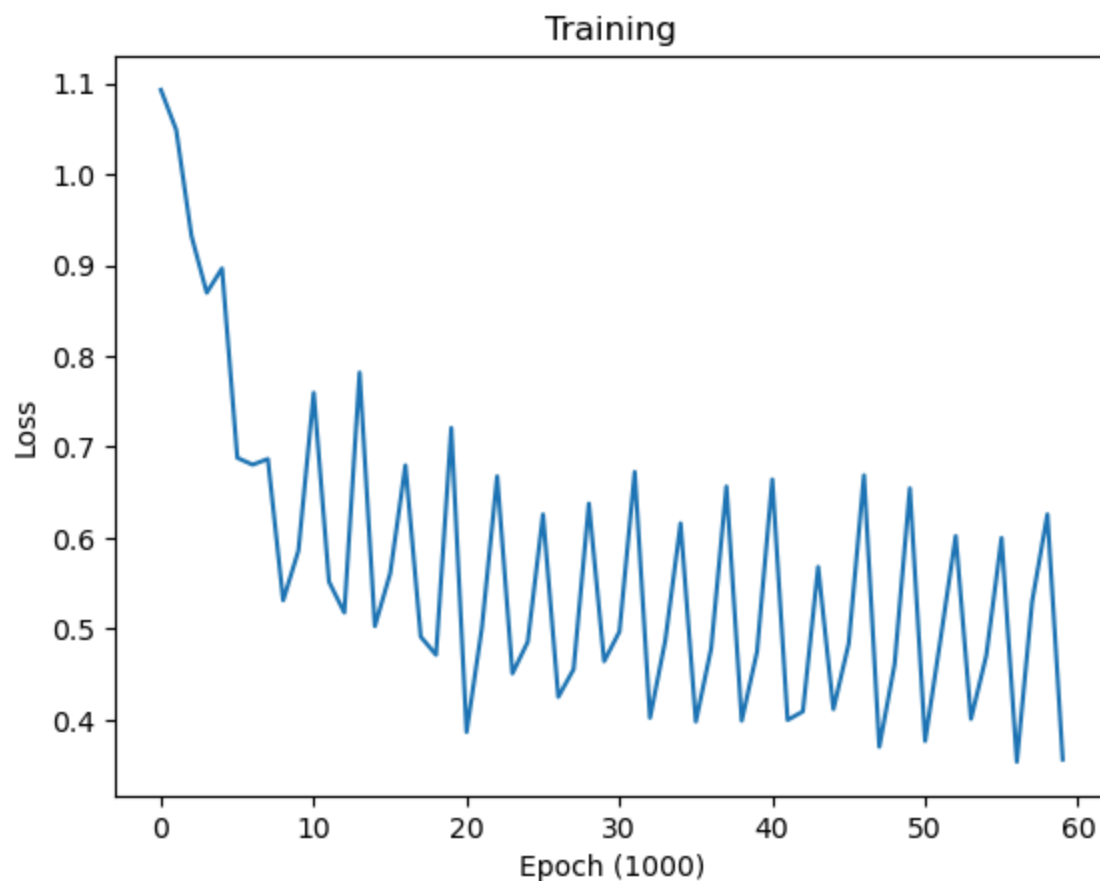
```
[codecarbon INFO @ 15:13:12] Energy consumed for all CPUs : 0.035432 kWh. Total CPU Power : 42.5
W
```

```
[codecarbon INFO @ 15:13:12] 0.040345 kWh of electricity used since the beginning.
```

```
[codecarbon INFO @ 15:13:12] 0.006106 g.CO2eq/s mean an estimation of 192.55602328089591 kg.CO2e
q/year
```

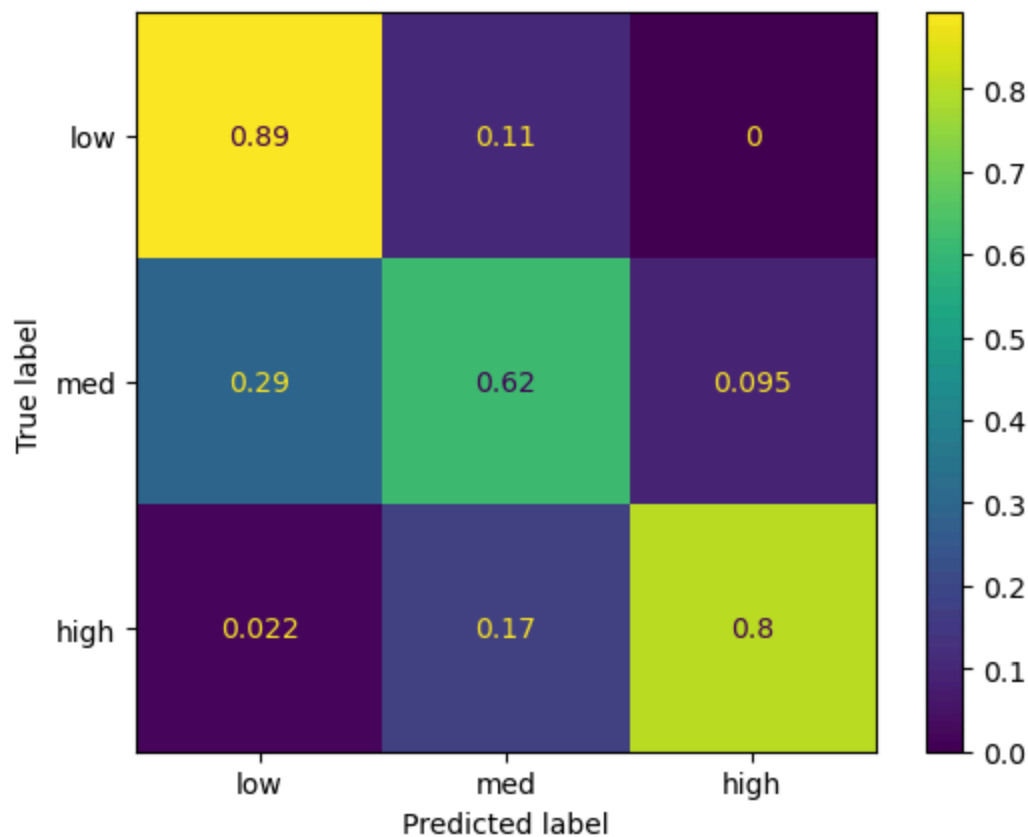
```
Loss after iteration 34000: 0.615984
Loss after iteration 35000: 0.397972
Loss after iteration 36000: 0.478280
Loss after iteration 37000: 0.656518
Loss after iteration 38000: 0.398895
Loss after iteration 39000: 0.474652
Loss after iteration 40000: 0.664267
Loss after iteration 41000: 0.399627
Loss after iteration 42000: 0.409015
Loss after iteration 43000: 0.568077
Loss after iteration 44000: 0.411560
Loss after iteration 45000: 0.483707
Loss after iteration 46000: 0.668879
Loss after iteration 47000: 0.370174
Loss after iteration 48000: 0.460681
Loss after iteration 49000: 0.654509
Loss after iteration 50000: 0.376518
Loss after iteration 51000: 0.487919
Loss after iteration 52000: 0.602000
Loss after iteration 53000: 0.400822
Loss after iteration 54000: 0.470511
Loss after iteration 55000: 0.599949
Loss after iteration 56000: 0.353616
Loss after iteration 57000: 0.529899
Loss after iteration 58000: 0.625933
Loss after iteration 59000: 0.356204
```

```
In [ ]: # Plot training loss
fig = plt.plot(np.array(nn.loss).squeeze())
plt.title(f"Training")
plt.xlabel("Epoch (1000)")
plt.ylabel("Loss")
plt.show()
```



```
[codecarbon INFO @ 15:13:27] Energy consumed for RAM : 0.004937 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:13:27] Energy consumed for all CPUs : 0.035609 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:13:27] 0.040546 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:13:42] Energy consumed for RAM : 0.004961 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:13:42] Energy consumed for all CPUs : 0.035787 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:13:42] 0.040748 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:13:57] Energy consumed for RAM : 0.004986 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:13:57] Energy consumed for all CPUs : 0.035964 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:13:57] 0.040950 kWh of electricity used since the beginning.
```

```
In [ ]: # Plot confusion matrix
y_true = np.argmax(y_test, axis=1)
y_pred = nn.predict(x_test)
display_labels = ["low", "med", "high"]
ConfusionMatrixDisplay.from_predictions(
    y_true, y_pred, normalize="true", display_labels=display_labels
)
plt.show()
```



```
[codecarbon INFO @ 15:14:12] Energy consumed for RAM : 0.005010 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:14:12] Energy consumed for all CPUs : 0.036141 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:14:12] 0.041152 kWh of electricity used since the beginning.
```

```
In [27]: # Total Loss
y_hat = nn.forward(x_test, use_dropout=False)
print("Cross entropy loss:", round(nn.cross_entropy_loss(y_test, y_hat), 3))
```

Cross entropy loss: 0.692

1.4 Loss plot and CE value for NN with Gradient Descent with Momentum [5pts Grad / 0.6% Bonus for Undergrad] [P]

Train your neural net implementation using gradient descent with momentum and print out the loss at every 1000th iteration (starting at iteration 0). The following cells will plot the loss vs epoch graph and calculate the final test CE.

```
In [28]: #####
### DO NOT CHANGE THIS CELL ###
#####
from NN import NeuralNet
from sklearn.metrics import ConfusionMatrixDisplay, confusion_matrix

x_train, y_train, x_test, y_test = get_housing_dataset()

nn = NeuralNet(
    y_train, lr=0.01, use_dropout=False, use_momentum=True
) # initialize neural net class
nn.gradient_descent(x_train, y_train, iter=60000, use_momentum=True) # train
```

Loss after iteration 0: 1.086276
Loss after iteration 1000: 0.904120
Loss after iteration 2000: 0.656440
Loss after iteration 3000: 0.598945
Loss after iteration 4000: 0.570251
Loss after iteration 5000: 0.553760
Loss after iteration 6000: 0.543053
Loss after iteration 7000: 0.536490
Loss after iteration 8000: 0.532221
Loss after iteration 9000: 0.529004
Loss after iteration 10000: 0.526308
Loss after iteration 11000: 0.523874
Loss after iteration 12000: 0.521527
Loss after iteration 13000: 0.519123
Loss after iteration 14000: 0.516534
Loss after iteration 15000: 0.513655
Loss after iteration 16000: 0.510424
Loss after iteration 17000: 0.506868
Loss after iteration 18000: 0.503132
Loss after iteration 19000: 0.499435
Loss after iteration 20000: 0.495933
Loss after iteration 21000: 0.492660
Loss after iteration 22000: 0.489586
Loss after iteration 23000: 0.486679
Loss after iteration 24000: 0.483932
Loss after iteration 25000: 0.481358
Loss after iteration 26000: 0.478988
Loss after iteration 27000: 0.476848

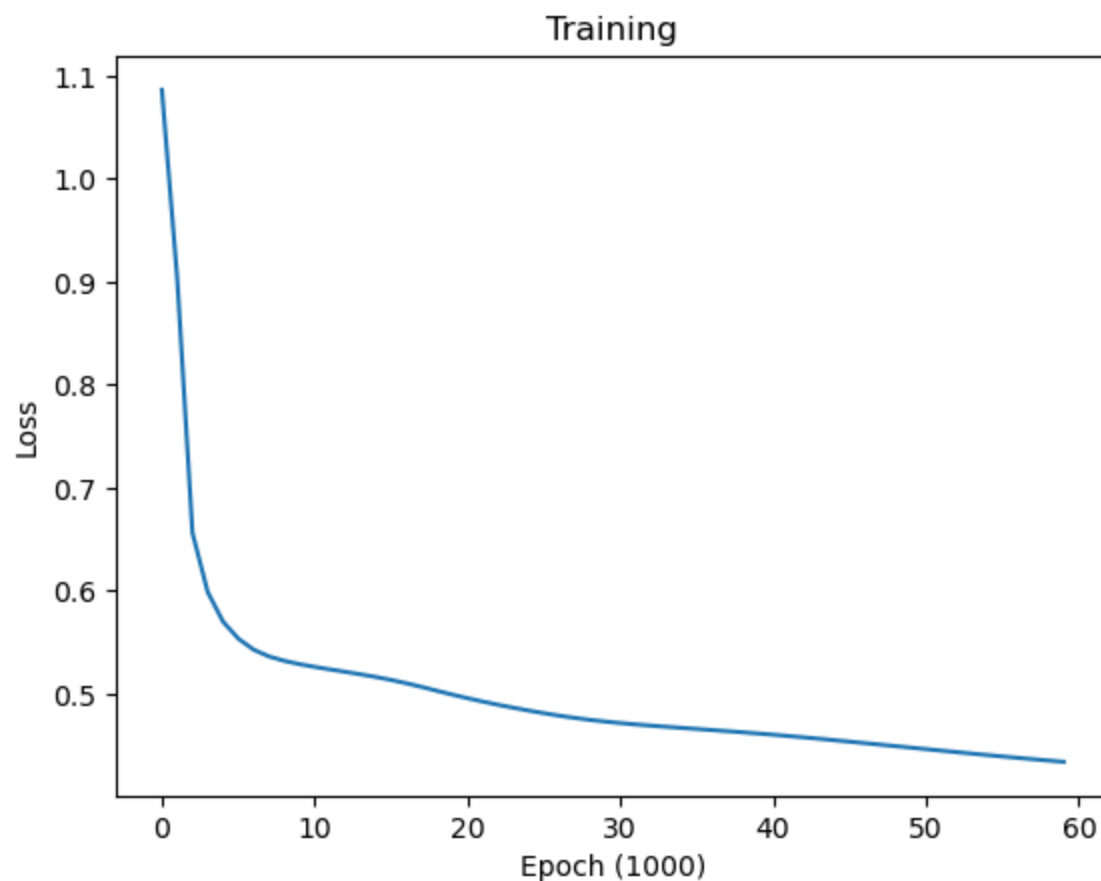
[codecarbon INFO @ 15:14:27] Energy consumed for RAM : 0.005035 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:14:27] Energy consumed for all CPUs : 0.036318 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:14:27] 0.041353 kWh of electricity used since the beginning.

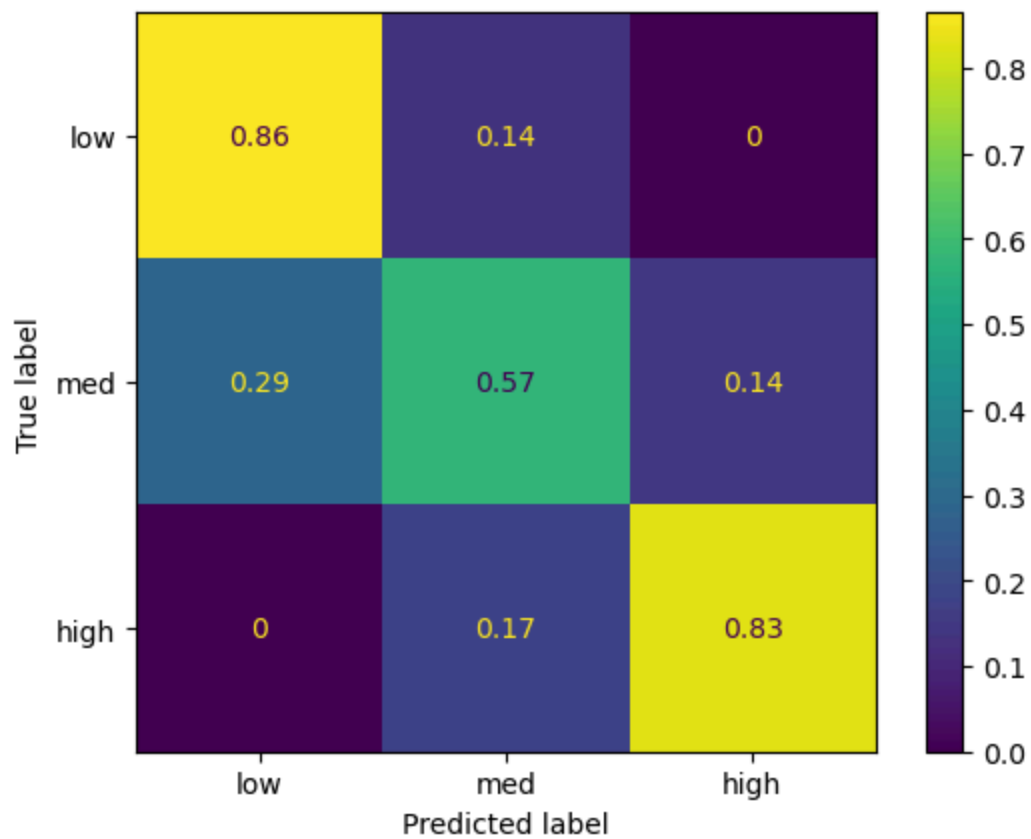
```
Loss after iteration 28000: 0.474951
Loss after iteration 29000: 0.473285
Loss after iteration 30000: 0.471821
Loss after iteration 31000: 0.470516
Loss after iteration 32000: 0.469328
Loss after iteration 33000: 0.468217
Loss after iteration 34000: 0.467149
Loss after iteration 35000: 0.466099
Loss after iteration 36000: 0.465045
Loss after iteration 37000: 0.463971
Loss after iteration 38000: 0.462864
Loss after iteration 39000: 0.461714
Loss after iteration 40000: 0.460514
Loss after iteration 41000: 0.459260
Loss after iteration 42000: 0.457953
Loss after iteration 43000: 0.456598
Loss after iteration 44000: 0.455204
Loss after iteration 45000: 0.453780
Loss after iteration 46000: 0.452338
Loss after iteration 47000: 0.450887
Loss after iteration 48000: 0.449431
Loss after iteration 49000: 0.447975
Loss after iteration 50000: 0.446521
Loss after iteration 51000: 0.445070
Loss after iteration 52000: 0.443628
Loss after iteration 53000: 0.442202
Loss after iteration 54000: 0.440797
Loss after iteration 55000: 0.439420
Loss after iteration 56000: 0.438073
Loss after iteration 57000: 0.436757
Loss after iteration 58000: 0.435469
Loss after iteration 59000: 0.434208
```

```
In [ ]: # Plot training loss
fig = plt.plot(np.array(nn.loss).squeeze())
plt.title(f"Training")
plt.xlabel("Epoch (1000)")
plt.ylabel("Loss")
plt.show()
```



[codecarbon INFO @ 15:14:42] Energy consumed for RAM : 0.005060 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:14:42] Energy consumed for all CPUs : 0.036495 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:14:42] 0.041555 kWh of electricity used since the beginning.

```
In [30]: # Plot confusion matrix
y_true = np.argmax(y_test, axis=1)
y_pred = nn.predict(x_test)
display_labels = ["low", "med", "high"]
ConfusionMatrixDisplay.from_predictions(
    y_true, y_pred, normalize="true", display_labels=display_labels
)
plt.show()
```



```
In [ ]: # Total Loss
y_hat = nn.forward(x_test, use_dropout=False)
print("Cross entropy loss:", round(nn.cross_entropy_loss(y_test, y_hat), 3))
```

Cross entropy loss: 0.696

[codecarbon INFO @ 15:14:57] Energy consumed for RAM : 0.005084 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:14:57] Energy consumed for all CPUs : 0.036672 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:14:57] 0.041756 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:15:12] Energy consumed for RAM : 0.005109 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:15:12] Energy consumed for all CPUs : 0.036850 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:15:12] 0.041958 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:15:12] 0.006107 g.CO2eq/s mean an estimation of 192.59440604535902 kg.CO2eq/year

[codecarbon INFO @ 15:15:27] Energy consumed for RAM : 0.005133 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:15:27] Energy consumed for all CPUs : 0.037027 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:15:27] 0.042160 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:15:42] Energy consumed for RAM : 0.005158 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:15:42] Energy consumed for all CPUs : 0.037204 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:15:42] 0.042362 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:15:57] Energy consumed for RAM : 0.005182 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:15:57] Energy consumed for all CPUs : 0.037381 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:15:57] 0.042564 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:16:12] Energy consumed for RAM : 0.005207 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:16:12] Energy consumed for all CPUs : 0.037558 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:16:12] 0.042765 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:16:27] Energy consumed for RAM : 0.005231 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:16:27] Energy consumed for all CPUs : 0.037736 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:16:27] 0.042967 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:16:42] Energy consumed for RAM : 0.005256 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:16:42] Energy consumed for all CPUs : 0.037913 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:16:42] 0.043169 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:16:57] Energy consumed for RAM : 0.005281 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:16:57] Energy consumed for all CPUs : 0.038090 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:16:57] 0.043371 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:17:12] Energy consumed for RAM : 0.005305 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:17:12] Energy consumed for all CPUs : 0.038267 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:17:12] 0.043572 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:17:12] 0.006107 g.CO2eq/s mean an estimation of 192.59237506338266 kg.CO2eq/year

[codecarbon INFO @ 15:17:27] Energy consumed for RAM : 0.005330 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:17:27] Energy consumed for all CPUs : 0.038444 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:17:27] 0.043774 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:17:42] Energy consumed for RAM : 0.005354 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:17:42] Energy consumed for all CPUs : 0.038621 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:17:42] 0.043976 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:17:57] Energy consumed for RAM : 0.005379 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:17:57] Energy consumed for all CPUs : 0.038799 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:17:57] 0.044177 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:18:12] Energy consumed for RAM : 0.005403 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:18:12] Energy consumed for all CPUs : 0.038976 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:18:12] 0.044379 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:18:27] Energy consumed for RAM : 0.005428 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:18:27] Energy consumed for all CPUs : 0.039153 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:18:27] 0.044581 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:18:42] Energy consumed for RAM : 0.005453 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:18:42] Energy consumed for all CPUs : 0.039330 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:18:42] 0.044782 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:18:57] Energy consumed for RAM : 0.005477 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:18:57] Energy consumed for all CPUs : 0.039507 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:18:57] 0.044984 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:19:12] Energy consumed for RAM : 0.005502 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:19:12] Energy consumed for all CPUs : 0.039684 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:19:12] 0.045186 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:19:12] 0.006107 g.CO2eq/s mean an estimation of 192.59570047861692 kg.CO2eq/year

[codecarbon INFO @ 15:19:27] Energy consumed for RAM : 0.005526 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:19:27] Energy consumed for all CPUs : 0.039861 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:19:27] 0.045387 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:19:42] Energy consumed for RAM : 0.005551 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:19:42] Energy consumed for all CPUs : 0.040038 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:19:42] 0.045589 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:19:57] Energy consumed for RAM : 0.005575 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:19:57] Energy consumed for all CPUs : 0.040216 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:19:57] 0.045791 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:20:12] Energy consumed for RAM : 0.005600 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:20:12] Energy consumed for all CPUs : 0.040393 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:20:12] 0.045993 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:20:27] Energy consumed for RAM : 0.005624 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:20:27] Energy consumed for all CPUs : 0.040570 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:20:27] 0.046194 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:20:42] Energy consumed for RAM : 0.005649 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:20:42] Energy consumed for all CPUs : 0.040747 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:20:42] 0.046396 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:20:57] Energy consumed for RAM : 0.005674 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:20:57] Energy consumed for all CPUs : 0.040924 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:20:57] 0.046598 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:21:12] Energy consumed for RAM : 0.005698 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:21:12] Energy consumed for all CPUs : 0.041102 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:21:12] 0.046800 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:21:12] 0.006107 g.CO2eq/s mean an estimation of 192.59011749156755 kg.CO2eq/year

[codecarbon INFO @ 15:21:27] Energy consumed for RAM : 0.005723 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:21:27] Energy consumed for all CPUs : 0.041279 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:21:27] 0.047001 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:21:42] Energy consumed for RAM : 0.005747 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:21:43] Energy consumed for all CPUs : 0.041456 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:21:43] 0.047203 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:21:58] Energy consumed for RAM : 0.005772 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:21:58] Energy consumed for all CPUs : 0.041633 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:21:58] 0.047405 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:22:13] Energy consumed for RAM : 0.005796 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:22:13] Energy consumed for all CPUs : 0.041810 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:22:13] 0.047607 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:22:28] Energy consumed for RAM : 0.005821 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:22:28] Energy consumed for all CPUs : 0.041988 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:22:28] 0.047809 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:22:43] Energy consumed for RAM : 0.005846 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:22:43] Energy consumed for all CPUs : 0.042165 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:22:43] 0.048010 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:22:58] Energy consumed for RAM : 0.005870 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:22:58] Energy consumed for all CPUs : 0.042342 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:22:58] 0.048212 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:23:13] Energy consumed for RAM : 0.005895 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:23:13] Energy consumed for all CPUs : 0.042519 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:23:13] 0.048414 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:23:13] 0.006107 g.CO2eq/s mean an estimation of 192.59134379308367 kg.CO2eq/year
[codecarbon INFO @ 15:23:28] Energy consumed for RAM : 0.005919 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:23:28] Energy consumed for all CPUs : 0.042696 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:23:28] 0.048615 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:23:43] Energy consumed for RAM : 0.005944 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:23:43] Energy consumed for all CPUs : 0.042873 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:23:43] 0.048817 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:23:58] Energy consumed for RAM : 0.005968 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:23:58] Energy consumed for all CPUs : 0.043050 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:23:58] 0.049019 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:24:13] Energy consumed for RAM : 0.005993 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:24:13] Energy consumed for all CPUs : 0.043228 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:24:13] 0.049220 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:24:28] Energy consumed for RAM : 0.006017 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:24:28] Energy consumed for all CPUs : 0.043405 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:24:28] 0.049422 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:24:43] Energy consumed for RAM : 0.006042 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:24:43] Energy consumed for all CPUs : 0.043582 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:24:43] 0.049624 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:24:58] Energy consumed for RAM : 0.006067 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:24:58] Energy consumed for all CPUs : 0.043759 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:24:58] 0.049826 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:25:13] Energy consumed for RAM : 0.006091 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:25:13] Energy consumed for all CPUs : 0.043936 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:25:13] 0.050027 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:25:13] 0.006107 g.CO2eq/s mean an estimation of 192.59210901373265 kg.CO2eq/year
[codecarbon INFO @ 15:25:28] Energy consumed for RAM : 0.006116 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:25:28] Energy consumed for all CPUs : 0.044113 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:25:28] 0.050229 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:25:43] Energy consumed for RAM : 0.006140 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:25:43] Energy consumed for all CPUs : 0.044291 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:25:43] 0.050431 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:25:58] Energy consumed for RAM : 0.006165 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 15:25:58] Energy consumed for all CPUs : 0.044468 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:25:58] 0.050633 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:26:13] Energy consumed for RAM : 0.006189 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:26:13] Energy consumed for all CPUs : 0.044645 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:26:13] 0.050834 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:26:28] Energy consumed for RAM : 0.006214 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:26:28] Energy consumed for all CPUs : 0.044822 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:26:28] 0.051036 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:26:43] Energy consumed for RAM : 0.006239 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:26:43] Energy consumed for all CPUs : 0.044999 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:26:43] 0.051238 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:26:58] Energy consumed for RAM : 0.006263 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:26:58] Energy consumed for all CPUs : 0.045176 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:26:58] 0.051439 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:27:13] Energy consumed for RAM : 0.006288 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:27:13] Energy consumed for all CPUs : 0.045354 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:27:13] 0.051641 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:27:13] 0.006107 g.CO2eq/s mean an estimation of 192.59793625924593 kg.CO2eq/year

[codecarbon INFO @ 15:27:28] Energy consumed for RAM : 0.006312 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:27:28] Energy consumed for all CPUs : 0.045531 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:27:28] 0.051843 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:27:43] Energy consumed for RAM : 0.006337 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:27:43] Energy consumed for all CPUs : 0.045708 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:27:43] 0.052045 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:27:58] Energy consumed for RAM : 0.006361 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:27:58] Energy consumed for all CPUs : 0.045885 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:27:58] 0.052246 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:28:13] Energy consumed for RAM : 0.006386 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:28:13] Energy consumed for all CPUs : 0.046062 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:28:13] 0.052448 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:28:28] Energy consumed for RAM : 0.006411 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:28:28] Energy consumed for all CPUs : 0.046239 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:28:28] 0.052650 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:28:43] Energy consumed for RAM : 0.006435 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:28:43] Energy consumed for all CPUs : 0.046417 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:28:43] 0.052852 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:28:58] Energy consumed for RAM : 0.006460 kWh. RAM Power : 5.8941850662231445 W

45 W
[codecarbon INFO @ 15:28:58] Energy consumed for all CPUs : 0.046594 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:28:58] 0.053053 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:29:13] Energy consumed for RAM : 0.006484 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:29:13] Energy consumed for all CPUs : 0.046771 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:29:13] 0.053255 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:29:13] 0.006107 g.CO2eq/s mean an estimation of 192.59985669627662 kg.CO2eq/year
[codecarbon INFO @ 15:29:28] Energy consumed for RAM : 0.006509 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:29:28] Energy consumed for all CPUs : 0.046948 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:29:28] 0.053457 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:29:43] Energy consumed for RAM : 0.006533 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:29:43] Energy consumed for all CPUs : 0.047125 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:29:43] 0.053659 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:29:58] Energy consumed for RAM : 0.006558 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:29:58] Energy consumed for all CPUs : 0.047302 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:29:58] 0.053860 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:30:13] Energy consumed for RAM : 0.006582 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:30:13] Energy consumed for all CPUs : 0.047479 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:30:13] 0.054062 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:30:28] Energy consumed for RAM : 0.006607 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:30:28] Energy consumed for all CPUs : 0.047656 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:30:28] 0.054263 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:30:43] Energy consumed for RAM : 0.006632 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:30:43] Energy consumed for all CPUs : 0.047834 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:30:43] 0.054465 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:30:58] Energy consumed for RAM : 0.006656 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:30:58] Energy consumed for all CPUs : 0.048011 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:30:58] 0.054667 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:31:13] Energy consumed for RAM : 0.006681 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:31:13] Energy consumed for all CPUs : 0.048188 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:31:13] 0.054869 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:31:13] 0.006107 g.CO2eq/s mean an estimation of 192.59469116731853 kg.CO2eq/year
[codecarbon INFO @ 15:31:28] Energy consumed for RAM : 0.006705 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:31:28] Energy consumed for all CPUs : 0.048365 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:31:28] 0.055070 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:31:43] Energy consumed for RAM : 0.006730 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 15:31:43] Energy consumed for all CPUs : 0.048542 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:31:43] 0.055272 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:31:58] Energy consumed for RAM : 0.006754 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:31:58] Energy consumed for all CPUs : 0.048720 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:31:58] 0.055474 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:32:13] Energy consumed for RAM : 0.006779 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:32:13] Energy consumed for all CPUs : 0.048897 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:32:13] 0.055676 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:32:28] Energy consumed for RAM : 0.006804 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:32:28] Energy consumed for all CPUs : 0.049074 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:32:28] 0.055878 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:32:43] Energy consumed for RAM : 0.006828 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:32:43] Energy consumed for all CPUs : 0.049251 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:32:43] 0.056079 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:32:58] Energy consumed for RAM : 0.006853 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:32:58] Energy consumed for all CPUs : 0.049428 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:32:58] 0.056281 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:33:13] Energy consumed for RAM : 0.006877 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:33:13] Energy consumed for all CPUs : 0.049606 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:33:13] 0.056483 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:33:13] 0.006107 g.CO2eq/s mean an estimation of 192.575606281738 kg.CO2eq/year
[codecarbon INFO @ 15:33:28] Energy consumed for RAM : 0.006902 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:33:28] Energy consumed for all CPUs : 0.049783 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:33:28] 0.056684 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:33:43] Energy consumed for RAM : 0.006926 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:33:43] Energy consumed for all CPUs : 0.049960 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:33:43] 0.056887 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:33:58] Energy consumed for RAM : 0.006951 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:33:58] Energy consumed for all CPUs : 0.050137 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:33:58] 0.057088 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:34:13] Energy consumed for RAM : 0.006975 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:34:13] Energy consumed for all CPUs : 0.050315 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:34:13] 0.057290 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:34:28] Energy consumed for RAM : 0.007000 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:34:28] Energy consumed for all CPUs : 0.050492 kWh. Total CPU Power : 42.5

W
[codecarbon INFO @ 15:34:28] 0.057492 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:34:43] Energy consumed for RAM : 0.007025 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:34:43] Energy consumed for all CPUs : 0.050669 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:34:43] 0.057694 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:34:58] Energy consumed for RAM : 0.007049 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:34:58] Energy consumed for all CPUs : 0.050846 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:34:58] 0.057895 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:35:13] Energy consumed for RAM : 0.007074 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:35:13] Energy consumed for all CPUs : 0.051023 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:35:13] 0.058097 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:35:13] 0.006106 g.CO2eq/s mean an estimation of 192.57306943290666 kg.CO2e
q/year
[codecarbon INFO @ 15:35:28] Energy consumed for RAM : 0.007098 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:35:28] Energy consumed for all CPUs : 0.051200 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:35:28] 0.058299 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:35:43] Energy consumed for RAM : 0.007123 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:35:43] Energy consumed for all CPUs : 0.051378 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:35:43] 0.058500 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:35:58] Energy consumed for RAM : 0.007147 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:35:58] Energy consumed for all CPUs : 0.051555 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:35:58] 0.058702 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:36:13] Energy consumed for RAM : 0.007172 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:36:13] Energy consumed for all CPUs : 0.051732 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:36:13] 0.058904 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:36:28] Energy consumed for RAM : 0.007197 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:36:28] Energy consumed for all CPUs : 0.051909 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:36:28] 0.059106 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:36:43] Energy consumed for RAM : 0.007221 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:36:43] Energy consumed for all CPUs : 0.052086 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:36:43] 0.059307 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:36:58] Energy consumed for RAM : 0.007246 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:36:58] Energy consumed for all CPUs : 0.052263 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:36:58] 0.059509 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:37:13] Energy consumed for RAM : 0.007270 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:37:13] Energy consumed for all CPUs : 0.052440 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:37:13] 0.059711 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:37:13] 0.006106 g.CO2eq/s mean an estimation of 192.5739719838551 kg.CO2eq/year

[codecarbon INFO @ 15:37:28] Energy consumed for RAM : 0.007295 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:37:28] Energy consumed for all CPUs : 0.052618 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:37:28] 0.059912 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:37:43] Energy consumed for RAM : 0.007319 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:37:43] Energy consumed for all CPUs : 0.052795 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:37:43] 0.060114 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:37:58] Energy consumed for RAM : 0.007344 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:37:58] Energy consumed for all CPUs : 0.052972 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:37:58] 0.060316 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:38:13] Energy consumed for RAM : 0.007369 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:38:13] Energy consumed for all CPUs : 0.053150 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:38:13] 0.060519 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:38:28] Energy consumed for RAM : 0.007393 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:38:28] Energy consumed for all CPUs : 0.053327 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:38:28] 0.060720 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:38:43] Energy consumed for RAM : 0.007418 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:38:43] Energy consumed for all CPUs : 0.053504 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:38:43] 0.060922 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:38:58] Energy consumed for RAM : 0.007442 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:38:58] Energy consumed for all CPUs : 0.053681 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:38:58] 0.061124 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:39:13] Energy consumed for RAM : 0.007467 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:39:13] Energy consumed for all CPUs : 0.053859 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:39:13] 0.061325 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:39:13] 0.006106 g.CO2eq/s mean an estimation of 192.55902466276902 kg.CO2eq/year

[codecarbon INFO @ 15:39:28] Energy consumed for RAM : 0.007491 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:39:28] Energy consumed for all CPUs : 0.054036 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:39:28] 0.061527 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:39:43] Energy consumed for RAM : 0.007516 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:39:43] Energy consumed for all CPUs : 0.054213 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:39:43] 0.061729 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:39:58] Energy consumed for RAM : 0.007540 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:39:58] Energy consumed for all CPUs : 0.054390 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:39:58] 0.061930 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:40:13] Energy consumed for RAM : 0.007565 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:40:13] Energy consumed for all CPUs : 0.054567 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:40:13] 0.062132 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:40:28] Energy consumed for RAM : 0.007590 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:40:28] Energy consumed for all CPUs : 0.054744 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:40:28] 0.062334 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:40:43] Energy consumed for RAM : 0.007614 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:40:43] Energy consumed for all CPUs : 0.054922 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:40:43] 0.062536 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:40:58] Energy consumed for RAM : 0.007639 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:40:58] Energy consumed for all CPUs : 0.055099 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:40:58] 0.062738 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:41:13] Energy consumed for RAM : 0.007663 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:41:13] Energy consumed for all CPUs : 0.055277 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:41:13] 0.062940 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:41:13] 0.006107 g.CO2eq/s mean an estimation of 192.5851510706965 kg.CO2eq/year

[codecarbon INFO @ 15:41:28] Energy consumed for RAM : 0.007688 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:41:28] Energy consumed for all CPUs : 0.055454 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:41:28] 0.063142 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:41:43] Energy consumed for RAM : 0.007712 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:41:43] Energy consumed for all CPUs : 0.055631 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:41:43] 0.063343 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:41:58] Energy consumed for RAM : 0.007737 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:41:58] Energy consumed for all CPUs : 0.055808 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:41:58] 0.063545 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:42:13] Energy consumed for RAM : 0.007762 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:42:13] Energy consumed for all CPUs : 0.055985 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:42:13] 0.063747 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:42:28] Energy consumed for RAM : 0.007786 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:42:28] Energy consumed for all CPUs : 0.056162 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:42:28] 0.063948 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:42:43] Energy consumed for RAM : 0.007811 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:42:43] Energy consumed for all CPUs : 0.056339 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:42:43] 0.064150 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:42:58] Energy consumed for RAM : 0.007835 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 15:42:58] Energy consumed for all CPUs : 0.056517 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:42:58] 0.064352 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:43:13] Energy consumed for RAM : 0.007860 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:43:13] Energy consumed for all CPUs : 0.056694 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:43:13] 0.064553 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:43:13] 0.006107 g.CO2eq/s mean an estimation of 192.57488180262462 kg.CO2eq/year

[codecarbon INFO @ 15:43:28] Energy consumed for RAM : 0.007884 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:43:29] Energy consumed for all CPUs : 0.056871 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:43:29] 0.064755 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:43:44] Energy consumed for RAM : 0.007909 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:43:44] Energy consumed for all CPUs : 0.057048 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:43:44] 0.064957 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:43:59] Energy consumed for RAM : 0.007933 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:43:59] Energy consumed for all CPUs : 0.057225 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:43:59] 0.065158 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:44:14] Energy consumed for RAM : 0.007958 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:44:14] Energy consumed for all CPUs : 0.057402 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:44:14] 0.065360 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:44:29] Energy consumed for RAM : 0.007983 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:44:29] Energy consumed for all CPUs : 0.057579 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:44:29] 0.065562 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:44:44] Energy consumed for RAM : 0.008007 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:44:44] Energy consumed for all CPUs : 0.057756 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:44:44] 0.065764 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:44:59] Energy consumed for RAM : 0.008032 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:44:59] Energy consumed for all CPUs : 0.057934 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:44:59] 0.065965 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:45:14] Energy consumed for RAM : 0.008056 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:45:14] Energy consumed for all CPUs : 0.058110 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:45:14] 0.066167 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:45:14] 0.006106 g.CO2eq/s mean an estimation of 192.54414992115335 kg.CO2eq/year

[codecarbon INFO @ 15:45:29] Energy consumed for RAM : 0.008081 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:45:29] Energy consumed for all CPUs : 0.058288 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:45:29] 0.066368 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:45:44] Energy consumed for RAM : 0.008105 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:45:44] Energy consumed for all CPUs : 0.058465 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:45:44] 0.066570 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:45:59] Energy consumed for RAM : 0.008130 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:45:59] Energy consumed for all CPUs : 0.058642 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:45:59] 0.066772 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:46:14] Energy consumed for RAM : 0.008154 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:46:14] Energy consumed for all CPUs : 0.058819 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:46:14] 0.066974 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:46:29] Energy consumed for RAM : 0.008179 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:46:29] Energy consumed for all CPUs : 0.058996 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:46:29] 0.067175 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:46:44] Energy consumed for RAM : 0.008204 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:46:44] Energy consumed for all CPUs : 0.059173 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:46:44] 0.067377 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:46:59] Energy consumed for RAM : 0.008228 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:46:59] Energy consumed for all CPUs : 0.059351 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:46:59] 0.067579 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:47:14] Energy consumed for RAM : 0.008253 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:47:14] Energy consumed for all CPUs : 0.059528 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:47:14] 0.067780 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:47:14] 0.006106 g.CO2eq/s mean an estimation of 192.57452196317175 kg.CO2eq/year
[codecarbon INFO @ 15:47:29] Energy consumed for RAM : 0.008277 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:47:29] Energy consumed for all CPUs : 0.059705 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:47:29] 0.067982 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:47:44] Energy consumed for RAM : 0.008302 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:47:44] Energy consumed for all CPUs : 0.059882 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:47:44] 0.068183 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:47:59] Energy consumed for RAM : 0.008326 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:47:59] Energy consumed for all CPUs : 0.060059 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:47:59] 0.068385 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:48:14] Energy consumed for RAM : 0.008351 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:48:14] Energy consumed for all CPUs : 0.060236 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:48:14] 0.068587 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:48:29] Energy consumed for RAM : 0.008375 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:48:29] Energy consumed for all CPUs : 0.060413 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:48:29] 0.068789 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:48:44] Energy consumed for RAM : 0.008400 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:48:44] Energy consumed for all CPUs : 0.060590 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:48:44] 0.068990 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:48:59] Energy consumed for RAM : 0.008425 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:48:59] Energy consumed for all CPUs : 0.060768 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:48:59] 0.069192 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:49:14] Energy consumed for RAM : 0.008449 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:49:14] Energy consumed for all CPUs : 0.060945 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:49:14] 0.069394 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:49:14] 0.006106 g.CO2eq/s mean an estimation of 192.56632819477045 kg.CO2e
q/year
[codecarbon INFO @ 15:49:29] Energy consumed for RAM : 0.008474 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:49:29] Energy consumed for all CPUs : 0.061122 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:49:29] 0.069595 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:49:44] Energy consumed for RAM : 0.008498 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:49:44] Energy consumed for all CPUs : 0.061299 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:49:44] 0.069797 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:49:59] Energy consumed for RAM : 0.008523 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:49:59] Energy consumed for all CPUs : 0.061476 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:49:59] 0.069999 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:50:14] Energy consumed for RAM : 0.008547 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:50:14] Energy consumed for all CPUs : 0.061653 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:50:14] 0.070201 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:50:29] Energy consumed for RAM : 0.008572 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:50:29] Energy consumed for all CPUs : 0.061831 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:50:29] 0.070402 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:50:44] Energy consumed for RAM : 0.008596 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:50:44] Energy consumed for all CPUs : 0.062008 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:50:44] 0.070604 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:50:59] Energy consumed for RAM : 0.008621 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:50:59] Energy consumed for all CPUs : 0.062185 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:50:59] 0.070806 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:51:14] Energy consumed for RAM : 0.008646 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:51:14] Energy consumed for all CPUs : 0.062362 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:51:14] 0.071008 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:51:14] 0.006106 g.CO2eq/s mean an estimation of 192.54921869033424 kg.CO2e

q/year

[codecarbon INFO @ 15:51:29] Energy consumed for RAM : 0.008670 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:51:29] Energy consumed for all CPUs : 0.062539 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:51:29] 0.071209 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:51:44] Energy consumed for RAM : 0.008695 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:51:44] Energy consumed for all CPUs : 0.062716 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:51:44] 0.071411 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:51:59] Energy consumed for RAM : 0.008719 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:51:59] Energy consumed for all CPUs : 0.062893 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:51:59] 0.071613 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:52:14] Energy consumed for RAM : 0.008744 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:52:14] Energy consumed for all CPUs : 0.063071 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:52:14] 0.071814 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:52:29] Energy consumed for RAM : 0.008768 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:52:29] Energy consumed for all CPUs : 0.063248 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:52:29] 0.072016 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:52:44] Energy consumed for RAM : 0.008793 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:52:44] Energy consumed for all CPUs : 0.063425 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:52:44] 0.072218 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:52:59] Energy consumed for RAM : 0.008817 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:52:59] Energy consumed for all CPUs : 0.063602 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:52:59] 0.072419 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:53:14] Energy consumed for RAM : 0.008842 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:53:14] Energy consumed for all CPUs : 0.063779 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:53:14] 0.072621 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:53:14] 0.006106 g.CO2eq/s mean an estimation of 192.56563694716354 kg.CO2eq/year

[codecarbon INFO @ 15:53:29] Energy consumed for RAM : 0.008867 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:53:29] Energy consumed for all CPUs : 0.063956 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:53:29] 0.072823 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:53:44] Energy consumed for RAM : 0.008891 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:53:44] Energy consumed for all CPUs : 0.064133 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:53:44] 0.073025 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:53:59] Energy consumed for RAM : 0.008916 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 15:53:59] Energy consumed for all CPUs : 0.064311 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 15:53:59] 0.073226 kWh of electricity used since the beginning.

[codecarbon INFO @ 15:54:14] Energy consumed for RAM : 0.008940 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 15:54:14] Energy consumed for all CPUs : 0.064488 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:54:14] 0.073428 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:54:29] Energy consumed for RAM : 0.008965 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:54:29] Energy consumed for all CPUs : 0.064665 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:54:29] 0.073630 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:54:44] Energy consumed for RAM : 0.008989 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:54:44] Energy consumed for all CPUs : 0.064842 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:54:44] 0.073831 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:54:59] Energy consumed for RAM : 0.009014 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:54:59] Energy consumed for all CPUs : 0.065019 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:54:59] 0.074033 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:55:14] Energy consumed for RAM : 0.009038 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:55:14] Energy consumed for all CPUs : 0.065196 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:55:14] 0.074235 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:55:14] 0.006106 g.CO2eq/s mean an estimation of 192.55428055210444 kg.CO2eq/year
[codecarbon INFO @ 15:55:29] Energy consumed for RAM : 0.009063 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:55:29] Energy consumed for all CPUs : 0.065373 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:55:29] 0.074437 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:55:44] Energy consumed for RAM : 0.009088 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:55:44] Energy consumed for all CPUs : 0.065551 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:55:44] 0.074638 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:55:59] Energy consumed for RAM : 0.009112 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:55:59] Energy consumed for all CPUs : 0.065728 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:55:59] 0.074840 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:56:14] Energy consumed for RAM : 0.009137 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:56:14] Energy consumed for all CPUs : 0.065905 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:56:14] 0.075042 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:56:29] Energy consumed for RAM : 0.009161 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:56:29] Energy consumed for all CPUs : 0.066082 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:56:29] 0.075243 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:56:44] Energy consumed for RAM : 0.009186 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:56:44] Energy consumed for all CPUs : 0.066259 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 15:56:44] 0.075445 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:56:59] Energy consumed for RAM : 0.009210 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 15:56:59] Energy consumed for all CPUs : 0.066436 kWh. Total CPU Power : 42.5

W
[codecarbon INFO @ 15:56:59] 0.075647 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:57:14] Energy consumed for RAM : 0.009235 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:57:14] Energy consumed for all CPUs : 0.066613 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:57:14] 0.075848 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:57:14] 0.006106 g.CO2eq/s mean an estimation of 192.56053974449537 kg.CO2e
q/year
[codecarbon INFO @ 15:57:29] Energy consumed for RAM : 0.009259 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:57:29] Energy consumed for all CPUs : 0.066791 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:57:29] 0.076050 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:57:44] Energy consumed for RAM : 0.009284 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:57:44] Energy consumed for all CPUs : 0.066968 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:57:44] 0.076252 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:57:59] Energy consumed for RAM : 0.009309 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:57:59] Energy consumed for all CPUs : 0.067145 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:57:59] 0.076453 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:58:14] Energy consumed for RAM : 0.009333 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:58:14] Energy consumed for all CPUs : 0.067322 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:58:14] 0.076655 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:58:29] Energy consumed for RAM : 0.009358 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:58:29] Energy consumed for all CPUs : 0.067499 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:58:29] 0.076857 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:58:44] Energy consumed for RAM : 0.009382 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:58:44] Energy consumed for all CPUs : 0.067676 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:58:44] 0.077058 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:58:59] Energy consumed for RAM : 0.009407 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:58:59] Energy consumed for all CPUs : 0.067853 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:58:59] 0.077260 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:59:14] Energy consumed for RAM : 0.009431 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:59:14] Energy consumed for all CPUs : 0.068030 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:59:14] 0.077462 kWh of electricity used since the beginning.
[codecarbon INFO @ 15:59:14] 0.006106 g.CO2eq/s mean an estimation of 192.5680590793397 kg.CO2eq/
year
[codecarbon INFO @ 15:59:29] Energy consumed for RAM : 0.009456 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 15:59:29] Energy consumed for all CPUs : 0.068208 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 15:59:29] 0.077663 kWh of electricity used since the beginning.
[codecarbon WARNING @ 16:34:41] Background scheduler didn't run for a long period (2111s), result
s might be inaccurate
[codecarbon INFO @ 16:34:41] Energy consumed for RAM : 0.012913 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 16:34:41] Energy consumed for all CPUs : 0.093138 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:34:41] 0.106051 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:34:56] Energy consumed for RAM : 0.012938 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:34:56] Energy consumed for all CPUs : 0.093315 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:34:56] 0.106253 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:35:11] Energy consumed for RAM : 0.012963 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:35:11] Energy consumed for all CPUs : 0.093492 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:35:11] 0.106455 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:35:26] Energy consumed for RAM : 0.012987 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:35:26] Energy consumed for all CPUs : 0.093669 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:35:26] 0.106656 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:35:41] Energy consumed for RAM : 0.013012 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:35:41] Energy consumed for all CPUs : 0.093846 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:35:41] 0.106858 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:35:56] Energy consumed for RAM : 0.013036 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:35:56] Energy consumed for all CPUs : 0.094024 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:35:56] 0.107060 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:36:11] Energy consumed for RAM : 0.013061 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:36:11] Energy consumed for all CPUs : 0.094201 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:36:11] 0.107261 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:36:11] 0.006108 g.CO2eq/s mean an estimation of 192.62768090801578 kg.CO2eq/year
[codecarbon INFO @ 16:36:26] Energy consumed for RAM : 0.013085 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:36:26] Energy consumed for all CPUs : 0.094378 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:36:26] 0.107463 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:36:41] Energy consumed for RAM : 0.013110 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:36:41] Energy consumed for all CPUs : 0.094555 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:36:41] 0.107665 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:36:56] Energy consumed for RAM : 0.013134 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:36:56] Energy consumed for all CPUs : 0.094732 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:36:56] 0.107866 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:37:11] Energy consumed for RAM : 0.013159 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:37:11] Energy consumed for all CPUs : 0.094910 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:37:11] 0.108069 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:37:26] Energy consumed for RAM : 0.013184 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:37:26] Energy consumed for all CPUs : 0.095087 kWh. Total CPU Power : 42.5

W
[codecarbon INFO @ 16:37:26] 0.108271 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:37:41] Energy consumed for RAM : 0.013208 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:37:41] Energy consumed for all CPUs : 0.095265 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:37:41] 0.108473 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:37:56] Energy consumed for RAM : 0.013233 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:37:56] Energy consumed for all CPUs : 0.095442 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:37:56] 0.108674 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:38:11] Energy consumed for RAM : 0.013257 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:38:11] Energy consumed for all CPUs : 0.095619 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:38:11] 0.108876 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:38:11] 0.006106 g.CO2eq/s mean an estimation of 192.55642819906447 kg.CO2e
q/year
[codecarbon INFO @ 16:38:26] Energy consumed for RAM : 0.013282 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:38:26] Energy consumed for all CPUs : 0.095796 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:38:26] 0.109078 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:38:41] Energy consumed for RAM : 0.013306 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:38:41] Energy consumed for all CPUs : 0.095973 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:38:41] 0.109279 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:38:56] Energy consumed for RAM : 0.013331 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:38:56] Energy consumed for all CPUs : 0.096150 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:38:56] 0.109481 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:39:11] Energy consumed for RAM : 0.013356 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:39:11] Energy consumed for all CPUs : 0.096327 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:39:11] 0.109683 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:39:26] Energy consumed for RAM : 0.013380 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:39:26] Energy consumed for all CPUs : 0.096504 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:39:26] 0.109884 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:39:41] Energy consumed for RAM : 0.013405 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:39:41] Energy consumed for all CPUs : 0.096681 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:39:41] 0.110086 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:39:56] Energy consumed for RAM : 0.013429 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:39:56] Energy consumed for all CPUs : 0.096859 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:39:56] 0.110288 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:40:11] Energy consumed for RAM : 0.013454 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:40:11] Energy consumed for all CPUs : 0.097036 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:40:11] 0.110489 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:40:11] 0.006107 g.CO2eq/s mean an estimation of 192.576185828638 kg.CO2eq/year

[codecarbon INFO @ 16:40:26] Energy consumed for RAM : 0.013478 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:40:26] Energy consumed for all CPUs : 0.097213 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:40:26] 0.110691 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:40:41] Energy consumed for RAM : 0.013503 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:40:41] Energy consumed for all CPUs : 0.097390 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:40:41] 0.110893 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:40:56] Energy consumed for RAM : 0.013527 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:40:56] Energy consumed for all CPUs : 0.097567 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:40:56] 0.111095 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:41:11] Energy consumed for RAM : 0.013552 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:41:11] Energy consumed for all CPUs : 0.097744 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:41:11] 0.111296 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:41:26] Energy consumed for RAM : 0.013577 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:41:26] Energy consumed for all CPUs : 0.097921 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:41:26] 0.111498 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:41:41] Energy consumed for RAM : 0.013601 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:41:41] Energy consumed for all CPUs : 0.098099 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:41:41] 0.111700 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:41:56] Energy consumed for RAM : 0.013626 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:41:56] Energy consumed for all CPUs : 0.098276 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:41:56] 0.111902 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:42:11] Energy consumed for RAM : 0.013650 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:42:11] Energy consumed for all CPUs : 0.098453 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:42:11] 0.112103 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:42:11] 0.006107 g.CO2eq/s mean an estimation of 192.57707711243611 kg.CO2eq/year

[codecarbon INFO @ 16:42:26] Energy consumed for RAM : 0.013675 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:42:26] Energy consumed for all CPUs : 0.098630 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:42:26] 0.112305 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:42:41] Energy consumed for RAM : 0.013699 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:42:41] Energy consumed for all CPUs : 0.098807 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:42:41] 0.112507 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:42:56] Energy consumed for RAM : 0.013724 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:42:56] Energy consumed for all CPUs : 0.098985 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:42:56] 0.112709 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:43:11] Energy consumed for RAM : 0.013748 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:43:11] Energy consumed for all CPUs : 0.099162 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:43:11] 0.112910 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:43:26] Energy consumed for RAM : 0.013773 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:43:26] Energy consumed for all CPUs : 0.099339 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:43:26] 0.113112 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:43:41] Energy consumed for RAM : 0.013798 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:43:41] Energy consumed for all CPUs : 0.099516 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:43:41] 0.113313 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:43:56] Energy consumed for RAM : 0.013822 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:43:56] Energy consumed for all CPUs : 0.099693 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:43:56] 0.113515 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:44:11] Energy consumed for RAM : 0.013847 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:44:11] Energy consumed for all CPUs : 0.099870 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:44:11] 0.113717 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:44:11] 0.006107 g.CO2eq/s mean an estimation of 192.57727884174102 kg.CO2eq/year

[codecarbon INFO @ 16:44:26] Energy consumed for RAM : 0.013871 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:44:26] Energy consumed for all CPUs : 0.100047 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:44:26] 0.113919 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:44:41] Energy consumed for RAM : 0.013896 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:44:41] Energy consumed for all CPUs : 0.100225 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:44:41] 0.114120 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:44:56] Energy consumed for RAM : 0.013920 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:44:56] Energy consumed for all CPUs : 0.100402 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:44:56] 0.114322 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:45:11] Energy consumed for RAM : 0.013945 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:45:11] Energy consumed for all CPUs : 0.100579 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:45:11] 0.114524 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:45:26] Energy consumed for RAM : 0.013969 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:45:26] Energy consumed for all CPUs : 0.100756 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:45:26] 0.114725 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:45:41] Energy consumed for RAM : 0.013994 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:45:41] Energy consumed for all CPUs : 0.100933 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:45:41] 0.114927 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:45:56] Energy consumed for RAM : 0.014019 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:45:56] Energy consumed for all CPUs : 0.101110 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:45:56] 0.115129 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:46:11] Energy consumed for RAM : 0.014043 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:46:11] Energy consumed for all CPUs : 0.101287 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:46:11] 0.115331 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:46:11] 0.006106 g.CO2eq/s mean an estimation of 192.57063265267573 kg.CO2eq/year

[codecarbon INFO @ 16:46:26] Energy consumed for RAM : 0.014068 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:46:26] Energy consumed for all CPUs : 0.101465 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:46:26] 0.115532 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:46:41] Energy consumed for RAM : 0.014092 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:46:41] Energy consumed for all CPUs : 0.101642 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:46:41] 0.115734 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:46:56] Energy consumed for RAM : 0.014117 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:46:56] Energy consumed for all CPUs : 0.101819 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:46:56] 0.115936 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:47:11] Energy consumed for RAM : 0.014141 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:47:11] Energy consumed for all CPUs : 0.101996 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:47:11] 0.116137 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:47:26] Energy consumed for RAM : 0.014166 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:47:26] Energy consumed for all CPUs : 0.102173 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:47:26] 0.116339 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:47:41] Energy consumed for RAM : 0.014191 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:47:41] Energy consumed for all CPUs : 0.102350 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:47:41] 0.116541 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:47:56] Energy consumed for RAM : 0.014215 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:47:56] Energy consumed for all CPUs : 0.102527 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:47:56] 0.116743 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:48:11] Energy consumed for RAM : 0.014240 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:48:11] Energy consumed for all CPUs : 0.102705 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:48:11] 0.116944 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:48:11] 0.006106 g.CO2eq/s mean an estimation of 192.5586994746152 kg.CO2eq/year

[codecarbon INFO @ 16:48:26] Energy consumed for RAM : 0.014264 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:48:26] Energy consumed for all CPUs : 0.102882 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:48:26] 0.117146 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:48:41] Energy consumed for RAM : 0.014289 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:48:41] Energy consumed for all CPUs : 0.103059 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:48:41] 0.117348 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:48:56] Energy consumed for RAM : 0.014313 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:48:57] Energy consumed for all CPUs : 0.103237 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:48:57] 0.117550 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:49:12] Energy consumed for RAM : 0.014338 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:49:12] Energy consumed for all CPUs : 0.103414 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:49:12] 0.117752 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:49:27] Energy consumed for RAM : 0.014363 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:49:27] Energy consumed for all CPUs : 0.103591 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:49:27] 0.117954 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:49:42] Energy consumed for RAM : 0.014387 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:49:42] Energy consumed for all CPUs : 0.103768 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:49:42] 0.118156 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:49:57] Energy consumed for RAM : 0.014412 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:49:57] Energy consumed for all CPUs : 0.103946 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:49:57] 0.118357 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:50:12] Energy consumed for RAM : 0.014436 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:50:12] Energy consumed for all CPUs : 0.104123 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:50:12] 0.118559 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:50:12] 0.006106 g.CO2eq/s mean an estimation of 192.56865026512418 kg.CO2eq/year

[codecarbon INFO @ 16:50:27] Energy consumed for RAM : 0.014461 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:50:27] Energy consumed for all CPUs : 0.104300 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:50:27] 0.118761 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:50:42] Energy consumed for RAM : 0.014485 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:50:42] Energy consumed for all CPUs : 0.104477 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:50:42] 0.118962 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:50:57] Energy consumed for RAM : 0.014510 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:50:57] Energy consumed for all CPUs : 0.104654 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:50:57] 0.119164 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:51:12] Energy consumed for RAM : 0.014534 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:51:12] Energy consumed for all CPUs : 0.104831 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:51:12] 0.119366 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:51:27] Energy consumed for RAM : 0.014559 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 16:51:27] Energy consumed for all CPUs : 0.105008 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:51:27] 0.119567 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:51:42] Energy consumed for RAM : 0.014584 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:51:42] Energy consumed for all CPUs : 0.105186 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:51:42] 0.119769 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:51:57] Energy consumed for RAM : 0.014608 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:51:57] Energy consumed for all CPUs : 0.105363 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:51:57] 0.119971 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:52:12] Energy consumed for RAM : 0.014633 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:52:12] Energy consumed for all CPUs : 0.105540 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:52:12] 0.120173 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:52:12] 0.006107 g.CO2eq/s mean an estimation of 192.57489183592256 kg.CO2e
q/year
[codecarbon INFO @ 16:52:27] Energy consumed for RAM : 0.014657 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:52:27] Energy consumed for all CPUs : 0.105717 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:52:27] 0.120374 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:52:42] Energy consumed for RAM : 0.014682 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:52:42] Energy consumed for all CPUs : 0.105894 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:52:42] 0.120576 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:52:57] Energy consumed for RAM : 0.014706 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:52:57] Energy consumed for all CPUs : 0.106071 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:52:57] 0.120778 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:53:12] Energy consumed for RAM : 0.014731 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:53:12] Energy consumed for all CPUs : 0.106249 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:53:12] 0.120980 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:53:27] Energy consumed for RAM : 0.014755 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:53:27] Energy consumed for all CPUs : 0.106426 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:53:27] 0.121181 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:53:42] Energy consumed for RAM : 0.014780 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:53:42] Energy consumed for all CPUs : 0.106603 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:53:42] 0.121383 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:53:57] Energy consumed for RAM : 0.014805 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:53:57] Energy consumed for all CPUs : 0.106780 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:53:57] 0.121584 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:54:12] Energy consumed for RAM : 0.014829 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 16:54:12] Energy consumed for all CPUs : 0.106957 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 16:54:12] 0.121786 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:54:12] 0.006107 g.CO2eq/s mean an estimation of 192.581075490124 kg.CO2eq/y

ear

[codecarbon INFO @ 16:54:27] Energy consumed for RAM : 0.014854 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:54:27] Energy consumed for all CPUs : 0.107136 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:54:27] 0.121990 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:54:42] Energy consumed for RAM : 0.014879 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:54:42] Energy consumed for all CPUs : 0.107314 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:54:42] 0.122192 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:54:57] Energy consumed for RAM : 0.014903 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:54:57] Energy consumed for all CPUs : 0.107491 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:54:57] 0.122394 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:55:12] Energy consumed for RAM : 0.014928 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:55:12] Energy consumed for all CPUs : 0.107668 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:55:12] 0.122596 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:55:27] Energy consumed for RAM : 0.014952 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:55:27] Energy consumed for all CPUs : 0.107845 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:55:27] 0.122797 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:55:42] Energy consumed for RAM : 0.014977 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:55:42] Energy consumed for all CPUs : 0.108022 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:55:42] 0.122999 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:55:57] Energy consumed for RAM : 0.015001 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:55:57] Energy consumed for all CPUs : 0.108199 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:55:57] 0.123201 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:56:12] Energy consumed for RAM : 0.015026 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:56:12] Energy consumed for all CPUs : 0.108377 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:56:12] 0.123403 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:56:12] 0.006107 g.CO2eq/s mean an estimation of 192.59761745004877 kg.CO2eq/year

[codecarbon INFO @ 16:56:27] Energy consumed for RAM : 0.015051 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:56:27] Energy consumed for all CPUs : 0.108554 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:56:27] 0.123605 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:56:42] Energy consumed for RAM : 0.015075 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:56:42] Energy consumed for all CPUs : 0.108731 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:56:42] 0.123806 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:56:57] Energy consumed for RAM : 0.015100 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 16:56:57] Energy consumed for all CPUs : 0.108908 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 16:56:57] 0.124008 kWh of electricity used since the beginning.

[codecarbon INFO @ 16:57:12] Energy consumed for RAM : 0.015124 kWh. RAM Power : 5.8941850662231445 W

45 W
[codecarbon INFO @ 16:57:12] Energy consumed for all CPUs : 0.109086 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:57:12] 0.124210 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:57:27] Energy consumed for RAM : 0.015149 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:57:27] Energy consumed for all CPUs : 0.109263 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:57:27] 0.124412 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:57:42] Energy consumed for RAM : 0.015173 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:57:42] Energy consumed for all CPUs : 0.109440 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:57:42] 0.124613 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:57:57] Energy consumed for RAM : 0.015198 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:57:57] Energy consumed for all CPUs : 0.109617 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:57:57] 0.124815 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:58:12] Energy consumed for RAM : 0.015223 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:58:12] Energy consumed for all CPUs : 0.109794 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:58:12] 0.125017 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:58:12] 0.006107 g.CO2eq/s mean an estimation of 192.59286374640186 kg.CO2eq/year
[codecarbon INFO @ 16:58:27] Energy consumed for RAM : 0.015247 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:58:27] Energy consumed for all CPUs : 0.109971 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:58:27] 0.125218 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:58:42] Energy consumed for RAM : 0.015272 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:58:42] Energy consumed for all CPUs : 0.110149 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:58:42] 0.125420 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:58:57] Energy consumed for RAM : 0.015296 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:58:57] Energy consumed for all CPUs : 0.110326 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:58:57] 0.125622 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:59:12] Energy consumed for RAM : 0.015321 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:59:12] Energy consumed for all CPUs : 0.110503 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:59:12] 0.125824 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:59:27] Energy consumed for RAM : 0.015345 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:59:27] Energy consumed for all CPUs : 0.110680 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:59:27] 0.126026 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:59:42] Energy consumed for RAM : 0.015370 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:59:42] Energy consumed for all CPUs : 0.110857 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 16:59:42] 0.126227 kWh of electricity used since the beginning.
[codecarbon INFO @ 16:59:57] Energy consumed for RAM : 0.015394 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 16:59:57] Energy consumed for all CPUs : 0.111034 kWh. Total CPU Power : 42.5

W
[codecarbon INFO @ 16:59:57] 0.126429 kWh of electricity used since the beginning.
[codecarbon INFO @ 17:00:12] Energy consumed for RAM : 0.015419 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 17:00:12] Energy consumed for all CPUs : 0.111212 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 17:00:12] 0.126631 kWh of electricity used since the beginning.
[codecarbon INFO @ 17:00:12] 0.006107 g.CO2eq/s mean an estimation of 192.59187436523376 kg.CO2e
q/year
[codecarbon INFO @ 17:00:27] Energy consumed for RAM : 0.015444 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 17:00:27] Energy consumed for all CPUs : 0.111389 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 17:00:27] 0.126832 kWh of electricity used since the beginning.
[codecarbon INFO @ 17:00:42] Energy consumed for RAM : 0.015468 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 17:00:42] Energy consumed for all CPUs : 0.111566 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 17:00:42] 0.127034 kWh of electricity used since the beginning.
[codecarbon INFO @ 17:00:57] Energy consumed for RAM : 0.015493 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 17:00:57] Energy consumed for all CPUs : 0.111743 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 17:00:57] 0.127236 kWh of electricity used since the beginning.
[codecarbon WARNING @ 18:31:43] Background scheduler didn't run for a long period (5445s), result
s might be inaccurate
[codecarbon INFO @ 18:31:43] Energy consumed for RAM : 0.024409 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:31:43] Energy consumed for all CPUs : 0.176034 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:31:43] 0.200443 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:31:58] Energy consumed for RAM : 0.024434 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:31:58] Energy consumed for all CPUs : 0.176211 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:31:58] 0.200644 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:32:13] Energy consumed for RAM : 0.024458 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:32:13] Energy consumed for all CPUs : 0.176388 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:32:13] 0.200846 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:32:28] Energy consumed for RAM : 0.024483 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:32:28] Energy consumed for all CPUs : 0.176565 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:32:28] 0.201048 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:32:43] Energy consumed for RAM : 0.024507 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:32:43] Energy consumed for all CPUs : 0.176742 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:32:43] 0.201250 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:32:43] 0.006108 g.CO2eq/s mean an estimation of 192.6291221278164 kg.CO2eq/
year
[codecarbon INFO @ 18:32:58] Energy consumed for RAM : 0.024532 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 18:32:58] Energy consumed for all CPUs : 0.176920 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 18:32:58] 0.201451 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:33:13] Energy consumed for RAM : 0.024556 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 18:33:13] Energy consumed for all CPUs : 0.177097 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 18:33:13] 0.201653 kWh of electricity used since the beginning.
[codecarbon INFO @ 18:33:28] Energy consumed for RAM : 0.024581 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 18:33:28] Energy consumed for all CPUs : 0.177274 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 18:33:28] 0.201855 kWh of electricity used since the beginning.
[codecarbon WARNING @ 19:47:03] Background scheduler didn't run for a long period (4414s), results might be inaccurate
[codecarbon INFO @ 19:47:03] Energy consumed for RAM : 0.031809 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:47:03] Energy consumed for all CPUs : 0.229394 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:47:03] 0.261203 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:47:18] Energy consumed for RAM : 0.031834 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:47:18] Energy consumed for all CPUs : 0.229571 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:47:18] 0.261405 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:47:33] Energy consumed for RAM : 0.031858 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:47:33] Energy consumed for all CPUs : 0.229748 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:47:33] 0.261607 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:47:48] Energy consumed for RAM : 0.031883 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:47:48] Energy consumed for all CPUs : 0.229925 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:47:48] 0.261808 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:48:03] Energy consumed for RAM : 0.031907 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:48:03] Energy consumed for all CPUs : 0.230103 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:48:03] 0.262010 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:48:03] 0.006108 g.CO2eq/s mean an estimation of 192.62873007367386 kg.CO2eq/year
[codecarbon INFO @ 19:48:18] Energy consumed for RAM : 0.031932 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:48:18] Energy consumed for all CPUs : 0.230280 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:48:18] 0.262212 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:48:33] Energy consumed for RAM : 0.031957 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:48:33] Energy consumed for all CPUs : 0.230457 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:48:33] 0.262413 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:48:48] Energy consumed for RAM : 0.031981 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:48:48] Energy consumed for all CPUs : 0.230634 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:48:48] 0.262615 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:49:03] Energy consumed for RAM : 0.032006 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 19:49:03] Energy consumed for all CPUs : 0.230811 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 19:49:03] 0.262817 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:49:18] Energy consumed for RAM : 0.032030 kWh. RAM Power : 5.89418506622314

```

45 W
[codecarbon INFO @ 19:49:18] Energy consumed for all CPUs : 0.230988 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:49:18] 0.263018 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:49:33] Energy consumed for RAM : 0.032055 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:49:33] Energy consumed for all CPUs : 0.231165 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:49:33] 0.263220 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:49:48] Energy consumed for RAM : 0.032079 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:49:48] Energy consumed for all CPUs : 0.231342 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:49:48] 0.263422 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:50:03] Energy consumed for RAM : 0.032104 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:50:03] Energy consumed for all CPUs : 0.231520 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:50:03] 0.263623 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:50:03] 0.006106 g.CO2eq/s mean an estimation of 192.574256154231 kg.CO2eq/y
ear
[codecarbon INFO @ 19:50:18] Energy consumed for RAM : 0.032128 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:50:18] Energy consumed for all CPUs : 0.231697 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:50:18] 0.263825 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:50:33] Energy consumed for RAM : 0.032153 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:50:33] Energy consumed for all CPUs : 0.231874 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:50:33] 0.264027 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:50:48] Energy consumed for RAM : 0.032178 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:50:48] Energy consumed for all CPUs : 0.232051 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:50:48] 0.264229 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:51:03] Energy consumed for RAM : 0.032202 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:51:03] Energy consumed for all CPUs : 0.232228 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:51:03] 0.264430 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:51:18] Energy consumed for RAM : 0.032227 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:51:18] Energy consumed for all CPUs : 0.232405 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:51:18] 0.264632 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:51:33] Energy consumed for RAM : 0.032251 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:51:33] Energy consumed for all CPUs : 0.232582 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:51:33] 0.264834 kWh of electricity used since the beginning.

```

2: Image Classification based on Convolutional Neural Networks [20pts: 20pts Grad / 2.7% Bonus for Undergrad + 1.1% Bonus for all] **[P][W]**

2.1 Image Classification using Pytorch and CNN [20pts Grad / 2.7% Bonus for Undergrad] [P][W]

Pytorch Description

[Pytorch](#) is a Machine Learning/Deep Learning tensor library based on Python and Torch that uses dynamic computation graphs. Pytorch is used for applications using GPUs and CPUs.

Helpful Links

- [Install Pytorch](#)
- [Pytorch Quickstart Tutorial](#)

Setup Pytorch

Make sure you installed pytorch and torchvision (directions [here](#)).

Please also see [Pytorch Quickstart Tutorial](#) to see how to load a data set, build a training loop, and test the model. Another good resource for building CNNs using Pytorch is [here](#).

Environment Setup

```
In [ ]: import torch
import torchvision
from torch.utils.data import non_deterministic
from torchvision.transforms import v2

%load_ext autoreload
%autoreload 2
%reload_ext autoreload
```

2.1.1 Load FashionMNIST Dataset and Data Augmentation [5pts - Bonus for Undergrad]

We use [Fashion-MNIST](#) dataset to train our model. This is a dataset of 70,000 28x28 grayscale images in 10 classes. There are 60,000 training images and 10,000 test images. We provide code for you to download Fashion-MNIST dataset below.

Data Augmentation [5pts]

Data augmentation is a technique to increase the diversity of your training set by applying random (but realistic) transformations such as image rotation and flipping the image around an axis. If the dataset in a machine learning model is rich and sufficient, the model performs better and more accurately. We will preprocess the training and testing set, but only the training set will undergo augmentation.

Go through the [Pytorch torchvision.transforms.v2 documentation](#) to see how to apply multiple transformations at once.

In the **cnn_image_transformations.py** file, complete the following functions to understand the common practices used for preprocessing and augmenting the image data:

- **create_training_transformations**

- In this function, you are going to preprocess and augment training data.
 - PREPROCESS: Convert the given PIL Images to Tensors
 - AUGMENTATION: Apply Random Horizontal Flip and Random Rotation

- **create_testing_transformations**

- In this function, you are going to only preprocess testing data.
 - PREPROCESS: Convert the given PIL Images to Tensors

Please note that the Gradescope only checks if expected preprocessing layers are existent.

References

[v2.Compose\(\)](#)

[v2.ToTensor\(\)](#) (Hint: Look at the warning)

[v2.RandomHorizontalFlip\(\)](#)

[v2.RandomApply\(\)](#)

[v2.RandomRotation\(\)](#)

[Article about performance regarding transformations](#)

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

from cnn_image_transformations import (
    create_testing_transformations,
    create_training_transformations,
)

# Create Transformations
training_transformations = create_training_transformations()
testing_transformation = create_testing_transformations()

# Load data
trainset = torchvision.datasets.FashionMNIST(
    root="./data", train=True, download=True, transform=training_transformations
)
testset = torchvision.datasets.FashionMNIST(
    root="./data", train=False, download=True, transform=testing_transformation
)

classes = (
    "Top",
    "Trouser",
    "Pullover",
    "Dress",
```

```

        "Coat",
        "Sandal",
        "Shirt",
        "Sneaker",
        "Bag",
        "Ankle boot",
    )

print(trainset.data.shape)
print(testset.data.shape)

```

Load some sample images from Fashion-MNIST [Setup - No points]

```

In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

import matplotlib.pyplot as plt
import numpy as np

trainloader = torch.utils.data.DataLoader(
    trainset, batch_size=32, shuffle=True, num_workers=2
)
testloader = torch.utils.data.DataLoader(
    testset, batch_size=32, shuffle=False, num_workers=2
)

# functions to show an image

def imshow(img):
    img = img / 2 + 0.5 # unnormalize
    npimg = img.numpy()
    plt.imshow(np.transpose(npimg, (1, 2, 0)))
    plt.show()

# get some random training images
dataiter = iter(trainloader)
images, labels = next(dataiter)

print("Image size")
print(v2.functional.get_size(images[0]))

# show images
imshow(torchvision.utils.make_grid(images))

```

As you can see from above, the FashionMNIST dataset contains different types of objects. The images have been size-normalized and objects remain centered in fixed-size images.

2.1.2 Build convolutional neural network model [5pts Grad / 0.7% Bonus for Undergrad] **[W]**

In this part, you need to build a convolutional neural network as described below. The architecture of the model is outlined.

In the **cnn.py** file, complete the following functions:

- **__init__**: See Defining Variables section
- **forward**: See Defining Model section

[INPUT - CONV - CONV - MAXPOOL - DROPOUT - CONV - CONV - MAXPOOL - DROPOUT - AVERAGEPOOL - FC1 - DROPOUT - FC2 - DROPOUT - FC3]

INPUT: $[28 \times 28 \times 1]$ will hold the raw pixel values of the image, in this case, an image of width 28, height 28. This layer should give 8 filters and have appropriate padding to maintain shape.

CONV: Conv. layer will compute the output of neurons that are connected to local regions in the input, each computing a dot product between their weights and a small region they are connected to the input volume. In our example architecture, we decide to set the `kernel_size` to be 3×3 . For example, the output of the Conv. layer may look like $[28 \times 28 \times 8]$ if we set `out_channels` to be 8 and use appropriate paddings to maintain shape.

CONV: Additional Conv. layer take outputs from above layers and applies more filters. We set the `kernel_size` to be 3×3 and `out_channels` to be 32.

MAXPOOL: MAXPOOL layer will perform a downsampling operation along the spatial dimensions (width, height). With pool size of 2×2 , resulting shape takes form 16×16 .

DROPOUT: DROPOUT layer with the dropout rate of 0.2 to prevent overfitting.

CONV: Additonal Conv. layer takes outputs from above layers and applies more filters. We set the `kernel_size` to be 3×3 and `out_channels` to be 32. Appropriate paddings are used to maintain shape.

CONV: Additonal Conv. layer takes outputs from above layers and applies more filters. We set the `kernel_size` to be 3×3 and `out_channels` to be 64. Appropriate paddings are used to maintain shape.

MAXPOOL: MAXPOOL layer will perform a downsampling operation along the spatial dimensions (width, height).

DROPOUT: Dropout layer with the dropout rate of 0.2 to prevent overfitting.

AVERAGEPOOL: AVERAGEPOOL layer will perform a downsampling operation along the spatial dimension (width, height). Checkout `AdaptiveAvgPool2d` below.

FC1: Dense layer which takes output from above layers, and has 256 neurons. `Flatten()` operations may be useful.

DROPOUT: Dropout layer with the dropout rate of 0.2 to prevent overfitting.

FC2: Dense layer which takes output from above layers, and has 128 neurons.

DROPOUT: Dropout layer with the dropout rate of 0.2 to prevent overfitting.

FC3: Dense layer with 10 neurons, and Softmax activation, is the final layer. The dimension of the output space is the number of classes.

Activation function: Use LeakyReLU with negative_slope 0.01 as the activation function for Conv. layers and Dense layers unless otherwise indicated to build your model architecture

Note that while this is a suggested model design, you may use other architectures and experiment with different layers for better results.

The following links are Pytorch documentation for the layers you are going to use to build the CNN.

- [Conv2d](#)
- [Dense](#)
- [MaxPool](#)
- [AdaptiveAvgPool2d](#)
- [Dropout](#)
- [LeakyReLU](#)
- [Flatten](#)

Lastly, if you would like to experiment with additional layers, explore the [torch.nn api](#).

```
In [ ]: #####  
### DO NOT CHANGE THIS CELL ###  
#####  
  
# Show the architecture of the model  
achi = plt.imread("./data/images/Architecture.png")  
fig = plt.figure(figsize=(10, 10))  
plt.imshow(achi)
```

Defining model [5pts Grad / 0.7% Bonus for Undergrad] **[W]**

You now need to complete the `__init__()` function and the `forward()` function in **cnn.py** to define your model structure.

Your model is required to have at least 2 convolutional layers and at least 2 dense layers. Ensuring that these requirements are met will earn you 5pts.

Once you have defined a model structure you may use the cell below to examine your architecture.

```
In [ ]: #####  
### DO NOT CHANGE THIS CELL ###  
#####  
  
# You can compare your architecture with the 'Architecture.png'  
  
from cnn import CNN  
  
net = CNN()  
print(net)
```

2.1.3 Train the network [8pts Grad / 1% Bonus for Undergrad] **[W]**

Tuning: Training the network is the next thing to try. You can set the hyperparameters in the cell below. If your hyperparameters are set properly, you should see the loss of the validation set decreased and the value of accuracy increased. **It may take more than 15 minutes to train your model.**

- Recommended Batch Sizes fall in the range 32-512 (use powers of 2)
- Recommended Epoch Counts fall in the range 5-20
- Recommended Learning Rates fall in the range .0001-.01

Expected Result: You should be able to achieve more than 90% accuracy on the test set to get full points. If you achieve accuracy between 75% to 84%, you will only get 3 points. An accuracy between 84% to 90% will earn an additional 3pts.

Note: If you would like to automate the tuning process, you can use a nested for loop to search for the hyperparameter that achieves the accuracy. You could also look into [grid search](#) for hyperparameter optimization.

- 75% to 84% earns 3pts
- 84% to 90% earns 3pts more (6pts total)
- 90%+ earns 2pts more (8pts total)

Train your own CNN model

```
In [ ]: from cnn import CNN
        from cnn_trainer import Trainer

        net = CNN()

        # TODO: Change hyperparameters here
        num_epochs = 10
        batch_size = 32
        init_lr = 5e-3

        # Choose best device to speed up training
        if torch.cuda.is_available():
            device = "cuda"
        elif torch.backends.mps.is_available():
            device = "mps"
        else:
            device = "cpu"
        print(f"Using {device} device")

        trainer = Trainer(
            net,
            trainset,
            testset,
            num_epochs=num_epochs,
            batch_size=batch_size,
            init_lr=init_lr,
            device=device,
        )
        trainer.train()
```

2.1.4 Examine accuracy and loss [2pts Grad / 0.3% Bonus for Undergrad] [W]

You should expect to see gradually decreasing loss and gradually increasing accuracy. Examine loss and accuracy by running the cell below, no editing is necessary. Having appropriate looking loss and accuracy plots will earn you the last 2pts for your convolutional neural net.

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####

# list all data in history
train_loss, train_accuracy, valid_loss, valid_accuracy = trainer.get_training_history()

# summarize history for accuracy and loss
plt.plot(train_accuracy)
plt.plot(valid_accuracy)
plt.title("model accuracy")
plt.ylabel("accuracy")
plt.xlabel("epoch")
plt.legend(["train", "valid"], loc="upper left")
plt.show()

plt.plot(train_loss)
plt.plot(valid_loss)
plt.title("model loss")
plt.ylabel("loss")
plt.xlabel("epoch")
plt.legend(["train", "valid"], loc="upper left")
plt.show()
```

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####

# make predictions
y_pred, y_pred_classes, y_gt_classes = trainer.predict(testloader)
y_pred_prob = torch.max(y_pred, dim=1).values

from sklearn.metrics import accuracy_score, confusion_matrix

plt.figure(figsize=(8, 7))
plt.imshow(confusion_matrix(y_gt_classes, y_pred_classes))
plt.title("Confusion matrix", fontsize=16)
plt.xticks(np.arange(10), classes, rotation=90, fontsize=12)
plt.yticks(np.arange(10), classes, fontsize=12)
plt.colorbar()
plt.show()
```

2.2 Exploring Deep CNN Architectures [1.1% Bonus for All] [W]

The network you have produced is rather simple relative to many of those used in industry and research. Researchers have worked to make CNN models deeper and deeper over the past years in an effort to gain

higher accuracy in predictions. While your model is only a handful of layers deep, some state of the art deep architectures may include up to 150 layers. However, this process has not been without challenges.

One such problem is the problem of the exploding gradient. The initial weights assigned to the neural nets creating large losses. Big gradient values can accumulate to the point where large parameter updates are observed, causing gradient descents to oscillate without coming to global minima. What's even worse is that these parameters can be so large that they overflow and return NaN values that cannot be updated anymore.

Many tactics have been used in an effort to solve this problem. One architecture, named Gradient Clipping, solves the vanishing gradient problem in a unique way. Researchers from Massachusetts Institute of Technology discussed about the mechanism and a theoretical explanation for the effectiveness of gradient clipping in training deep neural networks. Take a moment to explore how Gradient Clipping tackles the vanishing gradient problem by reading the original research paper here: <https://arxiv.org/pdf/1905.11881> (also included as PDF in papers directory).

Question: In your own words, explain how Gradient Clipping addresses the exploding gradient problem in 1-2 sentences below: (Please type answers directly in the cell below.)

We observe the gradient clipping prevents the gradients from becoming unnecessarily large by setting a maximum for their norm during training. These max thresholds avoids exploding gradients and allows the parameter updates to remain stable.

3: Random Forests [40pts + 2.1% Bonus for All] [P] [W]

NOTE: Please use sklearn's ExtraTreeClassifier in your Random Forest implementation. [You can find more details about this classifier here.](#)

For context, the general difference between an extra tree and decision tree classifier is that the decision tree optimizes which feature to reduce entropy on and at what value to split, while an extra tree randomly splits on the features given.

3.1 Random Forest Implementation [35pts] [P]

The decision boundaries drawn by decision or extra trees are very sharp, and fitting a tree of unbounded depth to a list of examples almost inevitably leads to **overfitting**. In an attempt to decrease the variance of an extra tree, we're going to use a technique called 'Bootstrap Aggregating' (often abbreviated 'bagging'). This stems from the idea that a collection of weak learners can learn decision boundaries as well as a strong learner. This is commonly called a Random Forest.

We can build a Random Forest as a collection of extra trees, as follows:

1. For every tree in the random forest, we're going to

a) Subsample the examples with replacement. Note that in this question, the size of the subsample data is equal to the original dataset.

b) From the subsamples in part a, choose attributes at random without replacement to learn on in accordance with a provided attribute subsampling rate. Based on what it was mentioned in the class, we randomly pick features in each split. We use a more general approach here to make the programming part easier. Let's randomly pick some features (65% percent of features) and grow the tree based on the pre-determined randomly selected features. Therefore, there is no need to find random features in each split.

c) Fit an extra tree to the subsample of data we've chosen to a certain depth.

You can refresh your understanding with the lecture notes on random forests.

Classification for a random forest is then done by taking a majority vote of the classifications yielded by each tree in the forest after it classifies an example.

In the **random_forest.py** file, complete the following functions:

- **_bootstrapping**: this function will be used in `bootstrapping()`
- **fit**: Fit the extra trees initialized in `__init__` with the datasets created in `bootstrapping()`. You will need to call `bootstrapping()`.

NOTES:

1. In the Random Forest Class, X is assumed to be a matrix with num_training rows and num_features columns where num_training is the number of total records and num_features is the number of features of each record. y is assumed to be a vector of labels of length num_training.
2. Look out for TODO's for the parts that need to be implemented
3. If you receive any `SettingWithCopyWarning` warnings from the Pandas library, you can safely ignore them.
4. Hint: when bootstrapping, set replace = False while creating col_idx

3.2 Hyperparameter Tuning with a Random Forest [5pts] [P]

In machine learning, hyperparameters are parameters that are set before the learning process begins. The max_depth, num_estimators, or max_features variables from 3.1 are examples of different hyperparameters for a random forest model. Let's first review the dataset in a bit more detail.

Dataset Objective

Imagine that we are a team of researchers working to track and document various information related to dry beans for a machine learning model that predicts what type of bean is represented. We know that there are multiple things to keep track of, such as the shapes and sizes that differentiate different types of beans. We will use the information we track and document in order to publish it for the general public.

After much reflection within the research team, we come to the conclusion that we can use past observations on bean images to create a model.

We will use our random forest algorithm from Q3.1 to predict the bean type.

You can find more information on the dataset [here](#).

The barbunya bean, also known as the cranberry bean, was first bred in Colombia.



Loading the dataset

The dataset that the company has collected has the following features:

There were 16 features used in this dataset.

Inputs:

1. Area: The area of a bean zone and the number of pixels within its boundaries
2. Perimeter: Bean circumference is defined as the length of its border
3. MajorAxisLength: The distance between the ends of the longest line that can be drawn from a bean
4. MinorAxisLength: The longest line that can be drawn from the bean while standing perpendicular to the main axis
5. AspectRatio: Defines the relationship between MajorAxisLength and MinorAxisLength
6. Eccentricity: Eccentricity of the ellipse having the same moments as the region
7. ConvexArea: Number of pixels in the smallest convex polygon that can contain the area of a bean seed
8. EquivDiameter Equivalent diameter, the diameter of a circle having the same area as a bean seed area
9. Extent Feature: The ratio of the pixels in the bounding box to the bean area
10. Solidity: Also known as convexity. The ratio of the pixels in the convex shell to those found in beans.
11. Roundness: Calculated with the following formula: $(4\pi A)/(P^2)$
12. Compactness: Measures the roundness of an object
13. ShapeFactor1
14. ShapeFactor2
15. ShapeFactor3
16. ShapeFactor4

Output:

17. Target value:
 - Seker
 - Barbunya
 - Bombay
 - Cali
 - Dermosan
 - Horoz
 - Sira

Your random forest model will try to predict this variable.

```
In [32]: import numpy as np
import pandas as pd
```

```
#####
### DO NOT CHANGE THIS CELL ###
#####
from sklearn import preprocessing

dry_bean_dataset = "./data/Dry_Bean_Dataset.csv"
df = pd.read_csv(dry_bean_dataset)

label_encoder = preprocessing.LabelEncoder()

X = df.drop(["Class"], axis=1)
y = label_encoder.fit_transform(df["Class"])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42
)
X_test = np.array(X_test)
X_train, y_train, X_test, y_test = (
    np.array(X_train),
    np.array(y_train),
    np.array(X_test),
    np.array(y_test),
)
```

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####
print(X_train.shape, y_train.shape, X_test.shape, y_test.shape)
assert X_train.shape == (9119, 16)
assert y_train.shape == (9119,)
assert X_test.shape == (4492, 16)
assert y_test.shape == (4492,)
```

```
(9119, 16) (9119,) (4492, 16) (4492,)
```

```
[codecarbon INFO @ 19:51:48] Energy consumed for RAM : 0.032276 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:51:48] Energy consumed for all CPUs : 0.232760 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:51:48] 0.265036 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:52:03] Energy consumed for RAM : 0.032300 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:52:03] Energy consumed for all CPUs : 0.232937 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:52:03] 0.265237 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:52:03] 0.006107 g.CO2eq/s mean an estimation of 192.58835291555627 kg.CO2e
q/year
[codecarbon INFO @ 19:52:18] Energy consumed for RAM : 0.032325 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:52:18] Energy consumed for all CPUs : 0.233114 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:52:18] 0.265439 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:52:33] Energy consumed for RAM : 0.032350 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:52:33] Energy consumed for all CPUs : 0.233291 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:52:33] 0.265641 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:52:48] Energy consumed for RAM : 0.032374 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:52:48] Energy consumed for all CPUs : 0.233468 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:52:48] 0.265842 kWh of electricity used since the beginning.
```

In the following codeblock, train your random forest model with different values for `max_depth`, `n_estimators`, or `max_features` and evaluate each model on the held-out test set. Try to choose a combination of hyperparameters that maximizes your prediction accuracy on the test set (aim for 85%+).

In **random_forest.py**, once you are satisfied with your chosen parameters, update the following function:

- **select_hyperparameters**: change the values for `max_depth`, `n_estimators`, and `max_features` to your chosen values

Submit this file to Gradescope. You must achieve at least a **85% accuracy** against the test set in Gradescope to receive full credit for this section.

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####
from utilities.localtests import TestRandomForest

"""
Once you have implemented Random forest, you can run this cell. If you implemented _bootStrapping
then this cell should execute without any errors.
"""
TestRandomForest("test_bootstrapping").test_bootstrapping()
```

test_bootstrapping passed!

```
[codecarbon INFO @ 19:53:03] Energy consumed for RAM : 0.032399 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:53:03] Energy consumed for all CPUs : 0.233645 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:53:03] 0.266044 kWh of electricity used since the beginning.
```

```
In [ ]: """
TODO:
n_estimators defines how many Extra trees are fitted for the random forest.
max_depth defines a stop condition when the tree reaches to a certain depth.
max_features controls the percentage of features that are used to fit each extra tree.

Tune these three parameters to achieve a better accuracy. n_estimators and max_depth must both
be at least 3 in value for moderately reliable answers. While you can use the provided test set
to evaluate your implementation, you will need to obtain 85% on the test set to receive full
credit for this section.
"""

import sklearn.ensemble
from random_forest import RandomForest
from sklearn import preprocessing

##### DO NOT CHANGE THIS RANDOM SEED #####
student_random_seed = 4641 + 7641
#####

##### CHANGE THESE VALUES #####

n_estimators = 15 # Hint: Consider values between 3-15.
max_depth = 15 # Hint: Consider values between 3-15.
max_features = 1.0 # Hint: Consider values between 0.3-1.0.
```

```
[codecarbon INFO @ 19:59:03] Energy consumed for RAM : 0.032988 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 19:59:03] Energy consumed for all CPUs : 0.237897 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:59:03] 0.270885 kWh of electricity used since the beginning.
[codecarbon INFO @ 19:59:03] Energy consumed for all CPUs : 0.237897 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 19:59:03] 0.270885 kWh of electricity used since the beginning.
```

```
In [ ]: #####
### DO NOT CHANGE THIS CELL ###
#####
random_forest = RandomForest(
    n_estimators, max_depth, max_features, random_seed=student_random_seed
)
random_forest.fit(X_train, y_train)
accuracy = random_forest.00B_score(X_test, y_test)
print("accuracy: %.4f" % accuracy)
```

```
accuracy: 0.8763
```

[codecarbon INFO @ 19:59:18] Energy consumed for RAM : 0.033013 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 19:59:18] Energy consumed for all CPUs : 0.238074 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 19:59:18] 0.271087 kWh of electricity used since the beginning.

[codecarbon INFO @ 19:59:33] Energy consumed for RAM : 0.033037 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 19:59:33] Energy consumed for all CPUs : 0.238251 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 19:59:33] 0.271288 kWh of electricity used since the beginning.

[codecarbon INFO @ 19:59:48] Energy consumed for RAM : 0.033062 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 19:59:48] Energy consumed for all CPUs : 0.238428 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 19:59:48] 0.271490 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:00:03] Energy consumed for RAM : 0.033086 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:00:03] Energy consumed for all CPUs : 0.238605 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:00:03] 0.271692 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:00:03] 0.006106 g.CO2eq/s mean an estimation of 192.57403051996448 kg.CO2eq/year

[codecarbon INFO @ 20:00:18] Energy consumed for RAM : 0.033111 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:00:18] Energy consumed for all CPUs : 0.238783 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:00:18] 0.271894 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:00:33] Energy consumed for RAM : 0.033135 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:00:33] Energy consumed for all CPUs : 0.238960 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:00:33] 0.272095 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:00:48] Energy consumed for RAM : 0.033160 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:00:48] Energy consumed for all CPUs : 0.239137 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:00:48] 0.272297 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:01:03] Energy consumed for RAM : 0.033185 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:01:03] Energy consumed for all CPUs : 0.239314 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:01:03] 0.272499 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:01:18] Energy consumed for RAM : 0.033209 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:01:18] Energy consumed for all CPUs : 0.239491 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:01:18] 0.272701 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:01:33] Energy consumed for RAM : 0.033234 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:01:33] Energy consumed for all CPUs : 0.239669 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:01:33] 0.272902 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:01:48] Energy consumed for RAM : 0.033258 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:01:48] Energy consumed for all CPUs : 0.239846 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:01:48] 0.273104 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:02:03] Energy consumed for RAM : 0.033283 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 20:02:03] Energy consumed for all CPUs : 0.240023 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:02:03] 0.273306 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:02:03] 0.006107 g.CO2eq/s mean an estimation of 192.5853669686615 kg.CO2eq/year
[codecarbon INFO @ 20:02:18] Energy consumed for RAM : 0.033307 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:02:18] Energy consumed for all CPUs : 0.240200 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:02:18] 0.273507 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:02:33] Energy consumed for RAM : 0.033332 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:02:33] Energy consumed for all CPUs : 0.240377 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:02:33] 0.273709 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:02:48] Energy consumed for RAM : 0.033356 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:02:48] Energy consumed for all CPUs : 0.240554 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:02:48] 0.273911 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:03:03] Energy consumed for RAM : 0.033381 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:03:03] Energy consumed for all CPUs : 0.240732 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:03:03] 0.274113 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:03:19] Energy consumed for RAM : 0.033406 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:03:19] Energy consumed for all CPUs : 0.240909 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:03:19] 0.274314 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:03:34] Energy consumed for RAM : 0.033430 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:03:34] Energy consumed for all CPUs : 0.241086 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:03:34] 0.274516 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:03:49] Energy consumed for RAM : 0.033455 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:03:49] Energy consumed for all CPUs : 0.241263 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:03:49] 0.274718 kWh of electricity used since the beginning.
```

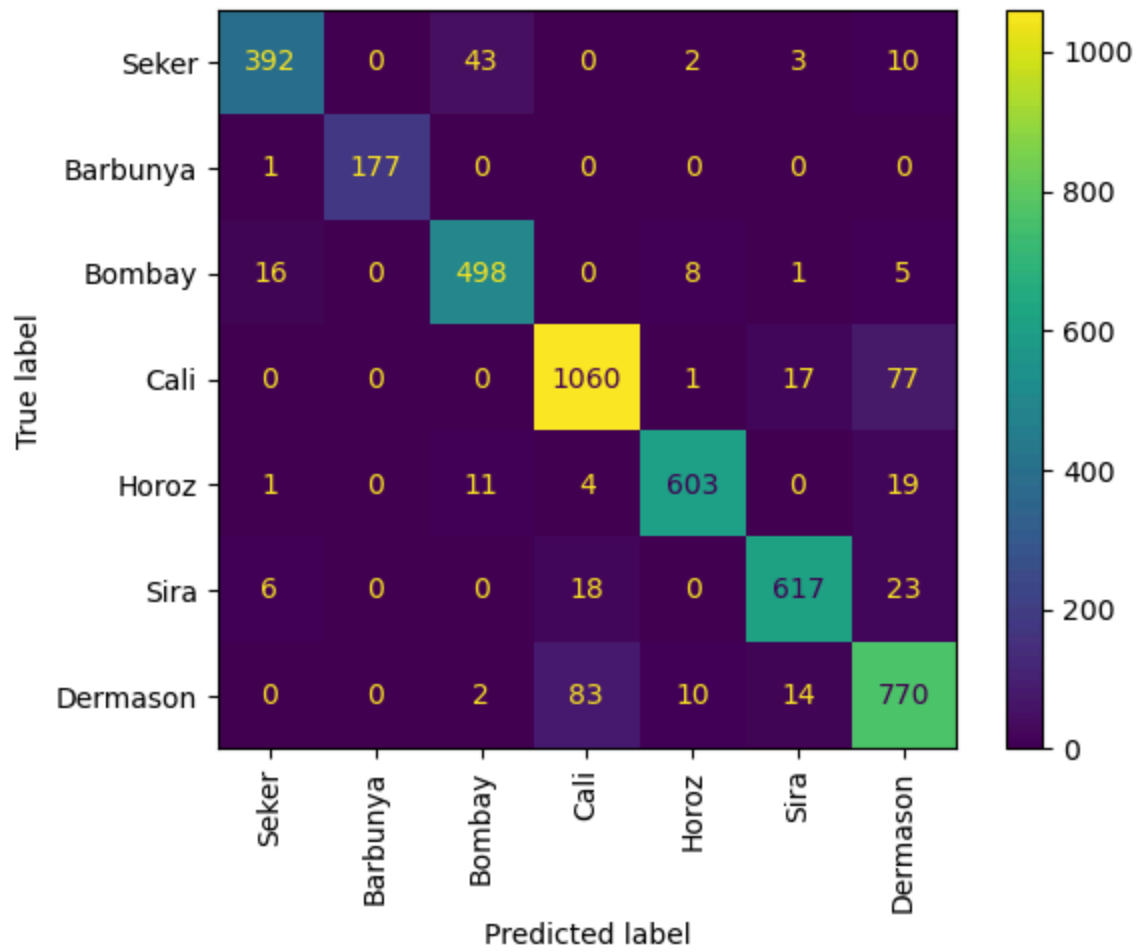
DON'T FORGET: Once you are satisfied with your chosen parameters, change the values for `max_depth`, `n_estimators`, and `max_features` in the `select_hyperparameters()` function of your RandomForest class in `random_forest.py` to your chosen values, and then submit this file to Gradescope. You must achieve at least a **85% accuracy** against the test set in Gradescope to receive full credit for this section.

Below is a code block that plots a confusion matrix for the classifier's predictions on the test set. A few things to think about: What are some trends seen in the matrix? Why do they happen?

```
In [ ]: from sklearn.metrics import ConfusionMatrixDisplay

pred = random_forest.predict(X_test)
labels = ["Seker", "Barbunya", "Bombay", "Cali", "Horoz", "Sira", "Dermason"]
ConfusionMatrixDisplay.from_predictions(
    y_test, pred, display_labels=labels, xticks_rotation="vertical")
```

```
)  
plt.show()
```



[codecarbon INFO @ 20:04:04] Energy consumed for RAM : 0.033479 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:04:04] Energy consumed for all CPUs : 0.241441 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:04:04] 0.274920 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:04:04] 0.006107 g.CO2eq/s mean an estimation of 192.5792343942583 kg.CO2eq/year

[codecarbon INFO @ 20:04:19] Energy consumed for RAM : 0.033504 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:04:19] Energy consumed for all CPUs : 0.241618 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:04:19] 0.275122 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:04:34] Energy consumed for RAM : 0.033528 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:04:34] Energy consumed for all CPUs : 0.241795 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:04:34] 0.275324 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:04:49] Energy consumed for RAM : 0.033553 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:04:49] Energy consumed for all CPUs : 0.241972 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:04:49] 0.275525 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:05:04] Energy consumed for RAM : 0.033578 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:05:04] Energy consumed for all CPUs : 0.242149 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:05:04] 0.275727 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:05:19] Energy consumed for RAM : 0.033602 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:05:19] Energy consumed for all CPUs : 0.242327 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:05:19] 0.275929 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:05:34] Energy consumed for RAM : 0.033627 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:05:34] Energy consumed for all CPUs : 0.242504 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:05:34] 0.276130 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:05:49] Energy consumed for RAM : 0.033651 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:05:49] Energy consumed for all CPUs : 0.242681 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:05:49] 0.276332 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:06:04] Energy consumed for RAM : 0.033676 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:06:04] Energy consumed for all CPUs : 0.242858 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:06:04] 0.276534 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:06:04] 0.006106 g.CO2eq/s mean an estimation of 192.5666385500809 kg.CO2eq/year

[codecarbon INFO @ 20:06:19] Energy consumed for RAM : 0.033700 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:06:19] Energy consumed for all CPUs : 0.243035 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:06:19] 0.276735 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:06:34] Energy consumed for RAM : 0.033725 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:06:34] Energy consumed for all CPUs : 0.243212 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:06:34] 0.276937 kWh of electricity used since the beginning.

```

[codecarbon INFO @ 20:06:49] Energy consumed for RAM : 0.033749 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:06:49] Energy consumed for all CPUs : 0.243389 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:06:49] 0.277139 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:07:04] Energy consumed for RAM : 0.033774 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:07:04] Energy consumed for all CPUs : 0.243566 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:07:04] 0.277340 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:07:19] Energy consumed for RAM : 0.033799 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:07:19] Energy consumed for all CPUs : 0.243743 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:07:19] 0.277542 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:07:34] Energy consumed for RAM : 0.033823 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:07:34] Energy consumed for all CPUs : 0.243921 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:07:34] 0.277744 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:07:49] Energy consumed for RAM : 0.033848 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:07:49] Energy consumed for all CPUs : 0.244098 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:07:49] 0.277945 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:08:04] Energy consumed for RAM : 0.033872 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 20:08:04] Energy consumed for all CPUs : 0.244275 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 20:08:04] 0.278147 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:08:04] 0.006107 g.CO2eq/s mean an estimation of 192.5778346983415 kg.CO2eq/
year

```

3.3 Plotting Feature Importance [1.1% Bonus for All] [W]

While building tree-based models, it's common to quantify how well splitting on a particular feature in an extra tree helps with predicting the target label in a dataset. Machine learning practitioners typically use "Gini importance", or the (normalized) total reduction in entropy brought by that feature to evaluate how important that feature is for predicting the target variable.

Gini importance is typically calculated as the reduction in entropy from reaching a split in an extra tree weighted by the probability of reaching that split in the extra tree. Sklearn internally computes the probability for reaching a split by finding the total number of samples that reaches it during the training phase divided by the total number of samples in the dataset. This weighted value is our feature importance.

Let's think about what this metric means with an example. A high probability of reaching a split on feature A in an extra tree trained on a dataset (many samples will reach this split for a decision) and a large reduction in entropy from splitting on feature A will result in a high feature importance value for feature A. This could mean feature A is a very important feature for predicting the probability of the target label. On the other hand, a low probability of reaching a split on feature B in an extra tree and a low reduction in entropy from splitting on feature B will result in a low feature importance value. This could mean feature B is not a very informative feature for predicting the target label. **Thus, the higher the feature importance value, the more important the feature is to predicting the target label.**

Fortunately for us, fitting a `sklearn.ExtraTreeClassifier` to a dataset automatically computes the Gini importance for every feature in the extra tree and stores these values in a **`feature_importances_`** variable. [Review the docs for more details on how to access this variable](#)

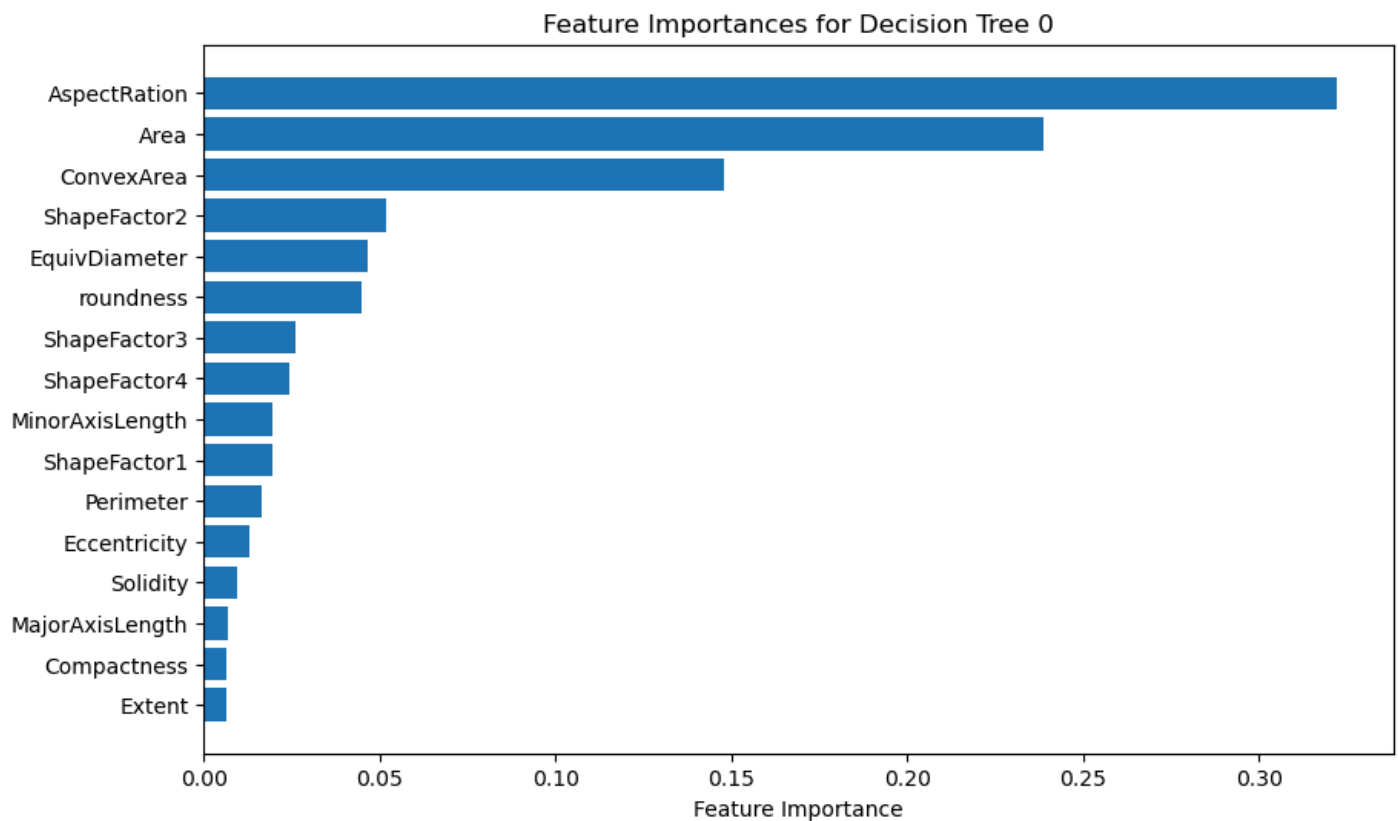
In the **`random_forest.py`** file, complete the following function:

- **`plot_feature_importance`**: Make sure to sort the bars in descending order and remove any features with feature importance of 0

In the cell below, call your implementation of `plot_feature_importance()` and display a bar plot that shows the feature importance values for at least one extra tree in your tuned random forest from Q3.2.

```
In [ ]: # TODO: Complete plot_feature_importance() in random_forest.py
```

```
random_forest.plot_feature_importance(X)
```



[codecarbon INFO @ 20:08:19] Energy consumed for RAM : 0.033897 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:08:19] Energy consumed for all CPUs : 0.244452 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:08:19] 0.278349 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:08:34] Energy consumed for RAM : 0.033921 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:08:34] Energy consumed for all CPUs : 0.244629 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:08:34] 0.278551 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:08:49] Energy consumed for RAM : 0.033946 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:08:49] Energy consumed for all CPUs : 0.244807 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:08:49] 0.278753 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:09:04] Energy consumed for RAM : 0.033971 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:09:04] Energy consumed for all CPUs : 0.244984 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:09:04] 0.278954 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:09:19] Energy consumed for RAM : 0.033995 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:09:19] Energy consumed for all CPUs : 0.245161 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:09:19] 0.279156 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:09:34] Energy consumed for RAM : 0.034020 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:09:34] Energy consumed for all CPUs : 0.245338 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:09:34] 0.279358 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:09:49] Energy consumed for RAM : 0.034044 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:09:49] Energy consumed for all CPUs : 0.245515 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:09:49] 0.279559 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:10:04] Energy consumed for RAM : 0.034069 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:10:04] Energy consumed for all CPUs : 0.245692 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:10:04] 0.279761 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:10:04] 0.006107 g.CO2eq/s mean an estimation of 192.58889385144568 kg.CO2eq/year

[codecarbon INFO @ 20:10:19] Energy consumed for RAM : 0.034093 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:10:19] Energy consumed for all CPUs : 0.245870 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:10:19] 0.279963 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:10:34] Energy consumed for RAM : 0.034118 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:10:34] Energy consumed for all CPUs : 0.246047 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:10:34] 0.280165 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:10:49] Energy consumed for RAM : 0.034142 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:10:49] Energy consumed for all CPUs : 0.246224 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:10:49] 0.280366 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:11:04] Energy consumed for RAM : 0.034167 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:11:04] Energy consumed for all CPUs : 0.246401 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:11:04] 0.280568 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:11:19] Energy consumed for RAM : 0.034192 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:11:19] Energy consumed for all CPUs : 0.246578 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:11:19] 0.280770 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:11:34] Energy consumed for RAM : 0.034216 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:11:34] Energy consumed for all CPUs : 0.246756 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:11:34] 0.280972 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:11:49] Energy consumed for RAM : 0.034241 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:11:49] Energy consumed for all CPUs : 0.246933 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:11:49] 0.281174 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:12:04] Energy consumed for RAM : 0.034265 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:12:04] Energy consumed for all CPUs : 0.247110 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:12:04] 0.281376 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:12:04] 0.006106 g.CO2eq/s mean an estimation of 192.57375560044582 kg.CO2eq/year

[codecarbon INFO @ 20:12:19] Energy consumed for RAM : 0.034290 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:12:19] Energy consumed for all CPUs : 0.247288 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:12:19] 0.281578 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:12:34] Energy consumed for RAM : 0.034314 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:12:34] Energy consumed for all CPUs : 0.247465 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:12:34] 0.281779 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:12:49] Energy consumed for RAM : 0.034339 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:12:49] Energy consumed for all CPUs : 0.247642 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:12:49] 0.281981 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:13:04] Energy consumed for RAM : 0.034364 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:13:04] Energy consumed for all CPUs : 0.247819 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:13:04] 0.282183 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:13:19] Energy consumed for RAM : 0.034388 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:13:19] Energy consumed for all CPUs : 0.247996 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:13:19] 0.282385 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:13:34] Energy consumed for RAM : 0.034413 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:13:34] Energy consumed for all CPUs : 0.248174 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:13:34] 0.282586 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:13:49] Energy consumed for RAM : 0.034437 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:13:49] Energy consumed for all CPUs : 0.248351 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:13:49] 0.282788 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:14:04] Energy consumed for RAM : 0.034462 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:14:04] Energy consumed for all CPUs : 0.248530 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:14:04] 0.282992 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:14:04] 0.006107 g.CO2eq/s mean an estimation of 192.5862446820556 kg.CO2eq/year

[codecarbon INFO @ 20:14:19] Energy consumed for RAM : 0.034487 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:14:19] Energy consumed for all CPUs : 0.248707 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:14:19] 0.283194 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:14:34] Energy consumed for RAM : 0.034511 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:14:34] Energy consumed for all CPUs : 0.248884 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:14:34] 0.283395 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:14:49] Energy consumed for RAM : 0.034536 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:14:49] Energy consumed for all CPUs : 0.249061 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:14:49] 0.283597 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:15:04] Energy consumed for RAM : 0.034560 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:15:04] Energy consumed for all CPUs : 0.249238 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:15:04] 0.283798 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:15:19] Energy consumed for RAM : 0.034585 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:15:19] Energy consumed for all CPUs : 0.249415 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:15:19] 0.284000 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:15:34] Energy consumed for RAM : 0.034609 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:15:34] Energy consumed for all CPUs : 0.249593 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:15:34] 0.284202 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:15:49] Energy consumed for RAM : 0.034634 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:15:49] Energy consumed for all CPUs : 0.249770 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:15:49] 0.284404 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:16:04] Energy consumed for RAM : 0.034659 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:16:04] Energy consumed for all CPUs : 0.249947 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:16:04] 0.284605 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:16:04] 0.006106 g.CO2eq/s mean an estimation of 192.5692919559966 kg.CO2eq/year

[codecarbon INFO @ 20:16:19] Energy consumed for RAM : 0.034683 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:16:19] Energy consumed for all CPUs : 0.250124 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:16:19] 0.284807 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:16:34] Energy consumed for RAM : 0.034708 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 20:16:34] Energy consumed for all CPUs : 0.250301 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:16:34] 0.285009 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:16:49] Energy consumed for RAM : 0.034732 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:16:49] Energy consumed for all CPUs : 0.250478 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:16:49] 0.285211 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:17:04] Energy consumed for RAM : 0.034757 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:17:04] Energy consumed for all CPUs : 0.250656 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:17:04] 0.285412 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:17:19] Energy consumed for RAM : 0.034781 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:17:19] Energy consumed for all CPUs : 0.250833 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:17:19] 0.285614 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:17:34] Energy consumed for RAM : 0.034806 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:17:34] Energy consumed for all CPUs : 0.251010 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:17:34] 0.285816 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:17:49] Energy consumed for RAM : 0.034831 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:17:49] Energy consumed for all CPUs : 0.251188 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:17:49] 0.286018 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:18:04] Energy consumed for RAM : 0.034855 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:18:04] Energy consumed for all CPUs : 0.251365 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:18:04] 0.286220 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:18:04] 0.006107 g.CO2eq/s mean an estimation of 192.578937537687 kg.CO2eq/year

[codecarbon INFO @ 20:18:19] Energy consumed for RAM : 0.034880 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:18:19] Energy consumed for all CPUs : 0.251542 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:18:19] 0.286422 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:18:34] Energy consumed for RAM : 0.034904 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:18:34] Energy consumed for all CPUs : 0.251719 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:18:34] 0.286623 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:18:49] Energy consumed for RAM : 0.034929 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:18:49] Energy consumed for all CPUs : 0.251896 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:18:49] 0.286825 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:19:04] Energy consumed for RAM : 0.034953 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:19:04] Energy consumed for all CPUs : 0.252073 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:19:04] 0.287027 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:19:19] Energy consumed for RAM : 0.034978 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 20:19:19] Energy consumed for all CPUs : 0.252250 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 20:19:19] 0.287228 kWh of electricity used since the beginning.

[codecarbon INFO @ 20:19:34] Energy consumed for RAM : 0.035002 kWh. RAM Power : 5.8941850662231445 W

45 W
[codecarbon INFO @ 20:19:34] Energy consumed for all CPUs : 0.252428 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:19:34] 0.287430 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:19:49] Energy consumed for RAM : 0.035027 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:19:49] Energy consumed for all CPUs : 0.252605 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:19:49] 0.287632 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:20:04] Energy consumed for RAM : 0.035052 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:20:04] Energy consumed for all CPUs : 0.252782 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:20:04] 0.287833 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:20:04] 0.006106 g.CO2eq/s mean an estimation of 192.57093267721154 kg.CO2eq/year
[codecarbon INFO @ 20:20:19] Energy consumed for RAM : 0.035076 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:20:19] Energy consumed for all CPUs : 0.252959 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:20:19] 0.288035 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:20:34] Energy consumed for RAM : 0.035101 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:20:34] Energy consumed for all CPUs : 0.253136 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:20:34] 0.288237 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:20:49] Energy consumed for RAM : 0.035125 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:20:49] Energy consumed for all CPUs : 0.253313 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:20:49] 0.288439 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:21:04] Energy consumed for RAM : 0.035150 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:21:04] Energy consumed for all CPUs : 0.253490 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:21:04] 0.288640 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:21:19] Energy consumed for RAM : 0.035174 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:21:20] Energy consumed for all CPUs : 0.253668 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:21:20] 0.288842 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:21:35] Energy consumed for RAM : 0.035199 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:21:35] Energy consumed for all CPUs : 0.253845 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:21:35] 0.289044 kWh of electricity used since the beginning.
[codecarbon INFO @ 20:21:50] Energy consumed for RAM : 0.035223 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 20:21:50] Energy consumed for all CPUs : 0.254022 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 20:21:50] 0.289246 kWh of electricity used since the beginning.
[codecarbon WARNING @ 21:51:19] Background scheduler didn't run for a long period (5369s), results might be inaccurate
[codecarbon INFO @ 21:51:19] Energy consumed for RAM : 0.044015 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:51:19] Energy consumed for all CPUs : 0.317411 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:51:19] 0.361426 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:51:19] 0.006108 g.CO2eq/s mean an estimation of 192.62896270455187 kg.CO2eq

q/year

[codecarbon INFO @ 21:51:34] Energy consumed for RAM : 0.044039 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:51:34] Energy consumed for all CPUs : 0.317588 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:51:34] 0.361627 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:51:49] Energy consumed for RAM : 0.044064 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:51:49] Energy consumed for all CPUs : 0.317765 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:51:49] 0.361829 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:52:04] Energy consumed for RAM : 0.044088 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:52:04] Energy consumed for all CPUs : 0.317942 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:52:04] 0.362031 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:52:19] Energy consumed for RAM : 0.044113 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:52:19] Energy consumed for all CPUs : 0.318120 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:52:19] 0.362233 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:52:34] Energy consumed for RAM : 0.044138 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:52:34] Energy consumed for all CPUs : 0.318297 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:52:34] 0.362434 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:52:49] Energy consumed for RAM : 0.044162 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:52:49] Energy consumed for all CPUs : 0.318474 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:52:49] 0.362636 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:53:04] Energy consumed for RAM : 0.044187 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:53:04] Energy consumed for all CPUs : 0.318651 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:53:04] 0.362838 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:53:19] Energy consumed for RAM : 0.044211 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:53:19] Energy consumed for all CPUs : 0.318828 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:53:19] 0.363039 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:53:19] 0.006106 g.CO2eq/s mean an estimation of 192.57213053284383 kg.CO2eq/year

[codecarbon INFO @ 21:53:34] Energy consumed for RAM : 0.044236 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:53:34] Energy consumed for all CPUs : 0.319005 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:53:34] 0.363241 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:53:49] Energy consumed for RAM : 0.044260 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:53:49] Energy consumed for all CPUs : 0.319182 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:53:49] 0.363443 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:54:04] Energy consumed for RAM : 0.044285 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 21:54:04] Energy consumed for all CPUs : 0.319360 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 21:54:04] 0.363644 kWh of electricity used since the beginning.

[codecarbon INFO @ 21:54:19] Energy consumed for RAM : 0.044309 kWh. RAM Power : 5.8941850662231445 W

45 W
[codecarbon INFO @ 21:54:19] Energy consumed for all CPUs : 0.319537 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:54:19] 0.363846 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:54:34] Energy consumed for RAM : 0.044334 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:54:34] Energy consumed for all CPUs : 0.319714 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:54:34] 0.364048 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:54:49] Energy consumed for RAM : 0.044359 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:54:49] Energy consumed for all CPUs : 0.319891 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:54:49] 0.364249 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:55:04] Energy consumed for RAM : 0.044383 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:55:04] Energy consumed for all CPUs : 0.320068 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:55:04] 0.364451 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:55:19] Energy consumed for RAM : 0.044408 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:55:19] Energy consumed for all CPUs : 0.320245 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:55:19] 0.364653 kWh of electricity used since the beginning.
[codecarbon INFO @ 21:55:19] 0.006106 g.CO2eq/s mean an estimation of 192.54357562790045 kg.CO2eq/year
[codecarbon INFO @ 21:55:34] Energy consumed for RAM : 0.044432 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 21:55:34] Energy consumed for all CPUs : 0.320422 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 21:55:34] 0.364854 kWh of electricity used since the beginning.
[codecarbon WARNING @ 10:10:38] Background scheduler didn't run for a long period (44104s), results might be inaccurate
[codecarbon INFO @ 10:10:38] Energy consumed for RAM : 0.116642 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:10:38] Energy consumed for all CPUs : 0.841095 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:10:38] 0.957737 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:10:53] Energy consumed for RAM : 0.116667 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:10:53] Energy consumed for all CPUs : 0.841272 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:10:53] 0.957939 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:11:08] Energy consumed for RAM : 0.116692 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:11:08] Energy consumed for all CPUs : 0.841449 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:11:08] 0.958140 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:11:23] Energy consumed for RAM : 0.116716 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:11:23] Energy consumed for all CPUs : 0.841626 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:11:23] 0.958342 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:11:38] Energy consumed for RAM : 0.116741 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:11:38] Energy consumed for all CPUs : 0.841803 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:11:38] 0.958544 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:11:53] Energy consumed for RAM : 0.116765 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 10:11:53] Energy consumed for all CPUs : 0.841980 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:11:53] 0.958745 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:12:08] Energy consumed for RAM : 0.116790 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:12:08] Energy consumed for all CPUs : 0.842157 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:12:08] 0.958947 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:12:08] 0.006108 g.CO2eq/s mean an estimation of 192.62994143770177 kg.CO2eq/year
[codecarbon INFO @ 10:12:23] Energy consumed for RAM : 0.116814 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:12:23] Energy consumed for all CPUs : 0.842334 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:12:23] 0.959149 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:12:38] Energy consumed for RAM : 0.116839 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:12:38] Energy consumed for all CPUs : 0.842512 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:12:38] 0.959350 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:12:53] Energy consumed for RAM : 0.116863 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:12:53] Energy consumed for all CPUs : 0.842689 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:12:53] 0.959552 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:13:08] Energy consumed for RAM : 0.116888 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:13:08] Energy consumed for all CPUs : 0.842866 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:13:08] 0.959754 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:13:23] Energy consumed for RAM : 0.116913 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:13:23] Energy consumed for all CPUs : 0.843043 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:13:23] 0.959955 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:13:38] Energy consumed for RAM : 0.116937 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:13:38] Energy consumed for all CPUs : 0.843220 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:13:38] 0.960157 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:13:53] Energy consumed for RAM : 0.116962 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:13:53] Energy consumed for all CPUs : 0.843397 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:13:53] 0.960359 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:14:08] Energy consumed for RAM : 0.116986 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:14:08] Energy consumed for all CPUs : 0.843574 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:14:08] 0.960561 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:14:08] 0.006106 g.CO2eq/s mean an estimation of 192.55701435105348 kg.CO2eq/year
[codecarbon INFO @ 10:14:23] Energy consumed for RAM : 0.117011 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:14:23] Energy consumed for all CPUs : 0.843751 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:14:23] 0.960762 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:14:38] Energy consumed for RAM : 0.117035 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 10:14:38] Energy consumed for all CPUs : 0.843928 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:14:38] 0.960964 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:14:53] Energy consumed for RAM : 0.117060 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:14:53] Energy consumed for all CPUs : 0.844106 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:14:53] 0.961165 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:15:08] Energy consumed for RAM : 0.117084 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:15:08] Energy consumed for all CPUs : 0.844283 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:15:08] 0.961367 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:15:23] Energy consumed for RAM : 0.117109 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:15:23] Energy consumed for all CPUs : 0.844460 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:15:23] 0.961569 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:15:38] Energy consumed for RAM : 0.117134 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:15:38] Energy consumed for all CPUs : 0.844637 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:15:38] 0.961770 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:15:53] Energy consumed for RAM : 0.117158 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:15:53] Energy consumed for all CPUs : 0.844814 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:15:53] 0.961972 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:16:08] Energy consumed for RAM : 0.117183 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:16:08] Energy consumed for all CPUs : 0.844991 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:16:08] 0.962174 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:16:08] 0.006106 g.CO2eq/s mean an estimation of 192.56769709330513 kg.CO2eq/year
[codecarbon INFO @ 10:16:23] Energy consumed for RAM : 0.117207 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:16:23] Energy consumed for all CPUs : 0.845168 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:16:23] 0.962376 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:16:38] Energy consumed for RAM : 0.117232 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:16:38] Energy consumed for all CPUs : 0.845346 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:16:38] 0.962577 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:16:53] Energy consumed for RAM : 0.117256 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:16:53] Energy consumed for all CPUs : 0.845523 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:16:53] 0.962779 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:17:08] Energy consumed for RAM : 0.117281 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:17:08] Energy consumed for all CPUs : 0.845700 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:17:08] 0.962981 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:17:23] Energy consumed for RAM : 0.117305 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:17:23] Energy consumed for all CPUs : 0.845877 kWh. Total CPU Power : 42.5

W
[codecarbon INFO @ 10:17:23] 0.963182 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:17:38] Energy consumed for RAM : 0.117330 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:17:38] Energy consumed for all CPUs : 0.846054 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:17:38] 0.963384 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:17:53] Energy consumed for RAM : 0.117355 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:17:53] Energy consumed for all CPUs : 0.846231 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:17:53] 0.963586 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:18:08] Energy consumed for RAM : 0.117379 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:18:08] Energy consumed for all CPUs : 0.846408 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:18:08] 0.963787 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:18:08] 0.006106 g.CO2eq/s mean an estimation of 192.55402970650024 kg.CO2e
q/year
[codecarbon INFO @ 10:18:23] Energy consumed for RAM : 0.117404 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:18:23] Energy consumed for all CPUs : 0.846585 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:18:23] 0.963989 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:18:38] Energy consumed for RAM : 0.117428 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:18:38] Energy consumed for all CPUs : 0.846763 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:18:38] 0.964191 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:18:53] Energy consumed for RAM : 0.117453 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:18:53] Energy consumed for all CPUs : 0.846939 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:18:53] 0.964392 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:19:08] Energy consumed for RAM : 0.117477 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:19:08] Energy consumed for all CPUs : 0.847117 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:19:08] 0.964594 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:19:23] Energy consumed for RAM : 0.117502 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:19:23] Energy consumed for all CPUs : 0.847294 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:19:23] 0.964796 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:19:38] Energy consumed for RAM : 0.117526 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:19:38] Energy consumed for all CPUs : 0.847471 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:19:38] 0.964998 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:19:53] Energy consumed for RAM : 0.117551 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:19:53] Energy consumed for all CPUs : 0.847648 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:19:53] 0.965200 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:20:08] Energy consumed for RAM : 0.117576 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:20:08] Energy consumed for all CPUs : 0.847826 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:20:08] 0.965401 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:20:08] 0.006106 g.CO2eq/s mean an estimation of 192.55435198877592 kg.CO2eq/year

[codecarbon INFO @ 10:20:23] Energy consumed for RAM : 0.117600 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:20:23] Energy consumed for all CPUs : 0.848003 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:20:23] 0.965603 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:20:38] Energy consumed for RAM : 0.117625 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:20:38] Energy consumed for all CPUs : 0.848180 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:20:38] 0.965805 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:20:53] Energy consumed for RAM : 0.117649 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:20:53] Energy consumed for all CPUs : 0.848357 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:20:53] 0.966006 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:21:08] Energy consumed for RAM : 0.117674 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:21:08] Energy consumed for all CPUs : 0.848534 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:21:08] 0.966208 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:21:23] Energy consumed for RAM : 0.117698 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:21:23] Energy consumed for all CPUs : 0.848711 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:21:23] 0.966410 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:21:38] Energy consumed for RAM : 0.117723 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:21:38] Energy consumed for all CPUs : 0.848888 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:21:38] 0.966611 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:21:54] Energy consumed for RAM : 0.117747 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:21:54] Energy consumed for all CPUs : 0.849066 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:21:54] 0.966813 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:22:09] Energy consumed for RAM : 0.117772 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:22:09] Energy consumed for all CPUs : 0.849243 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:22:09] 0.967015 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:22:09] 0.006106 g.CO2eq/s mean an estimation of 192.5634137587299 kg.CO2eq/year

[codecarbon INFO @ 10:22:24] Energy consumed for RAM : 0.117797 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:22:24] Energy consumed for all CPUs : 0.849420 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:22:24] 0.967216 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:22:39] Energy consumed for RAM : 0.117821 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:22:39] Energy consumed for all CPUs : 0.849597 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:22:39] 0.967418 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:22:54] Energy consumed for RAM : 0.117846 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:22:54] Energy consumed for all CPUs : 0.849774 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:22:54] 0.967620 kWh of electricity used since the beginning.

```
[codecarbon INFO @ 10:23:09] Energy consumed for RAM : 0.117870 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:23:09] Energy consumed for all CPUs : 0.849951 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:23:09] 0.967822 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:23:24] Energy consumed for RAM : 0.117895 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:23:24] Energy consumed for all CPUs : 0.850128 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:23:24] 0.968023 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:23:39] Energy consumed for RAM : 0.117919 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:23:39] Energy consumed for all CPUs : 0.850305 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:23:39] 0.968225 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:23:54] Energy consumed for RAM : 0.117944 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:23:54] Energy consumed for all CPUs : 0.850483 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:23:54] 0.968427 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:24:09] Energy consumed for RAM : 0.117968 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:24:09] Energy consumed for all CPUs : 0.850660 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:24:09] 0.968628 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:24:09] 0.006107 g.CO2eq/s mean an estimation of 192.57714571352224 kg.CO2e
q/year
[codecarbon INFO @ 10:24:24] Energy consumed for RAM : 0.117993 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:24:24] Energy consumed for all CPUs : 0.850837 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:24:24] 0.968830 kWh of electricity used since the beginning.
```

Note that there isn't one "correct" answer here. We simply want you to investigate how different features in your random forest contribute to predicting the target variable.

Also note that: the number of features can be different if you change `max_features` value since it ends up changing the number of features considered in bootstrapped datasets.

3.4 ADABOOST [1% Bonus for All] [\[P\]](#)

In lecture we learn how to implement bootstrapping, but there is another common method used to prevent overfitting which also incorporates multiple decision trees: boosting. For our implementation, boosting is where you assign importances to multiple models (weak learners) and return the result of their averaged output (a single strong learner). This is done sequentially, with importances being updated after the training of each new tree rather than in parallel like in bootstrapping

Specifically, you will be implementing adaptive boosting (ADABOOST) which reassigns importances to the randomized decision trees depending on their error weight.

Additional Resource: For a detailed walkthrough of ADABOOST implementation, see [Implementing the AdaBoost Algorithm from Scratch](#)

Using their error weights you will recalculate the importances AKA `alpha` values. You can use this formula to do it per model:

$$\alpha = \frac{1}{2} \log \left(\frac{1 - \text{error}}{\text{error} + 10^{-10}} \right)$$

Then you must update the error weights for all models for the next importance calculation by multiplying all weights using this formula:

$$\text{Updated Weights (Not Normalized)} = \text{Weights} \times e^{\alpha \times (\text{Number of Incorrect Classifications})}$$

We give you `predict_adaboost` which returns the prediction from the result of all the trees. You must implement `adaboost()` which iterates over the number of estimators, training a new tree and then updating all trees' weights depending on its error rate.

```
In [64]: # TODO: Complete adaboost() in random_forest.py
```

```
TestRandomForest("test_adaboost").test_adaboost()
```

NotImplementedError Traceback (most recent call last)

Cell In[64], line 3

```
1 # TODO: Complete adaboost() in random_forest.py
----> 3 TestRandomForest("test_adaboost").test_adaboost()
```

File `c:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4641\HW4\utilities\localtests.py:1211`, in `TestRandomForest.test_adaboost(self)`

```
1208 X_train = np.random.rand(100, num_feats)
1209 y_train = np.random.randint(0, 2, 100)
-> 1211 rf_test.adaboost(X_train, y_train)
1213 assert (
1214     len(rf_test.alphas) == 4
1215 ), "The number of alphas should match the number of estimators."
1217 predictions = rf_test.predict_adaboost(X_train)
```

File `c:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4641\HW4\random_forest.py:116`, in `RandomForest.adaboost(self, X, y)`

```
98 def adaboost(self, X, y):
99     """
100     TODO:
101     - Implement AdaBoost training by adjusting sample weights after each round of training a weak learner.
102     (...)
103     None. Trains the ensemble using AdaBoost's weighting mechanism.
104     """
--> 116     raise NotImplementedError()
```

NotImplementedError:

Now we can test the accuracy of our model using the helper function `predict_adaboost()` ! To pass Gradescope your accuracy should be above 85%!

```
In [ ]: adaboost_model = RandomForest(
        n_estimators, max_depth, max_features, random_seed=student_random_seed
    )
```

```
# Train the AdaBoost ensemble using adaboost method
adaboost_model.adaboost(X_train, y_train)
```

```
# Evaluate the accuracy on the test set
y_pred = adaboost_model.predict_adaboost(X_test)
accuracy = np.mean(y_pred == y_test)
print("AdaBoost accuracy: %.4f" % accuracy)
```

[codecarbon INFO @ 11:31:20] Energy consumed for RAM : 0.124561 kWh. RAM Power : 5.8941850662231445 W

NotImplementedError Traceback (most recent call last)

Cell In[65], line 6

```
1 adaboost_model = RandomForest(
2     n_estimators, max_depth, max_features, random_seed=student_random_seed
3 )
4 # Train the AdaBoost ensemble using adaboost method
----> 6 adaboost_model.adaboost(X_train, y_train)
7 # Evaluate the accuracy on the test set
8 y_pred = adaboost_model.predict_adaboost(X_test)
```

File c:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4641\HW4\random_forest.py:116, in RandomForest.adaboost(self, X, y)

```
98 def adaboost(self, X, y):
99     """
100     TODO:
101     - Implement AdaBoost training by adjusting sample weights after each round of training a weak learner.
102     (...)
103     None. Trains the ensemble using AdaBoost's weighting mechanism.
104     """
--> 116     raise NotImplementedError()
```

NotImplementedError:

[codecarbon INFO @ 11:31:20] Energy consumed for all CPUs : 0.898217 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 11:31:20] 1.022778 kWh of electricity used since the beginning.

```
[codecarbon INFO @ 11:31:35] Energy consumed for RAM : 0.124586 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 11:31:35] Energy consumed for all CPUs : 0.898394 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 11:31:35] 1.022980 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:31:50] Energy consumed for RAM : 0.124611 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 11:31:50] Energy consumed for all CPUs : 0.898571 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 11:31:50] 1.023181 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:32:05] Energy consumed for RAM : 0.124635 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 11:32:05] Energy consumed for all CPUs : 0.898748 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 11:32:05] 1.023383 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:32:20] Energy consumed for RAM : 0.124660 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 11:32:20] Energy consumed for all CPUs : 0.898925 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 11:32:20] 1.023585 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:32:20] 0.006107 g.CO2eq/s mean an estimation of 192.58241132790369 kg.CO2e
q/year
[codecarbon INFO @ 11:32:35] Energy consumed for RAM : 0.124684 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 11:32:35] Energy consumed for all CPUs : 0.899102 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 11:32:35] 1.023786 kWh of electricity used since the beginning.
```

4: SVM [15 pts] **[W]**

Consider a dataset with the following points in two-dimensional space:

x_1	x_2	y
-2.3	3	-1
-1.5	-1.5	-1
0.5	0.75	-1
1.9	2.25	-1
-2.0	6.75	1
-1.1	6.0	1
0.0	7.5	1
1.5	6.0	1

Here, x_1 and x_2 are features and y is the label.

Support Vector Machines (SVMs) aim to find a hyperplane that separates data points of different classes with the maximum margin. The larger the margin, the better the model can generalize to unseen data. Fortunately, scikit-learn's SVC class handles this computation for us programmatically.

4.1 Fitting an SVM classifier [10 pts] [W]

4.1.1 Fit the SVM Classifier [7 pts]

Since the points are already linearly separable, determine the separating hyperplane using a linear SVM programatically. Record the weights (theta) and bias (intercept) terms for this separating hyperplane, rounded to three decimal places.

Hint: To do this, you'll need to import the `SVC` class from the `sklearn.svm` module and initialize the SVC with a linear kernel. Then you can fit the data and find the separating hyperplane. Finally, you can get the needed values using `svm.coef_` and `svm.intercept_`.

```
In [ ]: from sklearn.svm import SVC
import numpy as np

# x-components
X = np.array([
    [-2.3, 3],
    [-1.5, -1.5],
    [0.5, 0.75],
    [1.9, 2.25],
    [-2.0, 6.75],
    [-1.1, 6.0],
    [0.0, 7.5],
    [1.5, 6.0]
])

y = np.array([-1, -1, -1, -1, 1, 1, 1, 1])

# initialize and fit SVM
svm = SVC(kernel='linear')
svm.fit(X, y)

# calculate weights and intercept
theta = svm.coef_
intercept = svm.intercept_

# roundddddddd
theta_rounded = np.round(theta, 3)
intercept_rounded = np.round(intercept, 3)

print("Weights (theta):", theta_rounded)
print("Intercept (bias):", intercept_rounded)
```

Weights (theta): [[0.111 0.622]]

Intercept (bias): [-2.611]

[codecarbon INFO @ 10:24:39] Energy consumed for RAM : 0.118018 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:24:39] Energy consumed for all CPUs : 0.851014 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:24:39] 0.969032 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:24:54] Energy consumed for RAM : 0.118042 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:24:54] Energy consumed for all CPUs : 0.851191 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:24:54] 0.969233 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:25:09] Energy consumed for RAM : 0.118067 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:25:09] Energy consumed for all CPUs : 0.851368 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:25:09] 0.969435 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:25:24] Energy consumed for RAM : 0.118091 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:25:24] Energy consumed for all CPUs : 0.851545 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:25:24] 0.969637 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:25:41] Energy consumed for RAM : 0.118116 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:25:41] Energy consumed for all CPUs : 0.851723 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:25:41] 0.969838 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:25:56] Energy consumed for RAM : 0.118140 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:25:56] Energy consumed for all CPUs : 0.851900 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:25:56] 0.970040 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:26:11] Energy consumed for RAM : 0.118165 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:26:11] Energy consumed for all CPUs : 0.852077 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:26:11] 0.970242 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:26:11] 0.006107 g.CO2eq/s mean an estimation of 192.58263607146824 kg.CO2eq/year

[codecarbon INFO @ 10:26:26] Energy consumed for RAM : 0.118189 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:26:26] Energy consumed for all CPUs : 0.852254 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:26:26] 0.970443 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:26:41] Energy consumed for RAM : 0.118214 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:26:41] Energy consumed for all CPUs : 0.852431 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:26:41] 0.970645 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:26:56] Energy consumed for RAM : 0.118239 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:26:56] Energy consumed for all CPUs : 0.852608 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:26:56] 0.970847 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:27:11] Energy consumed for RAM : 0.118263 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:27:11] Energy consumed for all CPUs : 0.852785 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:27:11] 0.971049 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:27:26] Energy consumed for RAM : 0.118288 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 10:27:26] Energy consumed for all CPUs : 0.852963 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:27:26] 0.971250 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:27:41] Energy consumed for RAM : 0.118312 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:27:41] Energy consumed for all CPUs : 0.853140 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:27:41] 0.971452 kWh of electricity used since the beginning.
```

4.1.2 Plot the SVM Classifier [3 pts]

Plot the features x_1 and x_2 using different colors to represent the labels y (e.g., red for -1, blue for 1). Include the separating hyperplane on the plot. Make sure your plot clearly distinguishes the two classes and visually demonstrates the hyperplane.

```
In [ ]: import matplotlib.pyplot as plt
import numpy as np

# plot
plt.figure(figsize=(8, 6))

# multiple colors
colors = ['red' if label == -1 else 'blue' for label in y]
plt.scatter(X[:, 0], X[:, 1], c=colors, s=100, edgecolors='k')

# mesh hyperplane
ax = plt.gca()
xlim = ax.get_xlim()
ylim = ax.get_ylim()

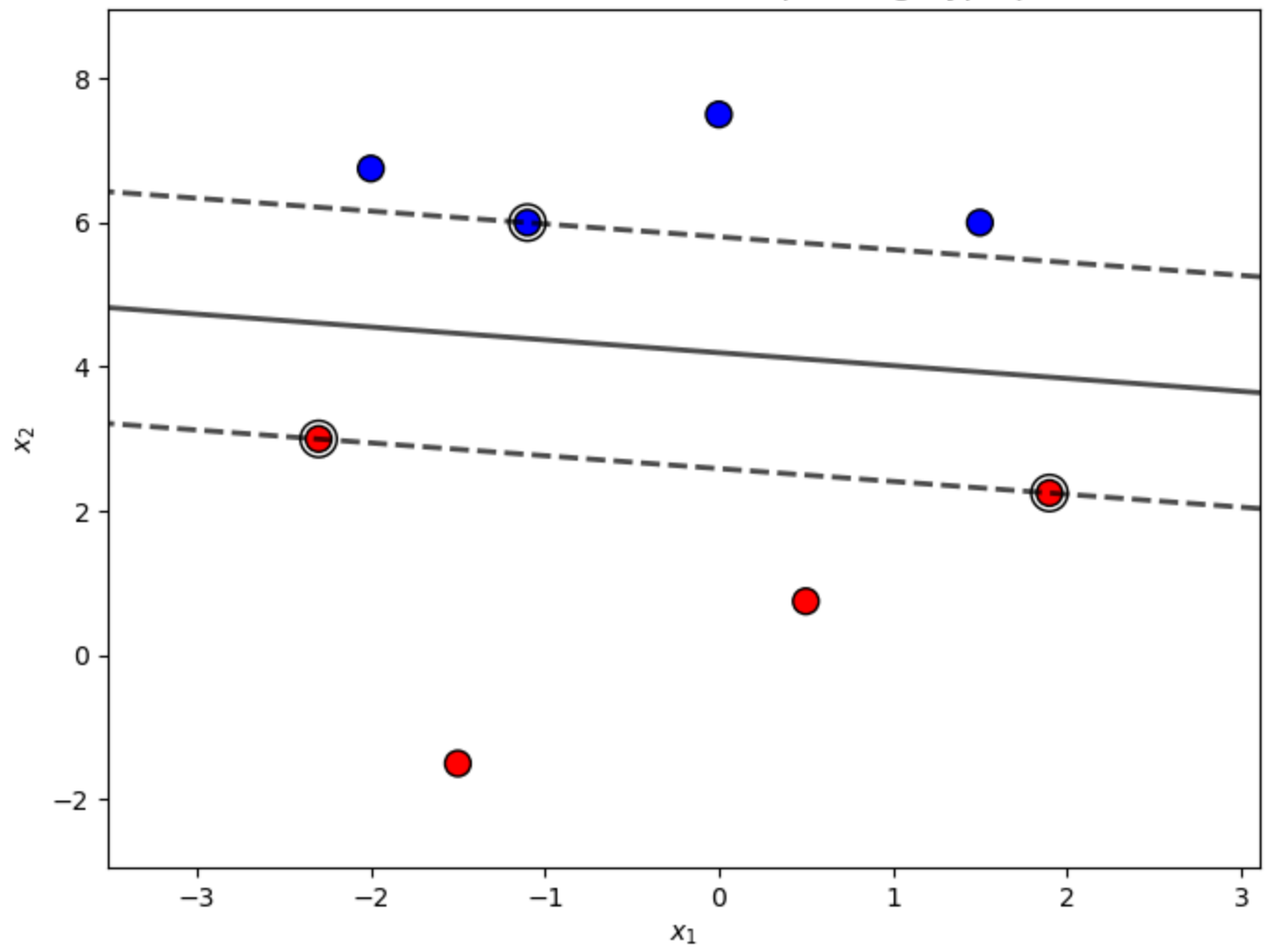
# gridddd
xx = np.linspace(xlim[0]-1, xlim[1]+1, 200)
yy = np.linspace(ylim[0]-1, ylim[1]+1, 200)
YY, XX = np.meshgrid(yy, xx)
xy = np.vstack([XX.ravel(), YY.ravel()]).T
Z = svm.decision_function(xy).reshape(XX.shape)

# decision boundaries
ax.contour(XX, YY, Z, colors='k', levels=[-1, 0, 1], alpha=0.7,
           linestyles=['--', '-', '--'], linewidths=2)

# support vector fancy stuff
ax.scatter(svm.support_vectors_[:, 0], svm.support_vectors_[:, 1], s=200,
           linewidth=1, facecolors='none', edgecolors='k')

plt.xlabel('$x_1$')
plt.ylabel('$x_2$')
plt.title('SVM Linear Classification with Separating Hyperplane')
plt.show()
```

SVM Linear Classification with Separating Hyperplane



[codecarbon INFO @ 10:27:56] Energy consumed for RAM : 0.118337 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:27:56] Energy consumed for all CPUs : 0.853317 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:27:56] 0.971654 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:28:11] Energy consumed for RAM : 0.118361 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:28:11] Energy consumed for all CPUs : 0.853494 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:28:11] 0.971855 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:28:11] 0.006107 g.CO2eq/s mean an estimation of 192.58450187405222 kg.CO2eq/year

[codecarbon INFO @ 10:28:26] Energy consumed for RAM : 0.118386 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:28:26] Energy consumed for all CPUs : 0.853671 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:28:26] 0.972057 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:28:41] Energy consumed for RAM : 0.118410 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:28:41] Energy consumed for all CPUs : 0.853848 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:28:41] 0.972259 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:28:56] Energy consumed for RAM : 0.118435 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:28:56] Energy consumed for all CPUs : 0.854025 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:28:56] 0.972460 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:29:11] Energy consumed for RAM : 0.118460 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:29:11] Energy consumed for all CPUs : 0.854203 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:29:11] 0.972662 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:29:26] Energy consumed for RAM : 0.118484 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:29:26] Energy consumed for all CPUs : 0.854380 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:29:26] 0.972864 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:29:41] Energy consumed for RAM : 0.118509 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:29:41] Energy consumed for all CPUs : 0.854557 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:29:41] 0.973065 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:29:56] Energy consumed for RAM : 0.118533 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:29:56] Energy consumed for all CPUs : 0.854734 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:29:56] 0.973267 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:30:11] Energy consumed for RAM : 0.118558 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:30:11] Energy consumed for all CPUs : 0.854911 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:30:11] 0.973469 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:30:11] 0.006107 g.CO2eq/s mean an estimation of 192.57782147966495 kg.CO2eq/year

[codecarbon INFO @ 10:30:26] Energy consumed for RAM : 0.118582 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:30:26] Energy consumed for all CPUs : 0.855088 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:30:26] 0.973671 kWh of electricity used since the beginning.

4.2 Using Kernels [5 pts] [W]

Suppose we added another point, (0.8, -6.5) with label 1, which causes the dataset to be non-linearly separable. This can be seen by plotting the new set of points, which is impossible to split by a hyperplane that achieves 100% accuracy. To solve this issue, SVM uses kernels, which transform the points in a dataset to a higher-dimensional space making them linearly separable. In section 4.1, you used a linear kernel, which didn't transform the dataset since it was already linearly separable.

Create a specific kernel (transformation function) that could be applied to the dataset with the new point that makes the data linearly separable again.

```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.svm import SVC

# x components again
X = np.array([
    [-2.3, 3],
    [-1.5, -1.5],
    [0.5, 0.75],
    [1.9, 2.25],
    [-2.0, 6.75],
    [-1.1, 6.0],
    [0.0, 7.5],
    [1.5, 6.0],
    [0.8, -6.5] # wit new point
])

y = np.array([-1, -1, -1, -1, 1, 1, 1, 1, 1])

# transform!
def transform(X):
    return np.column_stack((X[:, 0], X[:, 1]**2))

# Let's transform!
X_transformed = transform(X)

# initialize and fit model
svm = SVC(kernel='linear')
svm.fit(X_transformed, y)

# get weights and intercept
theta = svm.coef_[0]
intercept = svm.intercept_[0]

print("Weights (theta):", theta)
print("Intercept (bias):", intercept)

# plot
plt.figure(figsize=(8, 6))

# colors!!
colors = ['red' if label == -1 else 'blue' for label in y]
plt.scatter(X_transformed[:, 0], X_transformed[:, 1], c=colors, s=100, edgecolors='k')
```

```

# hyperplane
ax = plt.gca()
xlim = ax.get_xlim()
ylim = ax.get_ylim()

# gridddd
xx = np.linspace(xlim[0], xlim[1], 200)
yy = (-theta[0]*xx - intercept) / theta[1]

# hyperplane
plt.plot(xx, yy, 'k-')

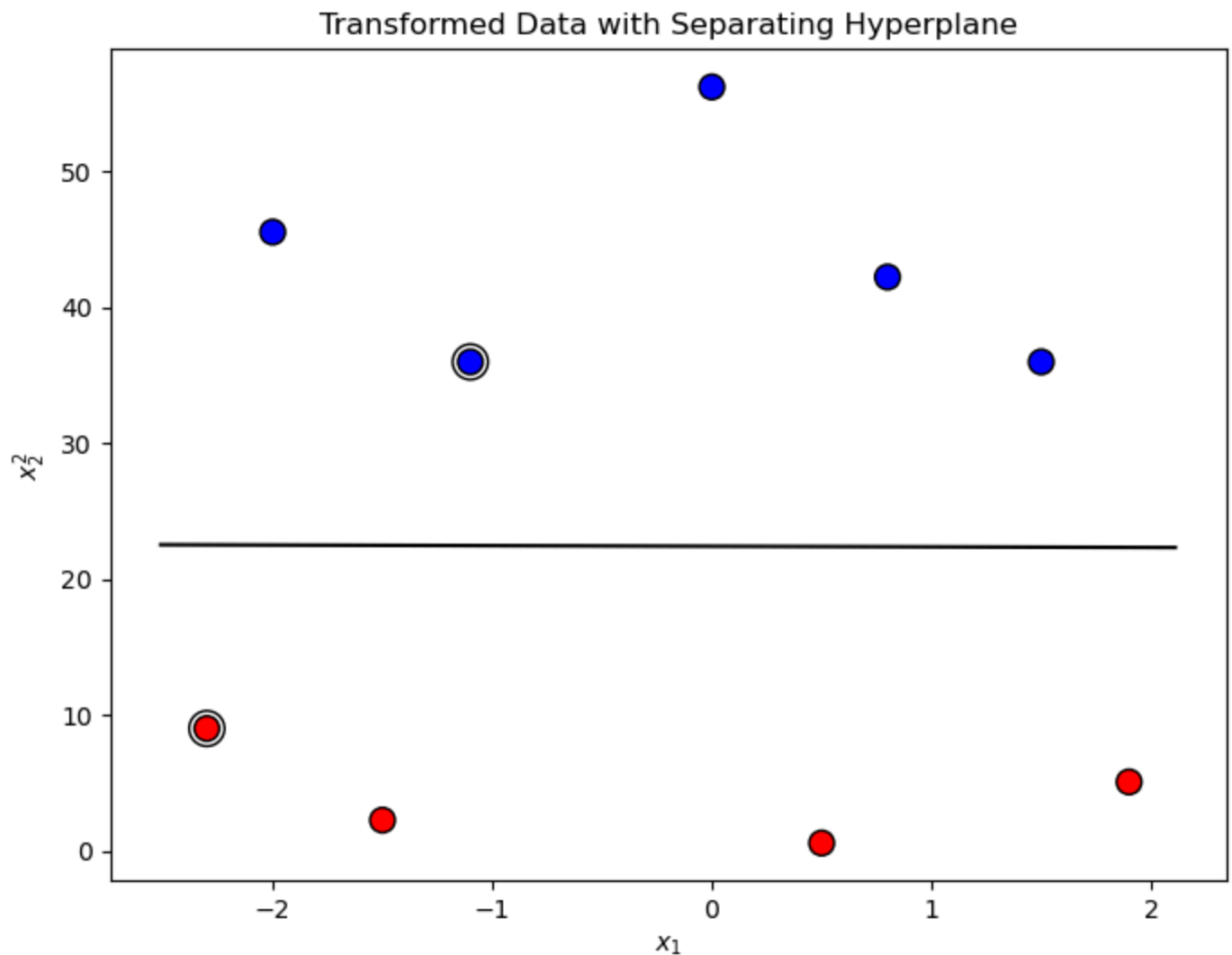
# supporting vectors
plt.scatter(svm.support_vectors_[:, 0], svm.support_vectors_[:, 1], s=200,
            linewidth=1, facecolors='none', edgecolors='k')

plt.xlabel('$x_1$')
plt.ylabel('$x_2^2$')
plt.title('Transformed Data with Separating Hyperplane')
plt.show()

```

Weights (theta): [0.00328569 0.07392805]

Intercept (bias): -1.6577953274853234



```
[codecarbon INFO @ 10:30:56] Energy consumed for RAM : 0.118632 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 10:30:56] Energy consumed for all CPUs : 0.855443 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:30:56] 0.974074 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:31:11] Energy consumed for RAM : 0.118656 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 10:31:11] Energy consumed for all CPUs : 0.855620 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:31:11] 0.974276 kWh of electricity used since the beginning.
```

5: Next Character Prediction using Recurrent Neural Networks (RNNs) [6.8% Bonus for All] [W] [P]

In this section, we'll compare two foundational types of recurrent neural network architectures: Simple Recurrent Neural Networks (Simple RNNs) and Long Short-Term Memory networks (LSTMs). The goal is to train these models to generate text in the style of Macbeth by predicting the next character in a given sequence. This exercise will highlight how each architecture manages sequential dependencies in text generation.

What are Recurrent Neural Networks?

Recurrent Neural Networks are a class of neural networks designed to handle sequential or time-series data, where the order of inputs matters. Unlike feedforward neural networks that treat each input independently, sequential networks maintain memory of previous inputs, making them ideal for tasks involving ordered data like text, time series, or video frames. These networks allow previous outputs to be used as inputs while having hidden states.

Common applications include:

- Text processing (language modeling, translation)
- Machine translation (translating from one language to the other)
- Time series prediction (stock prices, weather forecasting)

Types of Recurrent Neural Networks

We can divide the applications of RNNs into four main categories:

- **One-to-One**
 - Takes a single input and produces a single output
 - Like feedforward neural networks
 - Example: Image classification (1 image → 1 label)
- **Many-to-One**
 - Takes a sequence as input but produces a single output
 - Example: Sentiment analysis (many words → 1 sentiment)

- *This is what we'll be building for our character prediction task!*
- **One-to-Many**
 - Takes a single input and produces a sequence
 - Example: Image captioning (1 image → sequence of words)
- **Many-to-Many**
 - Takes a sequence and produces a sequence
 - Examples:
 - Translation (English sentence → French sentence)
 - Video frame prediction (sequence of frames → next frames)

Each type serves different purposes, and understanding which to use depends on your task's input and desired output structure.

In this section, our goal will be a **many-to-one** task: predicting the next character given a sequence of characters. Now let's take a look at the models we can use to accomplish this.

Simple Recurrent Neural Networks

Simple Recurrent Neural Networks (RNNs) are the simplest form of recurrent neural networks. They are similar to feedforward networks, but have connections that loop back on themselves, allowing information from previous steps to influence the current processing step. This looping mechanism enables Simple RNNs to capture dependencies over time, which is especially useful for tasks involving language, where the context of earlier words or characters affects the final prediction. Here's how Simple RNNs generally work:

1. At each timestep, the model:
 - Takes in a new input
 - Uses its current hidden state (memory of previous inputs)
 - Produces both an output and an updated hidden state
2. This process repeats for each element in the sequence, with the hidden state carrying information forward.

For our character prediction task, we'll use a many-to-one configuration:

- Input: We feed in characters one at a time (like "m", "a", "c", "b", "e")
- Hidden State: Gets updated with each new character
- Output: We only use the final prediction to guess the next character ("t")

Architecture



Simple RNNs often suffer from the vanishing gradient problem when performing backpropagation through time. This issue arises because the gradients are repeatedly multiplied by the weight matrix as they propagate through each time step. If the weights are small, these multiplications cause the gradients to decrease exponentially, effectively "vanishing" as they travel backward, while large weights can lead to gradient "explosion." As a result, the network struggles to learn and retain long-term dependencies,

meaning that early inputs in a sequence are gradually "forgotten," and training becomes ineffective for tasks requiring the processing of lengthy sequences.

For a detailed explanation of Simple RNNs and their limitations, see:

- [An Introduction to Recurrent Neural Networks \(RNNs\)](#)
- [The Vanishing Gradient Problem in RNNs](#)
- [Josh Starmer RNNs clearly explained](#)

Note: Simple/Vanilla RNN and RNN are often used interchangeably since Simple RNNs represent the most basic implementation of RNNs. However, RNN is actually a broader term that encompasses any neural network model which incorporates hidden states and uses its previous outputs as inputs.

Long Short-Term Memory Networks (LSTMs)

Long Short-Term Memory networks (LSTMs) are an advanced type of recurrent neural network designed to better handle long sequences. Unlike Simple RNNs that have a single memory channel, LSTMs have two memory channels that allows the network to selectively remember or forget information over long sequences. At each timestep, an LSTM:

- Takes in a new input
- Uses two types of state:
 - Cell State: Like a long-term memory that can preserve important information
 - Hidden State: Like a working memory for immediate processing
- Uses three gates to control information flow:
 - Forget gate: Decides what to remove from cell state
 - Input gate: Decides what new information to store from the current input
 - Output gate: Decides what parts of cell state to output

This dual-memory system allows LSTMs to maintain important information for longer periods while still processing new inputs effectively. Think of it like reading a book while taking notes - the cell state is like your notebook where you write down important information, while the hidden state is like your immediate attention to the current word and its context.

Architecture



For our character prediction task, we'll use a many-to-one LSTM configuration:

- Input: Characters fed in one at a time (like "m", "a", "c", "b", "e") as input
- Output: Get the predicted next character ("t") using the last hidden state

Simple RNNs struggle with long sequences because they try to pass all information through a single channel that gets increasingly noisy over time. LSTMs solve this by having two channels and specialized gates that allow the network to selectively update its memory state, rather than forcing all information through a single transformation. The memory cell pathway solves the vanishing gradient problem by providing a direct route for gradients to propagate backwards through time steps without being repeatedly multiplied by small

numbers that would cause them to vanish. The gating mechanisms ensure this pathway remains relatively stable while still allowing the network to selectively update its memory when needed.

For a deeper dive into LSTMs and their gate mechanisms, refer to:

- [Understanding LSTM Networks](#)
- [The Role of Gates in LSTMs](#)
- [Josh Starmer LSTM clearly explained](#)

Data Preparation

Loading the text

- We'll use Shakespeare's Macbeth from Project Gutenberg
- The text is first standardized by:
 - Converting to lowercase for consistency
 - Standardizing line endings and spacing
- This standardized text is referred to as the 'corpus' - our primary dataset for training and inference
- We vectorize the text by treating every character in our text as an individual unit (e.g., 'macbeth' -> ['m', 'a', 'c'...])

Character-to-Index Mapping

- Neural networks can't process raw text, so we need to convert characters to numbers.
- First, we find all the unique characters in our text (including alphabets and special characters like spaces and punctuations). This forms our vocabulary, whose size we call the vocabulary size
- Each character in the vocabulary receives a unique integer assignment using the default ASCII value
- We use a dictionary to store this mapping: {'a':1, 'b':2, 'c':3, ...}
- This mapping enables bidirectional conversion between characters and integers for model input and output interpretation
- We use this mapping to convert the vector of text into a vector of numerical values.

Index Representation

- You might be thinking that assigning numbers to letters based on alphabetical order seems rather arbitrary - and you'd be absolutely right! This is precisely why we need smarter index representations
- Using raw indices (like 1, 2, 3) implies an ordering and magnitude that doesn't exist
- For example, 'b' (2) is not "twice as much" as 'a' (1)
- We need a more sophisticated way to represent characters that avoids these misleading numerical relationships

a) **One-Hot Encoding:** Convert index to binary vector

- Each index becomes a vector of zeros with a single 1 at the index of the character's value/index
- Example:
 - index 1 ('a') → [0,1,0,0,...,0] (length 40)
 - index 5 ('e') → [0,0,0,0,1,0,...,0] (length 40)

- Sparse (mostly zeros)
- No relationship between characters (both vowels 'a' and 'e' are completely different vectors)

b) **Embeddings**: Convert index to learned dense vector

- Each character's index gets transformed into a list of numbers (a dense vector)
 - Think of it like assigning each character a unique point a higher dimension coordinate plane
 - The motivation is that we can position similar characters close to each other and transform discrete data like letters or words into a continuous vector space that the model can use to make predictions from
 - What makes embedding layers special is that they're learnable! The model figures out the best way to represent characters based on your specific data and task
 - This is super useful since what makes characters "similar" changes between tasks - the model learns what similarity means for your specific case and groups characters accordingly
 - The embedding layer's dimension parameter determines how many numbers we use to represent each character (the size of the vector space)
 - Let's look at a real example with embedding dimension 4:
 - The letter 'a' might become $\rightarrow [0.2, -0.5, 0.1, 0.8]$
 - The letter 'e' might become $\rightarrow [0.3, -0.4, 0.2, 0.7]$
 - In this example, model discovered that 'a' and 'e' are both vowels and represents them close to each other
 - Additionally, embeddings are more memory efficient because they are dense (non-sparse) unlike one-hot vectors
- For our model, we will be using an embedding layer in the model for index representations. Since this is a learnable layer, it will be part of our model architecture and not pre-processing.

Creating training data

- When preparing text data, we need to break it into fixed-length sequences for training. Each sequence becomes a training example where we try to predict the next character.
- Using a sliding window approach, we move through the text step by step. The sequence length (context window) is a key hyperparameter - too short and the model lacks context, too long and training becomes computationally intensive. For example, with text "macbeth" (context window=4):
 Window 1: "macb" $\rightarrow$ predict "e"
 Window 2: "acbe" $\rightarrow$ predict "t"
 Window 3: "cbet" $\rightarrow$ predict "h"
- Our input data is size (`NUM_SEQUENCES` , `SEQUENCE_LEN`), where `NUM_SEQUENCES` is the number of sequences of size `SEQUENCE_LEN` in the text
- Our target data is size (`NUM_SEQUENCES` , `1`) representing all the next characters for all sequences in the text
- NOTE: While we show characters in the examples above for clarity, remember that the actual input and target data are numerical indices, not the characters themselves

Please refer to **preprocess_text_data** located in utilities for more details.

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####
```

```

from utilities.utils import preprocess_text_data

# Load and preprocess text
text = requests.get("https://www.gutenberg.org/files/1533/1533-0.txt").text
DATA = preprocess_text_data(text)

# unpack processed data components
X, Y, TEXT, CHAR_INDICES, INDICES_CHAR, VOCAB, VOCAB_SIZE, SEQUENCE_LEN = (
    DATA["x"],
    DATA["y"],
    DATA["text"],
    DATA["char_indices"],
    DATA["indices_char"],
    DATA["vocab"],
    DATA["vocab_size"],
    DATA["sequence_len"],
)

print("Length of Corpus: ", len(TEXT))
print("Vocabulary Map: ", CHAR_INDICES)
print(f"Vocabulary Size: ", VOCAB_SIZE)
print(f"X shape: {X.shape}")
print(f"Y shape: {Y.shape}")

```

Length of Corpus: 102188

Vocabulary Map: {' ': 0, '!': 1, ',': 2, '-': 3, '.': 4, ':': 5, ';': 6, '?': 7, 'a': 8, 'b': 9, 'c': 10, 'd': 11, 'e': 12, 'f': 13, 'g': 14, 'h': 15, 'i': 16, 'j': 17, 'k': 18, 'l': 19, 'm': 20, 'n': 21, 'o': 22, 'p': 23, 'q': 24, 'r': 25, 's': 26, 't': 27, 'u': 28, 'v': 29, 'w': 30, 'x': 31, 'y': 32, 'z': 33}

Vocabulary Size: 34

X shape: (102158, 30)

Y shape: (102158, 1)

[codecarbon INFO @ 10:31:26] Energy consumed for RAM : 0.118681 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:31:26] Energy consumed for all CPUs : 0.855797 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:31:26] 0.974478 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:31:41] Energy consumed for RAM : 0.118705 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:31:41] Energy consumed for all CPUs : 0.855974 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:31:41] 0.974679 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:31:56] Energy consumed for RAM : 0.118730 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:31:56] Energy consumed for all CPUs : 0.856151 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:31:56] 0.974881 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:32:11] Energy consumed for RAM : 0.118754 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:32:11] Energy consumed for all CPUs : 0.856329 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:32:11] 0.975083 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:32:11] 0.006107 g.CO2eq/s mean an estimation of 192.5943983627131 kg.CO2eq/year

[codecarbon INFO @ 10:32:26] Energy consumed for RAM : 0.118779 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:32:26] Energy consumed for all CPUs : 0.856506 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:32:26] 0.975285 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:32:41] Energy consumed for RAM : 0.118803 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:32:41] Energy consumed for all CPUs : 0.856683 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:32:41] 0.975486 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:32:56] Energy consumed for RAM : 0.118828 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:32:56] Energy consumed for all CPUs : 0.856860 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:32:56] 0.975688 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:33:11] Energy consumed for RAM : 0.118853 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:33:11] Energy consumed for all CPUs : 0.857037 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:33:11] 0.975890 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:33:26] Energy consumed for RAM : 0.118877 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:33:26] Energy consumed for all CPUs : 0.857214 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:33:26] 0.976091 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:33:41] Energy consumed for RAM : 0.118902 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:33:41] Energy consumed for all CPUs : 0.857391 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:33:41] 0.976293 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:33:56] Energy consumed for RAM : 0.118926 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:33:56] Energy consumed for all CPUs : 0.857569 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:33:56] 0.976495 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:34:11] Energy consumed for RAM : 0.118951 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 10:34:11] Energy consumed for all CPUs : 0.857746 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:34:11] 0.976696 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:34:11] 0.006107 g.CO2eq/s mean an estimation of 192.5765873937271 kg.CO2eq/year
[codecarbon INFO @ 10:34:26] Energy consumed for RAM : 0.118975 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 10:34:26] Energy consumed for all CPUs : 0.857923 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:34:26] 0.976898 kWh of electricity used since the beginning.
```

Data Structure

Final preprocessed data includes:

- **X**: Input sequences
 - (shape: [NUM_SEQUENCES , SEQUENCE_LEN])
 - Contains all character sequences of length SEQUENCE_LEN
- **Y**: Target characters
 - (shape: [NUM_SEQUENCES , 1])
 - Contains the next character that follows each sequence in X
- **VOCABULARY MAP**: The mapping from all unique characters in the text and their numerical representations
- **VOCAB_SIZE**: Total number of unique characters
- **SEQUENCE_LEN**: Length of input sequences

5.1 Model Architecture [4.4% Bonus for All] [\[P\]](#)

Before diving into the specific architectures, let's understand how data shapes transform through the embedding layer.

Input Sequence Shape Flow:

1. Initial input: (BATCH_SIZE, SEQUENCE_LEN)

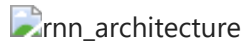
- BATCH_SIZE sequences containing SEQUENCE_LEN integers, where each integer represents a character from our vocabulary
- Example: If BATCH_SIZE=32 and SEQUENCE_LEN=15, shape is (32, 15)

2. Embedding Layer: (BATCH_SIZE, SEQUENCE_LEN, EMBEDDING_DIM)

- Transforms each integer into a vector of size EMBEDDING_DIM
- Example: If EMBEDDING_DIM=64:
 - Each character index becomes a vector of 64 numbers
 - Shape expands from (32, 15) to (32, 15, 64)
 - This means: 32 sequences, each 15 characters long, each character now represented by 64 numbers

5.1.1 Defining the Simple RNN Model [2.2% Bonus for All]

In this part, you need to build a simple recurrent neural network using tensorflow. The architecture of the model is outlined below:



[EMBEDDING - Simple RNN - FC - ACTIVATION]

EMBEDDING: The Embedding layer maps each integer (representing a character) in the input sequence to a dense vector representation. Each character index becomes a vector of **embedding dimension**. It has an input dimension of **vocab_size** (the total number of unique characters or tokens) and an output dimension defined by **embedding_dim**. This transformation allows the model to capture semantic relationships in the data.

- Input shape: (batch_size, sequence_length) - A sequence of character indices
- Output shape: (batch_size, sequence_length, embedding_dim) - Each character transformed into an embedding vector

Simple RNN: This layer processes the sequence data, passing information through time steps to learn temporal patterns. It has **rnn_units** neurons, determining the model's ability to capture dependencies in the sequential data.

- Input shape: (batch_size, sequence_length, embedding_dim) - Sequence of embedding vectors
- Output shape: (batch_size, rnn_units) - Final state output

FC (Dense Layer): A fully connected layer that transforms the RNN output to match the number of classes or possible output tokens. It has **vocab_size** neurons, ensuring that each output corresponds to a unique token or class.

- Input shape: (batch_size, rnn_units) - RNN final state
- Output shape: (batch_size, vocab_size) - Raw scores for each possible character

ACTIVATION (Softmax): The softmax activation function is applied to the output layer, producing a probability distribution over the vocabulary, allowing the model to interpret each output as a confidence level for each class.

- Input shape: (batch_size, vocab_size) - Raw scores
- Output shape: (batch_size, vocab_size) - Probability distribution over characters

You can refer to the following documentation on tensorflow layers for more details:

- [Embedding](#)
- [Simple RNN](#)
- [Dense](#)
- [Activation](#)

TODO: Implement the **define_model** function in **rnn.py**.

```
In [56]: #####  
### DO NOT CHANGE THIS CELL ###  
#####
```

```

from rnn import RNN

rnn_model = RNN(VOCAB_SIZE, SEQUENCE_LEN)
rnn_model.set_hyperparameters()
rnn_model.define_model()
rnn_model.build_model()
rnn_model.model.summary()

```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None , 34, 256)	8,704
simple_rnn (SimpleRNN)	(None , 128)	49,280
dense (Dense)	(None , 34)	4,386
activation (Activation)	(None , 34)	0

Total params: 62,370 (243.63 KB)

Trainable params: 62,370 (243.63 KB)

Non-trainable params: 0 (0.00 B)

```

In [ ]: from utilities.localtests import TestRNN

        tester = TestRNN("test_rnn_architecture").test_rnn_architecture()

```

test_rnn_architecture passed!

[codecarbon INFO @ 10:34:41] Energy consumed for RAM : 0.119000 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:34:41] Energy consumed for all CPUs : 0.858100 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:34:41] 0.977100 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:34:56] Energy consumed for RAM : 0.119025 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:34:56] Energy consumed for all CPUs : 0.858277 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:34:56] 0.977302 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:35:11] Energy consumed for RAM : 0.119049 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:35:11] Energy consumed for all CPUs : 0.858454 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:35:11] 0.977503 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:35:26] Energy consumed for RAM : 0.119074 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:35:26] Energy consumed for all CPUs : 0.858632 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:35:26] 0.977705 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:35:41] Energy consumed for RAM : 0.119098 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:35:41] Energy consumed for all CPUs : 0.858809 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:35:41] 0.977907 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:35:56] Energy consumed for RAM : 0.119123 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:35:56] Energy consumed for all CPUs : 0.858986 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:35:56] 0.978109 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:36:11] Energy consumed for RAM : 0.119147 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:36:11] Energy consumed for all CPUs : 0.859163 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:36:11] 0.978311 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:36:11] 0.006107 g.CO2eq/s mean an estimation of 192.59945719205578 kg.CO2eq/year

[codecarbon INFO @ 10:36:26] Energy consumed for RAM : 0.119172 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:36:26] Energy consumed for all CPUs : 0.859340 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:36:26] 0.978512 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:36:41] Energy consumed for RAM : 0.119196 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:36:41] Energy consumed for all CPUs : 0.859518 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:36:41] 0.978714 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:36:56] Energy consumed for RAM : 0.119221 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:36:56] Energy consumed for all CPUs : 0.859695 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:36:56] 0.978916 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:37:11] Energy consumed for RAM : 0.119246 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:37:11] Energy consumed for all CPUs : 0.859872 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:37:11] 0.979117 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:37:26] Energy consumed for RAM : 0.119270 kWh. RAM Power : 5.8941850662231445 W

```
[codecarbon INFO @ 10:37:26] Energy consumed for all CPUs : 0.860049 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:37:26] 0.979319 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:37:41] Energy consumed for RAM : 0.119295 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:37:41] Energy consumed for all CPUs : 0.860226 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:37:41] 0.979521 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:37:56] Energy consumed for RAM : 0.119319 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:37:56] Energy consumed for all CPUs : 0.860403 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:37:56] 0.979722 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:38:11] Energy consumed for RAM : 0.119344 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:38:11] Energy consumed for all CPUs : 0.860580 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:38:11] 0.979924 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:38:11] 0.006106 g.CO2eq/s mean an estimation of 192.56596844224782 kg.CO2eq/year
[codecarbon INFO @ 10:38:26] Energy consumed for RAM : 0.119368 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:38:26] Energy consumed for all CPUs : 0.860757 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:38:26] 0.980126 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:38:41] Energy consumed for RAM : 0.119393 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:38:41] Energy consumed for all CPUs : 0.860935 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:38:41] 0.980327 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:38:56] Energy consumed for RAM : 0.119417 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:38:56] Energy consumed for all CPUs : 0.861112 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:38:56] 0.980529 kWh of electricity used since the beginning.
```

5.1.2 Defining the LSTM Model [2.2% Bonus for All]

In this part, you need to build a long short-term memory (LSTM) network as described below. The architecture of the model is outlined below:



[EMBEDDING - LSTM - FC - ACTIVATION]

EMBEDDING: The Embedding layer maps each integer in the input sequence to a dense vector representation. It has an input dimension of **vocab_size** (the total number of unique characters or tokens) and an output dimension defined by **embedding_dim**. This transformation allows the model to capture semantic relationships in the data.

- Input shape: (batch_size, sequence_length) - A sequence of character indices
- Output shape: (batch_size, sequence_length, embedding_dim) - Each character transformed into an embedding vector

LSTM: This layer processes the sequence data, passing information through time steps to learn temporal patterns. It has **lstm_units** neurons, determining the LSTM ability to capture

dependencies in the sequential data.

- Input shape: (batch_size, sequence_length, embedding_dim) - Sequence of embedding vectors
- Output shape: (batch_size, lstm_units) - Final state output

FC (Dense Layer): A fully connected layer that transforms the LSTM output to match the number of classes or possible output tokens. It has **vocab_size** neurons, ensuring that each output corresponds to a unique token or class.

- Input shape: (batch_size, lstm_units) - LSTM final state
- Output shape: (batch_size, vocab_size) - Raw scores for each possible character

ACTIVATION (Softmax): The softmax activation function is applied to the output layer, producing a probability distribution over the vocabulary, allowing the model to interpret each output as a confidence level for each class.

- Input shape: (batch_size, vocab_size) - Raw scores
- Output shape: (batch_size, vocab_size) - Probability distribution over characters

You can refer to the following documentation on tensorflow layers for more details:

- [Embedding](#)
- [LSTM](#)
- [Dense](#)
- [Activation](#)

TODO: Implement the **define_model** function in **lstm.py**.

```
In [58]: #####  
### DO NOT CHANGE THIS CELL ###  
#####  
  
from lstm import LSTM  
  
lstm_model = LSTM(VOCAB_SIZE, SEQUENCE_LEN)  
lstm_model.set_hyperparameters()  
lstm_model.define_model()  
lstm_model.build_model()  
lstm_model.model.summary()
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
embedding_3 (Embedding)	(None, 34, 256)	8,704
lstm (LSTM)	(None, 128)	197,120
dense_3 (Dense)	(None, 34)	4,386
activation_3 (Activation)	(None, 34)	0

Total params: 210,210 (821.13 KB)

Trainable params: 210,210 (821.13 KB)

Non-trainable params: 0 (0.00 B)

In []:

```
#####  
### DO NOT CHANGE THIS CELL ###  
#####  
  
from utilities.localtests import TestLSTM  
  
tester = TestLSTM("test_lstm_architecture").test_lstm_architecture()
```

test_lstm_architecture passed!

[codecarbon INFO @ 10:39:11] Energy consumed for RAM : 0.119442 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:39:11] Energy consumed for all CPUs : 0.861289 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:39:11] 0.980731 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:39:26] Energy consumed for RAM : 0.119467 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:39:26] Energy consumed for all CPUs : 0.861466 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:39:26] 0.980933 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:39:41] Energy consumed for RAM : 0.119491 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:39:41] Energy consumed for all CPUs : 0.861643 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:39:41] 0.981134 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:39:56] Energy consumed for RAM : 0.119516 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:39:56] Energy consumed for all CPUs : 0.861820 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:39:56] 0.981336 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:40:11] Energy consumed for RAM : 0.119540 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:40:11] Energy consumed for all CPUs : 0.861998 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:40:11] 0.981538 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:40:11] 0.006107 g.CO2eq/s mean an estimation of 192.59215829135312 kg.CO2eq/year

[codecarbon INFO @ 10:40:26] Energy consumed for RAM : 0.119565 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:40:26] Energy consumed for all CPUs : 0.862175 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:40:26] 0.981740 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:40:41] Energy consumed for RAM : 0.119589 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:40:41] Energy consumed for all CPUs : 0.862352 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:40:41] 0.981942 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:40:56] Energy consumed for RAM : 0.119614 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:40:56] Energy consumed for all CPUs : 0.862529 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:40:56] 0.982143 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:41:11] Energy consumed for RAM : 0.119639 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:41:11] Energy consumed for all CPUs : 0.862706 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:41:11] 0.982345 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:41:26] Energy consumed for RAM : 0.119663 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:41:26] Energy consumed for all CPUs : 0.862884 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:41:26] 0.982547 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:41:41] Energy consumed for RAM : 0.119688 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:41:41] Energy consumed for all CPUs : 0.863061 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:41:41] 0.982748 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:41:56] Energy consumed for RAM : 0.119712 kWh. RAM Power : 5.8941850662231445 W

[codecarbon INFO @ 10:41:56] Energy consumed for all CPUs : 0.863238 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:41:56] 0.982950 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:42:11] Energy consumed for RAM : 0.119737 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:42:11] Energy consumed for all CPUs : 0.863415 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:42:11] 0.983152 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:42:11] 0.006107 g.CO2eq/s mean an estimation of 192.57750757765916 kg.CO2eq/year

[codecarbon INFO @ 10:42:26] Energy consumed for RAM : 0.119761 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:42:26] Energy consumed for all CPUs : 0.863592 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:42:26] 0.983353 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:42:41] Energy consumed for RAM : 0.119786 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:42:41] Energy consumed for all CPUs : 0.863769 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:42:41] 0.983555 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:42:56] Energy consumed for RAM : 0.119810 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:42:56] Energy consumed for all CPUs : 0.863947 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:42:56] 0.983757 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:43:11] Energy consumed for RAM : 0.119835 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:43:11] Energy consumed for all CPUs : 0.864124 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:43:11] 0.983959 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:43:26] Energy consumed for RAM : 0.119860 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:43:26] Energy consumed for all CPUs : 0.864301 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:43:26] 0.984160 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:43:41] Energy consumed for RAM : 0.119884 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:43:41] Energy consumed for all CPUs : 0.864478 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:43:41] 0.984362 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:43:56] Energy consumed for RAM : 0.119909 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:43:56] Energy consumed for all CPUs : 0.864655 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:43:56] 0.984564 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:44:11] Energy consumed for RAM : 0.119933 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:44:11] Energy consumed for all CPUs : 0.864833 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:44:11] 0.984766 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:44:11] 0.006106 g.CO2eq/s mean an estimation of 192.5736091739206 kg.CO2eq/year

[codecarbon INFO @ 10:44:26] Energy consumed for RAM : 0.119958 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:44:26] Energy consumed for all CPUs : 0.865010 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:44:26] 0.984968 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:44:41] Energy consumed for RAM : 0.119982 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:44:41] Energy consumed for all CPUs : 0.865187 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:44:41] 0.985169 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:44:56] Energy consumed for RAM : 0.120007 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:44:56] Energy consumed for all CPUs : 0.865364 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:44:56] 0.985371 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:45:11] Energy consumed for RAM : 0.120032 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:45:11] Energy consumed for all CPUs : 0.865541 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:45:11] 0.985573 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:45:26] Energy consumed for RAM : 0.120056 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:45:26] Energy consumed for all CPUs : 0.865718 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:45:26] 0.985774 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:45:41] Energy consumed for RAM : 0.120081 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:45:41] Energy consumed for all CPUs : 0.865895 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:45:41] 0.985976 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:45:56] Energy consumed for RAM : 0.120105 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:45:56] Energy consumed for all CPUs : 0.866073 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:45:56] 0.986178 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:46:11] Energy consumed for RAM : 0.120130 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:46:11] Energy consumed for all CPUs : 0.866250 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:46:11] 0.986380 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:46:11] 0.006107 g.CO2eq/s mean an estimation of 192.59307944857463 kg.CO2eq/year

[codecarbon INFO @ 10:46:26] Energy consumed for RAM : 0.120154 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:46:26] Energy consumed for all CPUs : 0.866427 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:46:26] 0.986581 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:46:41] Energy consumed for RAM : 0.120179 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:46:41] Energy consumed for all CPUs : 0.866604 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:46:41] 0.986783 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:46:56] Energy consumed for RAM : 0.120203 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:46:56] Energy consumed for all CPUs : 0.866781 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:46:56] 0.986985 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:47:11] Energy consumed for RAM : 0.120228 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:47:11] Energy consumed for all CPUs : 0.866958 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:47:11] 0.987186 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:47:26] Energy consumed for RAM : 0.120253 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:47:26] Energy consumed for all CPUs : 0.867136 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:47:26] 0.987388 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:47:41] Energy consumed for RAM : 0.120277 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:47:41] Energy consumed for all CPUs : 0.867313 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:47:41] 0.987590 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:47:56] Energy consumed for RAM : 0.120302 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:47:56] Energy consumed for all CPUs : 0.867490 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:47:56] 0.987791 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:48:11] Energy consumed for RAM : 0.120326 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:48:11] Energy consumed for all CPUs : 0.867667 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:48:11] 0.987993 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:48:11] 0.006106 g.CO2eq/s mean an estimation of 192.57237221655694 kg.CO2e
q/year
[codecarbon INFO @ 10:48:26] Energy consumed for RAM : 0.120351 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:48:26] Energy consumed for all CPUs : 0.867844 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:48:26] 0.988195 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:48:41] Energy consumed for RAM : 0.120375 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:48:41] Energy consumed for all CPUs : 0.868021 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:48:41] 0.988397 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:48:56] Energy consumed for RAM : 0.120400 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:48:56] Energy consumed for all CPUs : 0.868198 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:48:56] 0.988598 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:49:11] Energy consumed for RAM : 0.120424 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:49:11] Energy consumed for all CPUs : 0.868375 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:49:11] 0.988800 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:49:26] Energy consumed for RAM : 0.120449 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:49:26] Energy consumed for all CPUs : 0.868553 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:49:26] 0.989002 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:49:41] Energy consumed for RAM : 0.120474 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:49:41] Energy consumed for all CPUs : 0.868730 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:49:41] 0.989203 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:49:56] Energy consumed for RAM : 0.120498 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:49:56] Energy consumed for all CPUs : 0.868907 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:49:56] 0.989405 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:50:11] Energy consumed for RAM : 0.120523 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:50:11] Energy consumed for all CPUs : 0.869084 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:50:11] 0.989606 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:50:11] 0.006106 g.CO2eq/s mean an estimation of 192.5488962077181 kg.CO2eq/

year

[codecarbon INFO @ 10:50:26] Energy consumed for RAM : 0.120547 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:50:26] Energy consumed for all CPUs : 0.869261 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:50:26] 0.989808 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:50:41] Energy consumed for RAM : 0.120572 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:50:41] Energy consumed for all CPUs : 0.869438 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:50:41] 0.990010 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:50:56] Energy consumed for RAM : 0.120596 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:50:56] Energy consumed for all CPUs : 0.869615 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:50:56] 0.990211 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:51:11] Energy consumed for RAM : 0.120621 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:51:11] Energy consumed for all CPUs : 0.869792 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:51:12] 0.990413 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:51:27] Energy consumed for RAM : 0.120645 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:51:27] Energy consumed for all CPUs : 0.869969 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:51:27] 0.990615 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:51:42] Energy consumed for RAM : 0.120670 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:51:42] Energy consumed for all CPUs : 0.870147 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:51:42] 0.990816 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:51:57] Energy consumed for RAM : 0.120694 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:51:57] Energy consumed for all CPUs : 0.870324 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:51:57] 0.991018 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:52:12] Energy consumed for RAM : 0.120719 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:52:12] Energy consumed for all CPUs : 0.870501 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:52:12] 0.991220 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:52:12] 0.006106 g.CO2eq/s mean an estimation of 192.5586351141519 kg.CO2eq/year

[codecarbon INFO @ 10:52:27] Energy consumed for RAM : 0.120744 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:52:27] Energy consumed for all CPUs : 0.870678 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:52:27] 0.991421 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:52:42] Energy consumed for RAM : 0.120768 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:52:42] Energy consumed for all CPUs : 0.870855 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:52:42] 0.991623 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:52:57] Energy consumed for RAM : 0.120793 kWh. RAM Power : 5.89418506622314 45 W

[codecarbon INFO @ 10:52:57] Energy consumed for all CPUs : 0.871032 kWh. Total CPU Power : 42.5 W

[codecarbon INFO @ 10:52:57] 0.991825 kWh of electricity used since the beginning.

[codecarbon INFO @ 10:53:12] Energy consumed for RAM : 0.120817 kWh. RAM Power : 5.89418506622314

45 W
[codecarbon INFO @ 10:53:12] Energy consumed for all CPUs : 0.871209 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:53:12] 0.992027 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:53:27] Energy consumed for RAM : 0.120842 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:53:27] Energy consumed for all CPUs : 0.871386 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:53:27] 0.992228 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:53:42] Energy consumed for RAM : 0.120866 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:53:42] Energy consumed for all CPUs : 0.871564 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:53:42] 0.992430 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:53:57] Energy consumed for RAM : 0.120891 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:53:57] Energy consumed for all CPUs : 0.871741 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:53:57] 0.992632 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:54:12] Energy consumed for RAM : 0.120916 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:54:12] Energy consumed for all CPUs : 0.871918 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:54:12] 0.992834 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:54:12] 0.006107 g.CO2eq/s mean an estimation of 192.58439776199546 kg.CO2eq/year
[codecarbon INFO @ 10:54:27] Energy consumed for RAM : 0.120940 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:54:27] Energy consumed for all CPUs : 0.872095 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:54:27] 0.993035 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:54:42] Energy consumed for RAM : 0.120965 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:54:42] Energy consumed for all CPUs : 0.872272 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:54:42] 0.993237 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:54:57] Energy consumed for RAM : 0.120989 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:54:57] Energy consumed for all CPUs : 0.872449 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:54:57] 0.993438 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:55:12] Energy consumed for RAM : 0.121014 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:55:12] Energy consumed for all CPUs : 0.872626 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:55:12] 0.993640 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:55:27] Energy consumed for RAM : 0.121038 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:55:27] Energy consumed for all CPUs : 0.872804 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:55:27] 0.993842 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:55:42] Energy consumed for RAM : 0.121063 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:55:42] Energy consumed for all CPUs : 0.872981 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 10:55:42] 0.994044 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:55:57] Energy consumed for RAM : 0.121087 kWh. RAM Power : 5.89418506622314 45 W
[codecarbon INFO @ 10:55:57] Energy consumed for all CPUs : 0.873158 kWh. Total CPU Power : 42.5

```

W
[codecarbon INFO @ 10:55:57] 0.994246 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:56:12] Energy consumed for RAM : 0.121112 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:56:12] Energy consumed for all CPUs : 0.873335 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:56:12] 0.994447 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:56:12] 0.006107 g.CO2eq/s mean an estimation of 192.5926888920646 kg.CO2eq/
year
[codecarbon INFO @ 10:56:27] Energy consumed for RAM : 0.121137 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:56:27] Energy consumed for all CPUs : 0.873512 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:56:27] 0.994649 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:56:42] Energy consumed for RAM : 0.121161 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:56:42] Energy consumed for all CPUs : 0.873690 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:56:42] 0.994851 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:56:57] Energy consumed for RAM : 0.121186 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:56:57] Energy consumed for all CPUs : 0.873867 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:56:57] 0.995053 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:57:12] Energy consumed for RAM : 0.121210 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:57:12] Energy consumed for all CPUs : 0.874044 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:57:12] 0.995254 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:57:27] Energy consumed for RAM : 0.121235 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:57:27] Energy consumed for all CPUs : 0.874221 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:57:27] 0.995456 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:57:42] Energy consumed for RAM : 0.121259 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:57:42] Energy consumed for all CPUs : 0.874398 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:57:42] 0.995658 kWh of electricity used since the beginning.
[codecarbon INFO @ 10:57:57] Energy consumed for RAM : 0.121284 kWh. RAM Power : 5.89418506622314
45 W
[codecarbon INFO @ 10:57:57] Energy consumed for all CPUs : 0.874575 kWh. Total CPU Power : 42.5
W
[codecarbon INFO @ 10:57:57] 0.995859 kWh of electricity used since the beginning.

```

5.2 RNN vs LSTM Model Text Generation Training Comparison Analysis [2.4% Bonus for All] [W]

Next, we'll train our RNN and LSTM models with their respective hyperparameters for character-level text generation. The model architecture is designed to process fixed-length sequences and predict the next most likely character based on the learned patterns.

During training, we need one final transformation of our target data.

- As shown in both architectures, the model outputs a probability distribution over all possible characters (VOCAB_SIZE)

- To compare this with our target data, we convert our target indices into one-hot vectors
- For example, if our target is 'e' or 5 and VOCAB_SIZE is 34:
 - Target index: [5] → One-hot: [0, 0, 0, 0, 1, 0, ..., 0] (length 34)
- This transformation happens in the training loop, converting Y from shape (BATCH_SIZE, 1) to (BATCH_SIZE, VOCAB_SIZE)
- The categorical cross-entropy loss can then compare our predicted probability distribution with this one-hot target distribution

Training Configuration

- **Optimizer:** RMSprop (Root Mean Square Propagation) optimizer for efficient training of recurrent networks. Read more about it [here](#).
- **Loss Function:** Categorical crossentropy to measure accuracy of predicted probability distribution. Read more about it [here](#).
- **Training Parameters:** Number of epochs and batch size are defined in the hyperparameters section of `rnn.py` and `lstm.py` respectively

The core training loop implementation can be found in `base_sequential_model.py`.

NOTE: The initial model training may take 10-15 minutes per model. After that, the function will automatically load the saved weights stored in the `model_weights` directory, making subsequent runs much faster. If you want to retrain from scratch instead of using saved weights, you can either:

- Set `train_from_scratch=True` in the parameters
- Delete the existing weights from the `model_weights` directory

Training Visualization

To monitor the training progress:

- A plot showing loss metrics across epochs will be generated
- This helps identify potential overfitting or convergence issues
- The visualization will be displayed automatically after training completes

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

rnn_model.train(x=X, y=Y, train_from_scratch=False)
rnn_model.plot_loss()
```

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

lstm_model.train(x=X, y=Y, train_from_scratch=False)
lstm_model.plot_loss()
```

After training both models, we can now use them to generate text in the style of Macbeth. The generation process involves three key components:

1. Seed Selection

- Start with a random seed sequence from our text corpus
- This sequence provides initial context for the generation

2. Prediction Loop

- Model predicts probability over all characters based on current sequence and we sample next character
- Append generated character to the sequence
- Slide window forward by one character to maintain context length and predict using generated character

3. Temperature Parameter

- Controls the randomness of predictions:
 - Lower values (< 0.5): More conservative, deterministic text
 - Higher values (> 1.0): More diverse but potentially less coherent text
 - Default value of 0.5 balances coherence and creativity

Below, we use our TextGenerator class to generate samples from both models:

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

from text_generator import TextGenerator

generator = TextGenerator(CHAR_INDICES, INDICES_CHAR, SEQUENCE_LEN)
```

```
In [ ]: #####
      ### DO NOT CHANGE THIS CELL ###
      #####

start_index = random.randint(0, len(TEXT) - SEQUENCE_LEN - 1)
seed_text = TEXT[start_index : start_index + SEQUENCE_LEN]

generator.generate(model=rnn_model, seed_text=seed_text, length=150, temperature=0.75)
generator.generate(model=lstm_model, seed_text=seed_text, length=150, temperature=0.75)
```

Analyze your experience training Simple RNN and LSTM models for Macbeth character prediction and answer the following:

1. Identify and explain a real-world application where Simple RNN would be more suitable than LSTM
2. Identify and explain a real-world application where LSTM would be more suitable than Simple RNN

For each case, support your answer using evidence from at least 1 of these aspects from your training:

A. Generated Text Quality

- What patterns and quality differences did you observe in the text output?

B. Training Duration

- How long did the model take to train?

C. Loss Convergence

- How did the loss values change across training epochs?

D. Final Loss Achieved

- What was the final loss produced by the model?

You can research and add other considerations to strengthen your argument, but ensure you include at least 1 observed metric from your training. Some additional considerations include:

- Inference speed
- Memory requirements
- Ability to maintain context
- Simplicity of architecture

You will receive full credit if you use evidence from training and make a sound argument about why one model would be more suitable than the other for both cases.

Sample Response Format

A. Simple RNN Best Use Case: [Application Name]

- Choose 1+ observed metrics:
 - [Metric name]: [What you observed in your training]
- Why this matters: [Explain how your evidence supports using RNN over LSTM for this application]

B. LSTM Best Use Case: [Application Name]

- Choose 1+ observed metrics:
 - [Metric name]: [What you observed in your training]
- Why this matters: [Explain how your evidence supports using LSTM over RNN for this application]

Optional, but recommended: Add theoretical considerations to strengthen your argument

- [Theoretical difference & why it matters for your chosen application]

YOUR ANSWER HERE

```
In [66]: tracker.stop()
```

```
[codecarbon INFO @ 12:13:57] Energy consumed for RAM : 0.128746 kWh. RAM Power : 5.8941850662231445 W
[codecarbon INFO @ 12:13:57] Energy consumed for all CPUs : 0.928403 kWh. Total CPU Power : 42.5 W
[codecarbon INFO @ 12:13:57] 1.057149 kWh of electricity used since the beginning.
c:\Users\devin\anaconda3\envs\ml_hw4\Lib\site-packages\codecarbon\output_methods\file.py:52: FutureWarning: The behavior of DataFrame concatenation with empty or all-NA entries is deprecated. In a future version, this will no longer exclude empty or all-NA columns when determining the result dtypes. To retain the old behavior, exclude the relevant entries before the concat operation.
  df = pd.concat([df, pd.DataFrame.from_records([dict(total.values)])])
```

```
Out[66]: 0.4803557234720255
```

